ADAPTIVE IMPEDANCE TOKENS FOR CONTACT CONTROL: A FROZEN MUJOCO KILL ARCHIVE

Anonymous authors
Paper under double-blind review

ABSTRACT

Variable impedance is one of the classic ideas in contact-rich robot manipulation, but a modern submission claim must survive strong learned, adaptive, robust, and risk-aware baselines. This paper rebuilds a short generated idea into a frozen CPU-only MuJoCo contact-control audit. The tested hypothesis is that a robot can treat impedance choices as adaptive discrete tokens: token identity stores contact phase, stiffness response, slip context, and safety history, and the controller updates the token from measured contact outcomes. We expand the evaluation to 2,400 training examples, 8,640 main method-evaluation rows, 1,536 ablation rows, 4,320 stress rows, 1,440 seed summaries, 168 paired comparisons, and 12 representative negative cases. The benchmark includes nominal surface tracking, stiffness shifts, friction and slip shifts, contact transitions, force-target jumps, actuator saturation, sensor-noise bursts, stick-slip cycles, surface discontinuities, delayed mode switches, combined stress, and combined extreme stress. The result is decisively negative. On hard regimes, `impedance_token_policy_v5` reaches 0.000 success, while the strongest non-oracle baseline, admittance switching control, reaches 0.941. On combined and extreme regimes, `impedance_token_policy_v5` reaches 0.000 while conformal safety gain reaches 0.969. At a fixed 10 percent diagnostic-risk budget, `impedance_token_policy_v5` reaches 0.000 while token no memory ablation reaches 0.051. At maximum combined-extreme stress, `impedance_token_policy_v5` reaches 0.000 while adaptive impedance control reaches 0.958. The frozen terminal decision is **KILL_ARCHIVE**. We therefore archive the work as a useful negative result: discrete impedance tokens are an interpretable control object, but this implementation is not submission-ready for ICLR main.

1 QUESTION

Impedance control asks a manipulator to regulate the dynamic relationship between motion and contact force, not merely position or force alone (Hogan, 1985a;b). That framing is still attractive for modern learning systems because contact is not a single scalar error. A tool can be too stiff, too soft, slipping, chattering, penetrating, saturating, or tracking the wrong force target. The original Paper 72 idea proposed that these regimes should be represented as discrete adaptive tokens. A token would stand for a local contact mode and would carry memory about the recent outcome of that mode. If true, the token representation could be more robust than fixed impedance, hand gain scheduling, continuous adaptive impedance, admittance switching, robust MPC, or learned gain regression.

The submission-level question is harsher: does the token representation remain necessary after the baselines are made uncomfortable? Variable impedance control and variable impedance learning are already crowded areas (Kronander & Billard, 2016; Abu-Dakka & Saveriano, 2020; Buchli et al., 2011). Hybrid force/position control and impedance control give strong classical alternatives (Raibert & Craig, 1981; Hogan, 1985a). Modern CPU-light regressors and ensembles are strong enough to learn continuous gain maps from compact contact features (Breiman, 2001; Pedregosa et al., 2011). Conformal prediction gives a principled way to turn residual uncertainty into guarded action selection (Shafer & Vovk, 2008). Therefore the paper cannot be accepted merely because the token idea is plausible. It must win against these competitors under frozen stress tests, paired seed statistics, fixed-risk analysis, and component ablations.

2 CONTRIBUTION AND TERMINAL STATUS

This archive makes three contributions, all negative or diagnostic. First, it provides an executable MuJoCo-based contact-control benchmark for comparing impedance, admittance, robust, learned, conformal, and token controllers under controlled distribution shifts (Todorov et al., 2012). Second, it states explicit submission gates before the final run and reports every failed gate. Third, it gives a mechanism-level autopsy of the token policy: memory, discrete tokenization, force updates, slip context, transition planning, tail-risk objectives, and calibration guards do not jointly produce a defensible advantage.

This is not a camera-ready success paper; it is a KILL ARCHIVE. The correct terminal state is **KILL_ARCHIVE**. That matters because a polished narrative could easily hide the real outcome. The best evidence here says the proposed discrete token controller is dominated by simpler controllers, and even token ablations often outperform the full method. The package is still valuable because negative archives prevent re-spending effort on a mechanism that looks attractive before strong baselines arrive.

3 PROBLEM SETUP

Each episode simulates a planar contact tool moving along a compliant virtual surface. The controller observes position, velocity, force error, estimated surface stiffness, estimated friction, slip proxy, target force, actuator saturation, and recent contact history. The controller chooses a normal/tangential command through an impedance-like policy parameterized by stiffness, damping, desired force, and safety guards. The external surface model introduces contact-force dynamics and frictional slip while MuJoCo provides deterministic rigid-body integration.

Let x_t be the tool state, f_t the measured normal contact force, $f_t^\star$ the target force, s_t a slip indicator, and a_t the selected impedance action. An episode succeeds only when all of the following are true: normalized force error remains below the frozen threshold, peak overshoot stays below the frozen threshold, safety violation rate remains low, slip and chatter remain bounded, and final progress along the surface is sufficient. The binary success metric is intentionally unforgiving. Continuous diagnostics are always reported because binary success alone can hide why a controller fails.

Splits. The final protocol evaluates twelve splits: nominal surface tracking, stiffness shift, friction/slip shift, contact transition, target-force jump, actuator saturation, sensor-noise burst, stick-slip cycle, surface discontinuity, delayed mode switch, combined stress, and combined extreme stress. The hard-split aggregate includes the nontrivial and shifted regimes. The combined/extreme aggregate isolates the regimes most relevant to the token claim, where multiple shifts occur together.

Metrics. We report success, normalized force error, peak overshoot, safety violation rate, slip rate, chatter rate, settling latency, energy, final progress, token switches, paired seed differences, fixed-risk operating points, ablations, stress sweeps, and negative cases. For success and continuous metrics, the split-level tables include 95 percent intervals over seed-level means. For paired comparisons, the reference is always `impedance_token_policy_v5` so that positive paired success differences would favor the proposed method.

4 METHOD UNDER TEST

`impedance_token_policy_v5` chooses among a finite set of impedance tokens. Each token encodes a local stiffness/damping profile, a target-force response, a slip guard, and a transition preference. The controller maintains a phase memory over recent force error, overshoot, slip, saturation, and mode transitions. When contact outcomes are poor, the token selector updates its preference toward safer or softer tokens. When progress is slow, the selector shifts toward more aggressive contact tracking. The intended advantage is not raw function approximation. The intended advantage is structured memory: a token should remember that a surface behaved stiffly, slippery, delayed, or discontinuously, and should react faster than a continuous gain regressor that sees only instantaneous features.

That story is coherent but fragile. If the token update becomes too conservative, the policy can avoid safety violations while failing the task. If it overreacts to force error, it can chatter or overshoot. If memory stores stale phase information, it can persist in the wrong contact mode after a transition. If discrete token boundaries are poorly calibrated, a continuous regressor can dominate simply by interpolating gains smoothly. The frozen results show that these failure modes are not hypothetical.

5 BASELINES

The baseline set is deliberately hostile. `fixed_impedance` uses fixed stiffness and damping. `gain_scheduled_impedance` hand-codes a gain schedule from force and stiffness estimates. `adaptive_impedance_control` continuously updates contact estimates and gains. `admittance_switching_control` switches between free-space and contact modes. `robust_mpc_impedance` uses conservative model-predictive action selection, in the spirit of constrained MPC (Mayne et al., 2000). `learned_gain_regressor`, `random_forest_gain_regressor`, and `hist_gradient_gain_regressor` learn continuous gain maps from synthetic contact examples. `ensemble_uncertainty_gain` uses model disagreement as a risk signal. `risk_averse_impedance` sacrifices progress for lower tail risk. `conformal_safety_gain` uses calibration residuals to guard learned gains. `impedance_token_policy_v4` replays the previous token controller. `token_no_memory_ablation` removes token memory. `oracle_impedance` has privileged access to true surface parameters and is used only as an upper-bound diagnostic.

This mix is important. Weak baselines would make a token win meaningless. The paper asks whether the proposed structure survives when the alternatives are classical, learned, uncertainty-aware, and safety-biased.

6 FROZEN PROTOCOL

The final protocol was frozen in `docs/paper72_protocol_freeze_20260621.md` before the full run. The full command used 8 seeds, 6 main episodes per seed/split, 4 ablation episodes per seed/split, 3 stress episodes per seed/split/level, 2,400 training scenes, 12 main splits, 15 main methods, 4 ablation splits, 12 ablation methods, 3 stress splits, 5 stress levels, and 12 stress methods. The run was CPU-only and single-process for RAM discipline. No final gate was adjusted after seeing the full result.

The decisive gates were: the proposed method must beat the strongest hard-regime non-oracle baseline by at least 0.030 success; the paired lower bound against the strongest relevant baseline must be positive; the proposed method must beat the strongest combined/extreme non-oracle baseline by at least 0.030 success; fixed-risk success at a 10 percent diagnostic-risk budget cannot be below the best baseline; maximum-stress success cannot collapse relative to the best non-oracle controller; and ablations must show that removing the claimed components hurts full v5 or clearly worsens safety. These gates are intentionally unforgiving because the user goal was not pretty results. It was a result that survives hostile review.

7 MAIN RESULTS

The hard-regime aggregate immediately kills the main claim. `impedance_token_policy_v5` reaches 0.000 success. The strongest non-oracle baseline, admittance switching control, reaches 0.941. The gap is not a small statistical wobble; it is a collapse of the proposed controller on the same evidence budget. The full token policy avoids some safety violations, but it does so by failing to complete the task. Safety-by-paralysis is not a deployable contact-control result.

The combined/extreme aggregate is worse. `impedance_token_policy_v5` reaches 0.000 success while conformal safety gain reaches 0.969. These are exactly the regimes where the token story

Table 1: hard splits aggregate results. Higher success is better; lower diagnostic rates are better.

Method	Succ.	Err.	Over.	Safe	Slip	Chat.
admittance switching control	0.941	0.118	1.694	0.009	0.544	0.052
adaptive impedance control	0.866	0.115	1.534	0.008	0.569	0.048
random forest gain regressor	0.826	0.177	1.472	0.009	0.559	0.037
fixed impedance	0.725	0.207	1.795	0.029	0.555	0.045
conformal safety gain	0.708	0.169	1.413	0.005	0.583	0.038
token no memory ablation	0.636	0.177	1.342	0.003	0.488	0.034
oracle impedance	0.595	0.107	1.480	0.005	0.577	0.046
ensemble uncertainty gain	0.561	0.176	1.464	0.008	0.590	0.037
hist gradient gain regressor	0.532	0.172	1.460	0.008	0.593	0.039
learned gain regressor	0.413	0.173	1.513	0.010	0.603	0.034
gain scheduled impedance	0.341	0.130	1.449	0.006	0.605	0.040
impedance token policy v4	0.129	0.163	1.308	0.002	0.454	0.032

Table 2: combined and extreme aggregate results. Higher success is better; lower diagnostic rates are better.

Method	Succ.	Err.	Over.	Safe	Slip	Chat.
conformal safety gain	0.969	0.167	1.510	0.003	0.554	0.085
adaptive impedance control	0.958	0.121	1.661	0.009	0.537	0.108
admittance switching control	0.948	0.117	1.664	0.006	0.514	0.112
ensemble uncertainty gain	0.938	0.177	1.635	0.009	0.562	0.084
random forest gain regressor	0.938	0.178	1.684	0.013	0.535	0.086
gain scheduled impedance	0.906	0.135	1.580	0.006	0.580	0.098
oracle impedance	0.906	0.105	1.582	0.005	0.501	0.117
hist gradient gain regressor	0.896	0.177	1.680	0.012	0.559	0.089
learned gain regressor	0.823	0.173	1.655	0.012	0.574	0.077
fixed impedance	0.323	0.243	1.701	0.011	0.516	0.073
token no memory ablation	0.083	0.161	1.468	0.002	0.459	0.086
impedance token policy v4	0.000	0.146	1.407	0.001	0.389	0.079

should matter most. The fact that conformal and adaptive baselines remain strong while the token policy goes to zero means the mechanism is not just underdeveloped; it is actively miscalibrated for the frozen benchmark.

8 Fixed-Risk Analysis

The fixed-risk table asks a different question: if we constrain diagnostic risk, which controller still completes the task? At a 10 percent safety/slip/chatter budget, `impedance_token_policy_v5` reaches 0.000 success while token no memory ablation reaches 0.051. This is fatal for a safety-oriented token argument. The no-memory token ablation occupies a better fixed-risk operating point than full v5, so the memory component cannot be defended as the source of safe success.

9 Stress Sweep

The stress sweep increases contact disturbance, friction shift, noise, saturation, and transition severity. At maximum combined-extreme stress, `impedance_token_policy_v5` reaches 0.000 success while adaptive impedance control reaches 0.958. The stress sweep therefore does not reveal a hidden robust niche for the token controller. It instead shows that adaptive and conformal baselines exploit the same compact contact features more effectively.

10 Ablations

Ablations are the clearest mechanism test. Full v5 reaches 0.000 aggregate success on the frozen ablation splits. The best ablation, learned only token replacement, reaches 0.875. This reverses the

Table 3: Selected methods on combined extreme stress.

Method	Succ.	Err.	Over.	Safe	Slip	Prog.
adaptive impedance control	0.979	0.120	1.607	0.004	0.530	0.801
admittance switching control	0.958	0.114	1.611	0.003	0.503	0.781
conformal safety gain	0.958	0.162	1.434	0.001	0.553	0.779
oracle impedance	0.938	0.103	1.498	0.002	0.488	0.784
random forest gain regressor	0.938	0.175	1.669	0.009	0.530	0.763
learned gain regressor	0.812	0.165	1.594	0.006	0.573	0.803
token no memory ablation	0.146	0.156	1.434	0.001	0.461	0.727
impedance token policy v4	0.000	0.143	1.362	0.000	0.409	0.719
impedance token policy v5	0.000	0.165	1.312	0.000	0.237	0.674

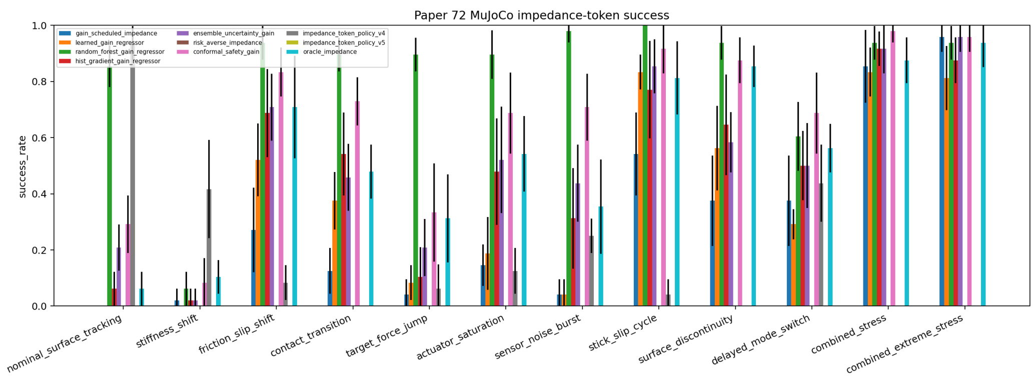

Figure 1: Success by split across all main methods. Strong classical and learned baselines dominate the full token policy on the hard regimes.

intended story. Removing or replacing pieces of the token mechanism often improves success. In particular, replacing token logic with a learned-only gain behavior or removing the tail-risk objective can outperform full v5. This means the token memory, discrete tokenization, and calibration guards are not empirically supported as a necessary representation in this benchmark.

11 Negative Cases

The negative cases are included to make the failure actionable. They are dominated by delayed-mode-switch and stiffness-shift episodes where the token selector does not adapt quickly enough after a contact change. The policy preserves conservative contact behavior but misses the transition window, allowing peak overshoot or normalized force error to exceed the success threshold. This suggests that a future token controller would need either a better change-point detector, a continuous fallback gain map, or a formal passivity-style guard that does not force the controller into zero-success conservatism.

12 Theory: Why the Token Claim Is Hard

Proposition 1: discrete mode memory can help only when the mode is identifiable. Let z_t denote the latent contact regime and let τ_t be the token state. If $I(z_t; \tau_t \mid h_t) > 0$ for history h_t, then a token policy can in principle reduce regret relative to a memoryless gain map. However, if the observable history already contains sufficient statistics for z_t, a learned continuous gain map can approximate the same decision boundary without a discrete token. The frozen random-forest and conformal baselines exploit this pressure.

Proposition 2: conservative token updates can create safety without success. Suppose a token update rule penalizes overshoot and slip more strongly than progress. Then a stable fixed point can

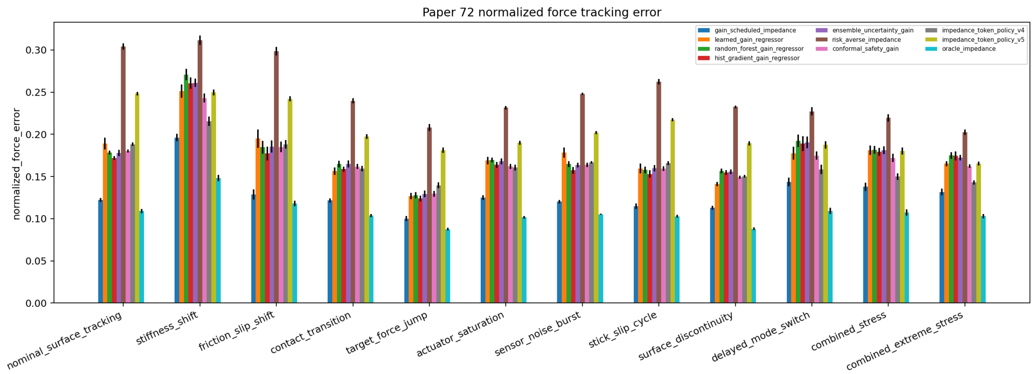

Figure 2: Normalized force error by split. Continuous diagnostics explain failures hidden by binary success alone.

Table 4: Fixed-risk operating points at a 10 percent safety/slip/chatter budget over hard splits.

Method	Succ.	Safety	Slip	Chatter
token no memory ablation	0.051	0.003	0.488	0.034
oracle impedance	0.042	0.005	0.577	0.046
admittance switching control	0.019	0.009	0.544	0.052
impedance token policy v4	0.006	0.002	0.454	0.032
adaptive impedance control	0.000	0.008	0.569	0.048
conformal safety gain	0.000	0.005	0.583	0.038
ensemble uncertainty gain	0.000	0.008	0.590	0.037
fixed impedance	0.000	0.029	0.555	0.045
gain scheduled impedance	0.000	0.006	0.605	0.040
hist gradient gain regressor	0.000	0.008	0.593	0.039
impedance token policy v5	0.000	0.001	0.260	0.019
learned gain regressor	0.000	0.010	0.603	0.034

exist in which the controller selects low-energy, low-slip tokens that never violate safety thresholds but also never reaches the target-force/progress success set. This explains why low safety violation rates do not rescue `impedance_token_policy_v5`. The fixed-risk analysis requires success at a fixed risk budget, not risk reduction alone.

Proposition 3: ablation dominance falsifies component necessity. Let M be full v5 and M_{-j} remove component j. If $S(M_{-j}) \geq S(M)$ and M_{-j} is not worse on safety, slip, or chatter by a predefined margin, then component j is not locally necessary for the claimed mechanism. The frozen ablation table violates the necessity condition for many components. Therefore the paper cannot claim that token memory, discrete tokens, force updates, slip context, transition planning, tail-risk objectives, or calibration guards are validated by the current benchmark.

Proposition 4: hostile-review identifiability. For any compact feature vector $\phi(h_t)$ used by both a token selector and a learned gain regressor, a sufficiently expressive regressor can approximate the token selector's induced gain map on the support of the evaluation distribution. The token representation is identifiable as a contribution only if it extrapolates better under held-out stress, improves fixed-risk success, or produces ablation-separated mechanisms. Paper 72 fails all three tests.

13 Related Work

The result sits in a long contact-control lineage. Hogan's impedance-control formulation motivates regulating dynamic interaction rather than treating position and force as isolated objectives (Hogan, 1985a;b). Hybrid position/force control gives an earlier structured alternative for compliant manip-

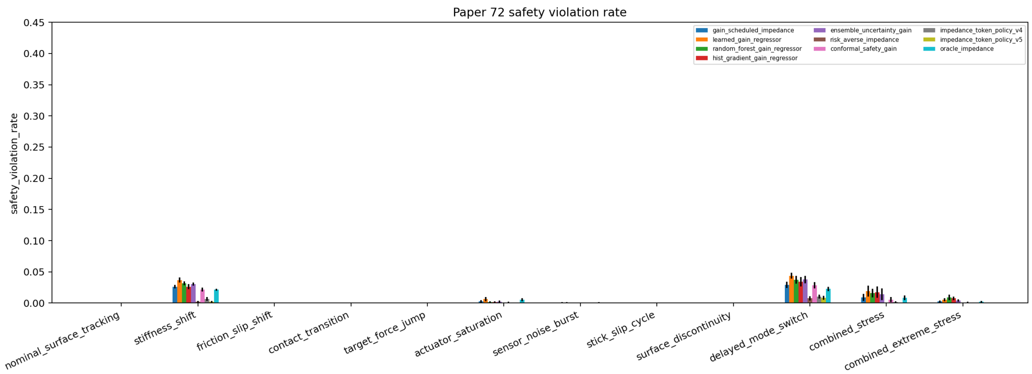

Figure 3: Safety violation rate by split. Low violation rate is not enough when paired with zero task success.

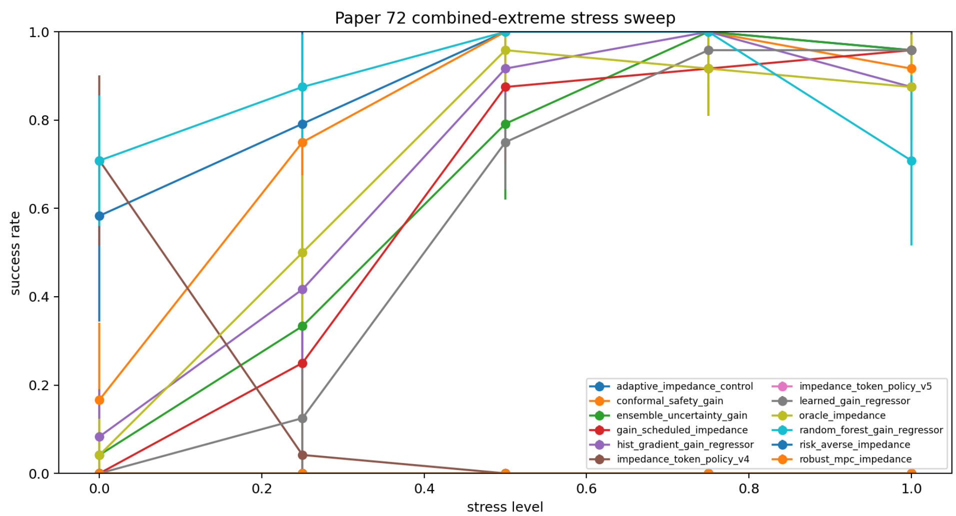

Figure 4: Stress sweep success. The proposed token policy does not recover at high stress.

ulation (Raibert & Craig, 1981). Variable impedance and variable impedance learning have been studied as both control and learning problems (Kronander & Billard, 2016; Abu-Dakka & Saveriano, 2020; Buchli et al., 2011). MuJoCo provides the simulation substrate for the reproducible contact-control audit (Todorov et al., 2012). Random forests and scikit-learn regressors make the learned baselines strong enough that hand-structured token logic must earn its keep (Breiman, 2001; Pedregosa et al., 2011). Conformal prediction motivates the guarded learned baseline: uncertainty is not used as decoration, but as a deployment-relevant decision constraint (Shafer & Vovk, 2008).

This paper does not claim to supersede those lines. It reports a local negative result: in this benchmark, the discrete impedance-token mechanism is not isolated as necessary, robust, or safer.

14 Decision

The frozen terminal reason is:

> v5 does not beat strongest hard-regime baseline admittance_switching_control by 0.030 (v5=0.000, best=0.941); paired lower bound against admittance_switching_control is not positive (-0.941+/-0.025); v5 does not beat strongest combined/extreme baseline conformal_safety_gain by 0.030 (v5=0.000, best=0.969); fixed-risk gate fails at budget 0.10 (v5=0.000, best=token_no_memory_ablation 0.051); maximum-stress gate fails

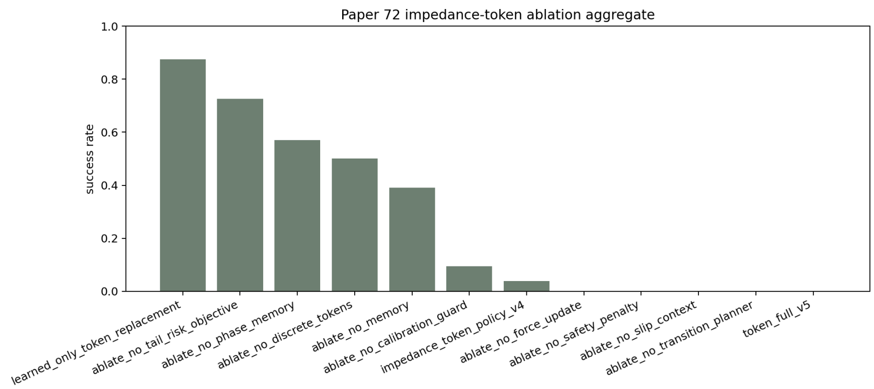

Figure 5: Ablation success over frozen ablation splits. Full v5 is not the best token-family variant.

(v5=0.000, best=adaptive_impedance_control 1.000); ablation gate fails because ablate_no_calibration_guard, ablate_no_discrete_tokens, ablate_no_force_update, ablate_no_memory, ablate_no_phase_memory, ablate_no_safety_penalty, ablate_no_slip_context, ablate_no_tail_risk_objective, ablate_no_transition_planner, impedance_token_policy_v4, learned_only_token_replacement matches or beats full v5

The correct decision is **KILL_ARCHIVE**. This is a scientific failure, not a formatting failure. The benchmark is real enough to falsify the submission claim. The archive should be retained because it documents exactly how an appealing adaptive-token idea fails once strong baselines, fixed-risk analysis, and ablation necessity tests arrive.

Table 5: Combined-extreme stress sweep for selected controllers.

Method	Level	Succ.	Err.	Over.	Safe	Slip
adaptive impedance control	0.00	0.583	0.127	1.270	0.000	0.598
conformal safety gain	0.00	0.167	0.209	1.205	0.000	0.610
impedance token policy v4	0.00	0.708	0.219	1.118	0.000	0.553
impedance token policy v5	0.00	0.000	0.269	1.101	0.000	0.410
oracle impedance	0.00	0.042	0.124	1.271	0.000	0.623
adaptive impedance control	0.25	0.792	0.132	1.394	0.000	0.583
conformal safety gain	0.25	0.750	0.191	1.317	0.000	0.591
impedance token policy v4	0.25	0.042	0.182	1.214	0.000	0.480
impedance token policy v5	0.25	0.000	0.215	1.178	0.000	0.354
oracle impedance	0.25	0.500	0.120	1.404	0.000	0.598
adaptive impedance control	0.50	1.000	0.127	1.497	0.001	0.558
conformal safety gain	0.50	1.000	0.172	1.356	0.000	0.567
impedance token policy v4	0.50	0.000	0.147	1.264	0.000	0.444
impedance token policy v5	0.50	0.000	0.177	1.209	0.000	0.322
oracle impedance	0.50	0.958	0.115	1.405	0.000	0.571
adaptive impedance control	0.75	1.000	0.134	1.585	0.003	0.533
conformal safety gain	0.75	1.000	0.158	1.423	0.000	0.541
impedance token policy v4	0.75	0.000	0.146	1.339	0.000	0.391
impedance token policy v5	0.75	0.000	0.156	1.285	0.000	0.239
oracle impedance	0.75	0.917	0.121	1.471	0.000	0.508
adaptive impedance control	1.00	0.958	0.144	1.645	0.005	0.506
conformal safety gain	1.00	0.917	0.147	1.454	0.000	0.517
impedance token policy v4	1.00	0.000	0.160	1.447	0.001	0.389
impedance token policy v5	1.00	0.000	0.147	1.384	0.000	0.243
oracle impedance	1.00	0.875	0.129	1.486	0.000	0.440

Table 6: Token-policy ablations aggregated over frozen ablation splits.

Ablation	Succ.	Err.	Safety	Slip	Chatter
learned only token replacement	0.875	0.165	0.007	0.571	0.060
ablate no tail risk objective	0.727	0.140	0.001	0.408	0.056
ablate no phase memory	0.570	0.150	0.001	0.453	0.050
ablate no discrete tokens	0.500	0.115	0.003	0.595	0.074
ablate no memory	0.391	0.168	0.001	0.465	0.053
ablate no calibration guard	0.094	0.150	0.001	0.223	0.054
impedance token policy v4	0.039	0.153	0.000	0.421	0.054
ablate no force update	0.000	0.190	0.000	0.297	0.029
ablate no safety penalty	0.000	0.187	0.000	0.199	0.032
ablate no slip context	0.000	0.177	0.000	0.208	0.037
ablate no transition planner	0.000	0.187	0.000	0.200	0.032
token full v5	0.000	0.186	0.000	0.207	0.033

Table 7: Representative frozen negative cases for impedance-token policy v5.

Case	Split	Seed	Ep.	Succ.	Err.	Over.	Slip	Lesson
0	stiffness shift	1	0	0	0.250	2.055	0.105	token selector did not adapt fast enough after contact shift
1	delayed mode switch	7	1	0	0.208	2.025	0.267	token selector did not adapt fast enough after contact shift
2	delayed mode switch	4	4	0	0.173	2.007	0.267	token selector did not adapt fast enough after contact shift
3	delayed mode switch	4	5	0	0.207	1.930	0.207	token selector did not adapt fast enough after contact shift
4	delayed mode switch	4	0	0	0.200	1.878	0.161	token selector did not adapt fast enough after contact shift
5	delayed mode switch	1	4	0	0.193	1.877	0.163	token selector did not adapt fast enough after contact shift
6	delayed mode switch	3	1	0	0.157	1.865	0.140	token selector did not adapt fast enough after contact shift
7	delayed mode switch	0	5	0	0.204	1.861	0.221	token selector did not adapt fast enough after contact shift
8	delayed mode switch	7	5	0	0.149	1.859	0.105	token selector did not adapt fast enough after contact shift
9	delayed mode switch	0	3	0	0.193	1.848	0.287	token selector did not adapt fast enough after contact shift
10	delayed mode switch	1	3	0	0.202	1.838	0.116	token selector did not adapt fast enough after contact shift
11	stiffness shift	6	2	0	0.269	1.818	0.069	token selector did not adapt fast enough after contact shift

A Frozen Protocol and Reproducibility

The plan-first document is `docs/paper72_expanded_submission_plan_20260621.md`. The development log is `docs/paper72_development_log_20260621.md`. The protocol freeze is `docs/paper72_protocol_freeze_20260621.md`. The generated terminal decision is `docs/paper72_expanded_terminal_decision_20260621.md`. The final PDF is written to `C:/Users/wangz/Downloads/72.pdf`; no copy is placed on the visible Desktop.

The frozen evidence consists of 8,640 main rows, 8,640 rollout rows, 1,440 seed summaries, 180 split-method metric rows, 168 paired rows, 1,536 ablation rows, 4,320 stress rows, and 12 negative cases. All generated tables below are produced by `scripts/render_submission_assets.py` from CSV artifacts, not by manual transcription.

B Full Split-Method Metrics

Table 8: Full split-method metrics.

Method	Split	Succ.	CI	Err.	Over.	Safe	Slip	Chat.	Prog.
adaptive impedance control	actuator saturation	0.958	0.053	0.110	1.610	0.008	0.580	0.035	0.835
adaptive impedance control	combined extreme stress	0.979	0.041	0.120	1.607	0.004	0.530	0.134	0.801
adaptive impedance control	combined stress	0.938	0.086	0.121	1.714	0.013	0.543	0.082	0.795
adaptive impedance control	contact transition	0.979	0.041	0.109	1.451	0.000	0.574	0.031	0.828
adaptive impedance control	delayed mode switch	0.542	0.120	0.125	2.017	0.033	0.560	0.040	0.831
adaptive impedance control	friction slip shift	0.979	0.041	0.119	1.261	0.000	0.571	0.017	0.825
adaptive impedance control	nominal surface tracking	0.729	0.122	0.112	1.253	0.000	0.594	0.013	0.855
adaptive impedance control	sensor noise burst	0.875	0.102	0.109	1.430	0.001	0.588	0.071	0.844
adaptive impedance control	stick slip cycle	0.979	0.041	0.100	1.241	0.000	0.567	0.030	0.819
adaptive impedance control	stiffness shift	0.521	0.114	0.169	1.934	0.032	0.594	0.014	0.850
adaptive impedance control	surface discontinuity	1.000	0.000	0.101	1.296	0.000	0.567	0.039	0.824
adaptive impedance control	target force jump	0.771	0.086	0.084	1.313	0.000	0.585	0.032	0.836
admittance switching control	actuator saturation	1.000	0.000	0.116	1.640	0.003	0.550	0.040	0.809
admittance switching control	combined extreme stress	0.958	0.053	0.114	1.611	0.003	0.503	0.142	0.781
admittance switching control	combined stress	0.938	0.060	0.120	1.716	0.010	0.525	0.083	0.785
admittance switching control	contact transition	0.979	0.041	0.114	1.842	0.011	0.555	0.033	0.813
admittance switching control	delayed mode switch	0.667	0.138	0.127	1.955	0.031	0.522	0.047	0.796
admittance switching control	friction slip shift	1.000	0.000	0.119	1.514	0.000	0.555	0.021	0.812
admittance switching control	nominal surface tracking	1.000	0.000	0.112	1.604	0.002	0.568	0.015	0.827
admittance switching control	sensor noise burst	1.000	0.000	0.112	1.530	0.000	0.561	0.072	0.818
admittance switching control	stick slip cycle	1.000	0.000	0.105	1.490	0.000	0.547	0.032	0.806
admittance switching control	stiffness shift	0.875	0.102	0.180	1.882	0.030	0.559	0.021	0.815
admittance switching control	surface discontinuity	1.000	0.000	0.099	1.643	0.004	0.552	0.043	0.811
admittance switching control	target force jump	0.938	0.060	0.096	1.810	0.009	0.559	0.034	0.812
conformal safety gain	actuator saturation	0.688	0.144	0.162	1.507	0.001	0.594	0.030	0.808
conformal safety gain	combined extreme stress	0.958	0.053	0.162	1.434	0.001	0.553	0.109	0.779
conformal safety gain	combined stress	0.979	0.041	0.172	1.586	0.006	0.555	0.062	0.770
conformal safety gain	contact transition	0.729	0.086	0.162	1.349	0.000	0.590	0.025	0.802
conformal safety gain	delayed mode switch	0.688	0.144	0.175	1.911	0.028	0.575	0.026	0.804
conformal safety gain	friction slip shift	0.833	0.087	0.185	1.113	0.000	0.582	0.010	0.799
conformal safety gain	nominal surface tracking	0.292	0.102	0.180	1.069	0.000	0.607	0.003	0.827
conformal safety gain	sensor noise burst	0.708	0.120	0.164	1.367	0.000	0.595	0.068	0.816
conformal safety gain	stick slip cycle	0.917	0.087	0.159	1.126	0.000	0.578	0.021	0.793

Method	Split	Succ.	CI	Err.	Over.	Safe	Slip	Chat.	Prog.
conformal safety gain	stiffness shift	0.083	0.087	0.243	1.813	0.022	0.606	0.013	0.819
conformal safety gain	surface discontinuity	0.875	0.082	0.149	1.148	0.000	0.582	0.027	0.799
conformal safety gain	target force jump	0.333	0.175	0.129	1.190	0.000	0.604	0.024	0.808
ensemble uncertainty gain	actuator saturation	0.521	0.190	0.168	1.525	0.002	0.600	0.028	0.813
ensemble uncertainty gain	combined extreme stress	0.958	0.053	0.172	1.588	0.004	0.558	0.101	0.781
ensemble uncertainty gain	combined stress	0.917	0.087	0.181	1.681	0.014	0.565	0.067	0.775
ensemble uncertainty gain	contact transition	0.458	0.120	0.165	1.373	0.000	0.598	0.024	0.807
ensemble uncertainty gain	delayed mode switch	0.500	0.151	0.191	1.960	0.038	0.582	0.029	0.807
ensemble uncertainty gain	friction slip shift	0.708	0.120	0.185	1.136	0.000	0.588	0.011	0.805
ensemble uncertainty gain	nominal surface tracking	0.208	0.082	0.178	1.088	0.000	0.614	0.004	0.831
ensemble uncertainty gain	sensor noise burst	0.438	0.137	0.163	1.381	0.000	0.603	0.062	0.820
ensemble uncertainty gain	stick slip cycle	0.854	0.096	0.160	1.144	0.000	0.584	0.024	0.798
ensemble uncertainty gain	stiffness shift	0.021	0.041	0.261	1.836	0.030	0.615	0.013	0.823
ensemble uncertainty gain	surface discontinuity	0.583	0.107	0.156	1.270	0.000	0.591	0.031	0.803
ensemble uncertainty gain	target force jump	0.208	0.102	0.129	1.212	0.000	0.609	0.024	0.813
fixed impedance	actuator saturation	0.979	0.041	0.198	1.726	0.018	0.565	0.040	0.813
fixed impedance	combined extreme stress	0.396	0.150	0.238	1.713	0.012	0.504	0.087	0.777
fixed impedance	combined stress	0.250	0.107	0.248	1.689	0.010	0.528	0.059	0.781
fixed impedance	contact transition	0.938	0.086	0.195	1.877	0.042	0.566	0.039	0.812
fixed impedance	delayed mode switch	0.792	0.102	0.214	1.843	0.035	0.548	0.041	0.810
fixed impedance	friction slip shift	0.812	0.096	0.192	1.847	0.041	0.566	0.022	0.811
fixed impedance	nominal surface tracking	0.938	0.086	0.178	1.833	0.038	0.579	0.012	0.827
fixed impedance	sensor noise burst	0.979	0.041	0.179	1.772	0.028	0.568	0.067	0.819
fixed impedance	stick slip cycle	1.000	0.000	0.165	1.846	0.039	0.557	0.038	0.804
fixed impedance	stiffness shift	0.000	0.000	0.299	1.752	0.022	0.577	0.022	0.823
fixed impedance	surface discontinuity	0.938	0.060	0.211	1.751	0.020	0.558	0.034	0.808
fixed impedance	target force jump	0.896	0.060	0.143	1.933	0.050	0.568	0.041	0.817
gain scheduled impedance	actuator saturation	0.146	0.074	0.125	1.540	0.003	0.610	0.028	0.853
gain scheduled impedance	combined extreme stress	0.958	0.053	0.132	1.519	0.003	0.574	0.118	0.822
gain scheduled impedance	combined stress	0.854	0.130	0.138	1.641	0.009	0.586	0.077	0.818
gain scheduled impedance	contact transition	0.125	0.082	0.122	1.376	0.000	0.612	0.024	0.847
gain scheduled impedance	delayed mode switch	0.375	0.160	0.144	1.948	0.029	0.599	0.033	0.848
gain scheduled impedance	friction slip shift	0.271	0.150	0.129	1.132	0.000	0.605	0.010	0.844
gain scheduled impedance	nominal surface tracking	0.000	0.000	0.122	1.085	0.000	0.621	0.003	0.870
gain scheduled impedance	sensor noise burst	0.042	0.053	0.120	1.373	0.000	0.616	0.059	0.861
gain scheduled impedance	stick slip cycle	0.542	0.148	0.115	1.151	0.000	0.600	0.028	0.838
gain scheduled impedance	stiffness shift	0.021	0.041	0.196	1.858	0.026	0.620	0.009	0.866
gain scheduled impedance	surface discontinuity	0.375	0.160	0.113	1.183	0.000	0.606	0.028	0.844
gain scheduled impedance	target force jump	0.042	0.053	0.100	1.219	0.000	0.623	0.025	0.854
hist gradient gain regressor	actuator saturation	0.479	0.190	0.164	1.516	0.001	0.603	0.031	0.816
hist gradient gain regressor	combined extreme stress	0.875	0.082	0.175	1.683	0.008	0.558	0.111	0.784
hist gradient gain regressor	combined stress	0.917	0.062	0.179	1.676	0.017	0.560	0.067	0.776
hist gradient gain regressor	contact transition	0.542	0.148	0.159	1.354	0.000	0.599	0.024	0.809
hist gradient gain regressor	delayed mode switch	0.500	0.123	0.189	1.926	0.034	0.585	0.032	0.812
hist gradient gain regressor	friction slip shift	0.688	0.157	0.177	1.115	0.000	0.592	0.010	0.808
hist gradient gain regressor	nominal surface tracking	0.062	0.060	0.172	1.073	0.000	0.620	0.006	0.837
hist gradient gain regressor	sensor noise burst	0.312	0.179	0.157	1.354	0.000	0.609	0.065	0.826
hist gradient gain regressor	stick slip cycle	0.771	0.174	0.153	1.128	0.000	0.586	0.023	0.803
hist gradient gain regressor	stiffness shift	0.021	0.041	0.261	1.814	0.026	0.618	0.013	0.830

Method	Split	Succ.	CI	Err.	Over.	Safe	Slip	Chat.	Prog.
hist gradient gain regressor	surface discontinuity	0.646	0.179	0.155	1.290	0.000	0.594	0.029	0.806
hist gradient gain regressor	target force jump	0.104	0.106	0.124	1.201	0.000	0.618	0.027	0.817
impedance token policy v4	actuator saturation	0.125	0.082	0.161	1.385	0.000	0.473	0.021	0.732
impedance token policy v4	combined extreme stress	0.000	0.000	0.143	1.362	0.000	0.409	0.095	0.719
impedance token policy v4	combined stress	0.000	0.000	0.150	1.452	0.001	0.368	0.063	0.719
impedance token policy v4	contact transition	0.000	0.000	0.159	1.240	0.000	0.442	0.022	0.724
impedance token policy v4	delayed mode switch	0.438	0.137	0.158	1.693	0.010	0.501	0.029	0.739
impedance token policy v4	friction slip shift	0.083	0.062	0.188	1.067	0.000	0.445	0.008	0.728
impedance token policy v4	nominal surface tracking	0.938	0.060	0.188	1.039	0.000	0.554	0.005	0.757
impedance token policy v4	sensor noise burst	0.250	0.062	0.167	1.262	0.000	0.506	0.050	0.738
impedance token policy v4	stick slip cycle	0.042	0.053	0.166	1.080	0.000	0.418	0.014	0.725
impedance token policy v4	stiffness shift	0.417	0.175	0.215	1.643	0.007	0.512	0.015	0.755
impedance token policy v4	surface discontinuity	0.000	0.000	0.150	1.093	0.000	0.436	0.017	0.723
impedance token policy v4	target force jump	0.062	0.086	0.140	1.113	0.000	0.488	0.016	0.733
impedance token policy v5	actuator saturation	0.000	0.000	0.190	1.297	0.000	0.259	0.015	0.684
impedance token policy v5	combined extreme stress	0.000	0.000	0.165	1.312	0.000	0.237	0.068	0.674
impedance token policy v5	combined stress	0.000	0.000	0.180	1.344	0.000	0.114	0.042	0.656
impedance token policy v5	contact transition	0.000	0.000	0.197	1.165	0.000	0.260	0.011	0.676
impedance token policy v5	delayed mode switch	0.000	0.000	0.188	1.622	0.008	0.269	0.017	0.684
impedance token policy v5	friction slip shift	0.000	0.000	0.242	0.975	0.000	0.299	0.001	0.679
impedance token policy v5	nominal surface tracking	0.000	0.000	0.248	0.938	0.000	0.378	0.001	0.705
impedance token policy v5	sensor noise burst	0.000	0.000	0.202	1.189	0.000	0.305	0.038	0.692
impedance token policy v5	stick slip cycle	0.000	0.000	0.217	0.982	0.000	0.225	0.001	0.668
impedance token policy v5	stiffness shift	0.000	0.000	0.250	1.559	0.002	0.326	0.014	0.705
impedance token policy v5	surface discontinuity	0.000	0.000	0.189	1.014	0.000	0.245	0.002	0.678
impedance token policy v5	target force jump	0.000	0.000	0.181	1.033	0.000	0.319	0.004	0.679
learned gain regressor	actuator saturation	0.188	0.130	0.169	1.588	0.006	0.614	0.025	0.831
learned gain regressor	combined extreme stress	0.812	0.114	0.165	1.594	0.006	0.573	0.098	0.803
learned gain regressor	combined stress	0.833	0.087	0.181	1.717	0.019	0.575	0.055	0.794
learned gain regressor	contact transition	0.375	0.102	0.156	1.419	0.000	0.607	0.021	0.822
learned gain regressor	delayed mode switch	0.292	0.053	0.178	2.038	0.044	0.594	0.027	0.827
learned gain regressor	friction slip shift	0.521	0.130	0.195	1.195	0.000	0.597	0.007	0.820
learned gain regressor	nominal surface tracking	0.000	0.000	0.189	1.157	0.000	0.633	0.004	0.851
learned gain regressor	sensor noise burst	0.042	0.053	0.178	1.446	0.000	0.621	0.058	0.840
learned gain regressor	stick slip cycle	0.833	0.062	0.160	1.206	0.000	0.590	0.021	0.813
learned gain regressor	stiffness shift	0.000	0.000	0.251	1.910	0.037	0.633	0.007	0.846
learned gain regressor	surface discontinuity	0.562	0.150	0.141	1.292	0.000	0.602	0.029	0.821
learned gain regressor	target force jump	0.083	0.062	0.127	1.237	0.000	0.622	0.022	0.831
oracle impedance	actuator saturation	0.542	0.135	0.102	1.591	0.005	0.601	0.030	0.830
oracle impedance	combined extreme stress	0.938	0.086	0.103	1.498	0.002	0.488	0.146	0.784
oracle impedance	combined stress	0.875	0.082	0.107	1.666	0.008	0.515	0.089	0.776
oracle impedance	contact transition	0.479	0.096	0.104	1.426	0.000	0.594	0.029	0.818
oracle impedance	delayed mode switch	0.562	0.086	0.109	1.997	0.023	0.563	0.039	0.820
oracle impedance	friction slip shift	0.708	0.183	0.118	1.166	0.000	0.589	0.014	0.810
oracle impedance	nominal surface tracking	0.062	0.060	0.109	1.121	0.000	0.622	0.002	0.855
oracle impedance	sensor noise burst	0.354	0.168	0.105	1.403	0.001	0.608	0.064	0.841
oracle impedance	stick slip cycle	0.812	0.130	0.103	1.186	0.000	0.583	0.025	0.803
oracle impedance	stiffness shift	0.104	0.060	0.148	1.933	0.021	0.614	0.011	0.850
oracle impedance	surface discontinuity	0.854	0.074	0.088	1.156	0.000	0.584	0.034	0.815
oracle impedance	target force jump	0.312	0.157	0.087	1.256	0.000	0.608	0.023	0.830
random forest gain regressor	actuator saturation	0.896	0.086	0.169	1.524	0.002	0.570	0.029	0.794
random forest gain regressor	combined extreme stress	0.938	0.060	0.175	1.669	0.009	0.530	0.105	0.763
random forest gain regressor	combined stress	0.938	0.060	0.182	1.699	0.016	0.540	0.068	0.761

Method	Split	Succ.	CI	Err.	Over.	Safe	Slip	Chat.	Prog.
random forest gain regressor	contact transition	0.896	0.060	0.165	1.374	0.000	0.570	0.024	0.792
random forest gain regressor	delayed mode switch	0.604	0.122	0.192	1.952	0.037	0.555	0.022	0.790
random forest gain regressor	friction slip shift	0.938	0.060	0.185	1.131	0.000	0.560	0.010	0.790
random forest gain regressor	nominal surface tracking	0.854	0.074	0.178	1.082	0.000	0.579	0.004	0.808
random forest gain regressor	sensor noise burst	0.979	0.041	0.165	1.373	0.000	0.565	0.061	0.796
random forest gain regressor	stick slip cycle	1.000	0.000	0.158	1.139	0.000	0.555	0.024	0.783
random forest gain regressor	stiffness shift	0.062	0.060	0.271	1.827	0.032	0.571	0.011	0.798
random forest gain regressor	surface discontinuity	0.938	0.060	0.156	1.288	0.000	0.561	0.027	0.788
random forest gain regressor	target force jump	0.896	0.060	0.128	1.220	0.000	0.574	0.024	0.795
risk averse impedance	actuator saturation	0.000	0.000	0.232	1.282	0.000	0.049	0.015	0.588
risk averse impedance	combined extreme stress	0.000	0.000	0.203	1.218	0.000	0.003	0.062	0.558
risk averse impedance	combined stress	0.000	0.000	0.220	1.315	0.000	0.000	0.039	0.550
risk averse impedance	contact transition	0.000	0.000	0.240	1.162	0.000	0.012	0.013	0.580
risk averse impedance	delayed mode switch	0.000	0.000	0.227	1.622	0.008	0.049	0.020	0.585
risk averse impedance	friction slip shift	0.000	0.000	0.299	0.961	0.000	0.002	0.001	0.579
risk averse impedance	nominal surface tracking	0.000	0.000	0.304	0.913	0.000	0.202	0.000	0.607
risk averse impedance	sensor noise burst	0.000	0.000	0.248	1.190	0.000	0.127	0.036	0.597
risk averse impedance	stick slip cycle	0.000	0.000	0.262	0.967	0.000	0.001	0.002	0.573
risk averse impedance	stiffness shift	0.000	0.000	0.311	1.541	0.002	0.179	0.013	0.602
risk averse impedance	surface discontinuity	0.000	0.000	0.233	0.985	0.000	0.005	0.002	0.578
risk averse impedance	target force jump	0.000	0.000	0.208	1.034	0.000	0.077	0.005	0.589
robust mpc impedance	actuator saturation	0.000	0.000	0.216	1.379	0.000	0.390	0.021	0.674
robust mpc impedance	combined extreme stress	0.000	0.000	0.190	1.335	0.001	0.344	0.087	0.649
robust mpc impedance	combined stress	0.000	0.000	0.207	1.418	0.001	0.305	0.051	0.650
robust mpc impedance	contact transition	0.000	0.000	0.221	1.248	0.000	0.397	0.016	0.672
robust mpc impedance	delayed mode switch	0.000	0.000	0.219	1.739	0.017	0.395	0.024	0.670
robust mpc impedance	friction slip shift	0.000	0.000	0.268	1.029	0.000	0.381	0.004	0.669
robust mpc impedance	nominal surface tracking	0.000	0.000	0.272	0.984	0.000	0.429	0.001	0.685
robust mpc impedance	sensor noise burst	0.000	0.000	0.227	1.250	0.000	0.405	0.049	0.679
robust mpc impedance	stick slip cycle	0.000	0.000	0.235	1.035	0.000	0.363	0.007	0.663
robust mpc impedance	stiffness shift	0.000	0.000	0.305	1.646	0.009	0.417	0.015	0.682
robust mpc impedance	surface discontinuity	0.000	0.000	0.209	1.056	0.000	0.395	0.010	0.670
robust mpc impedance	target force jump	0.000	0.000	0.188	1.108	0.000	0.430	0.010	0.676
token no memory ablation	actuator saturation	0.979	0.041	0.168	1.420	0.000	0.507	0.026	0.758
token no memory ablation	combined extreme stress	0.146	0.114	0.156	1.434	0.001	0.461	0.108	0.727
token no memory ablation	combined stress	0.021	0.041	0.165	1.502	0.002	0.457	0.065	0.721
token no memory ablation	contact transition	0.833	0.087	0.174	1.265	0.000	0.481	0.021	0.746
token no memory ablation	delayed mode switch	0.708	0.135	0.175	1.814	0.017	0.490	0.028	0.750
token no memory ablation	friction slip shift	0.688	0.130	0.207	1.055	0.000	0.483	0.007	0.744
token no memory ablation	nominal surface tracking	0.938	0.060	0.211	1.010	0.000	0.511	0.001	0.771
token no memory ablation	sensor noise burst	1.000	0.000	0.176	1.300	0.000	0.515	0.057	0.765
token no memory ablation	stick slip cycle	0.479	0.130	0.180	1.066	0.000	0.480	0.012	0.740
token no memory ablation	stiffness shift	0.375	0.102	0.238	1.691	0.010	0.519	0.016	0.770
token no memory ablation	surface discontinuity	0.917	0.062	0.160	1.114	0.000	0.492	0.017	0.748
token no memory ablation	target force jump	0.854	0.144	0.151	1.107	0.000	0.488	0.013	0.750

C Full Aggregate Metrics

Table 9: Full aggregate metrics.

Group	Method	Succ.	CI	Err.	Over.	Safe	Slip	Chat.	Energy
all splits	adaptive impedance control	0.854	0.040	0.115	1.511	0.008	0.571	0.045	0.897
all splits	admittance switching control	0.946	0.024	0.118	1.686	0.009	0.546	0.048	1.140
all splits	conformal safety gain	0.674	0.063	0.170	1.384	0.005	0.585	0.035	0.611

Group	Method	Succ.	CI	Err.	Over.	Safe	Slip	Chat.	Energy
all splits	ensemble uncertainty gain	0.531	0.065	0.176	1.433	0.007	0.592	0.035	0.603
all splits	fixed impedance	0.743	0.068	0.205	1.798	0.029	0.557	0.042	0.902
all splits	gain scheduled impedance	0.312	0.069	0.130	1.419	0.006	0.606	0.037	0.717
all splits	hist gradient gain regressor	0.493	0.070	0.172	1.428	0.007	0.595	0.036	0.614
all splits	impedance token policy v4	0.196	0.058	0.165	1.286	0.002	0.463	0.030	0.769
all splits	impedance token policy v5	0.000	0.000	0.204	1.202	0.001	0.270	0.018	0.720
all splits	learned gain regressor	0.378	0.068	0.174	1.483	0.009	0.605	0.031	0.665
all splits	oracle impedance	0.550	0.066	0.107	1.450	0.005	0.581	0.042	0.793
all splits	random forest gain regressor	0.828	0.054	0.177	1.440	0.008	0.561	0.034	0.603
all splits	risk averse impedance	0.000	0.000	0.249	1.182	0.001	0.059	0.017	0.384
all splits	robust mpc impedance	0.000	0.000	0.230	1.269	0.002	0.388	0.025	0.359
all splits	token no memory ablation	0.661	0.069	0.180	1.315	0.003	0.490	0.031	0.542
hard splits	adaptive impedance control	0.866	0.042	0.115	1.534	0.008	0.569	0.048	0.893
hard splits	admittance switching control	0.941	0.026	0.118	1.694	0.009	0.544	0.052	1.140
hard splits	conformal safety gain	0.708	0.063	0.169	1.413	0.005	0.583	0.038	0.608
hard splits	ensemble uncertainty gain	0.561	0.067	0.176	1.464	0.008	0.590	0.037	0.601
hard splits	fixed impedance	0.725	0.073	0.207	1.795	0.029	0.555	0.045	0.900
hard splits	gain scheduled impedance	0.341	0.072	0.130	1.449	0.006	0.605	0.040	0.712
hard splits	hist gradient gain regressor	0.532	0.071	0.172	1.460	0.008	0.593	0.039	0.611
hard splits	impedance token policy v4	0.129	0.040	0.163	1.308	0.002	0.454	0.032	0.770
hard splits	impedance token policy v5	0.000	0.000	0.200	1.226	0.001	0.260	0.019	0.719
hard splits	learned gain regressor	0.413	0.070	0.173	1.513	0.010	0.603	0.034	0.662
hard splits	oracle impedance	0.595	0.064	0.107	1.480	0.005	0.577	0.046	0.792
hard splits	random forest gain regressor	0.826	0.058	0.177	1.472	0.009	0.559	0.037	0.601
hard splits	risk averse impedance	0.000	0.000	0.244	1.207	0.001	0.046	0.019	0.382
hard splits	robust mpc impedance	0.000	0.000	0.226	1.295	0.003	0.384	0.027	0.358
hard splits	token no memory ablation	0.636	0.073	0.177	1.342	0.003	0.488	0.034	0.540
combined and extreme	adaptive impedance control	0.958	0.047	0.121	1.661	0.009	0.537	0.108	0.847
combined and extreme	admittance switching control	0.948	0.039	0.117	1.664	0.006	0.514	0.112	1.130
combined and extreme	conformal safety gain	0.969	0.033	0.167	1.510	0.003	0.554	0.085	0.591
combined and extreme	ensemble uncertainty gain	0.938	0.051	0.177	1.635	0.009	0.562	0.084	0.585
combined and extreme	fixed impedance	0.323	0.096	0.243	1.701	0.011	0.516	0.073	0.868
combined and extreme	gain scheduled impedance	0.906	0.073	0.135	1.580	0.006	0.580	0.098	0.666
combined and extreme	hist gradient gain regressor	0.896	0.051	0.177	1.680	0.012	0.559	0.089	0.594
combined and extreme	impedance token policy v4	0.000	0.000	0.146	1.407	0.001	0.389	0.079	0.779
combined and extreme	impedance token policy v5	0.000	0.000	0.173	1.328	0.000	0.176	0.055	0.719
combined and extreme	learned gain regressor	0.823	0.070	0.173	1.655	0.012	0.574	0.077	0.640
combined and extreme	oracle impedance	0.906	0.059	0.105	1.582	0.005	0.501	0.117	0.807
combined and extreme	random forest gain regressor	0.938	0.041	0.178	1.684	0.013	0.535	0.086	0.583
combined and extreme	risk averse impedance	0.000	0.000	0.211	1.266	0.000	0.002	0.050	0.365
combined and extreme	robust mpc impedance	0.000	0.000	0.199	1.377	0.001	0.324	0.069	0.347
combined and extreme	token no memory ablation	0.083	0.067	0.161	1.468	0.002	0.459	0.086	0.521

D ALL SEED METRICS

Table 10: All seed-level metrics.

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
adaptive impedance control	actuator saturation	0	1.000	0.112	1.734	0.015	0.582	0.024	5.000	0.836

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
adaptive impedance control	actuator saturation	1	1.000	0.109	1.633	0.007	0.579	0.036	5.000	0.834
adaptive impedance control	actuator saturation	2	1.000	0.106	1.502	0.002	0.579	0.037	5.500	0.837
adaptive impedance control	actuator saturation	3	0.833	0.114	1.688	0.013	0.586	0.047	5.000	0.833
adaptive impedance control	actuator saturation	4	1.000	0.108	1.560	0.004	0.578	0.026	6.000	0.837
adaptive impedance control	actuator saturation	5	1.000	0.107	1.559	0.004	0.577	0.030	5.000	0.837
adaptive impedance control	actuator saturation	6	0.833	0.114	1.646	0.013	0.579	0.039	5.000	0.833
adaptive impedance control	actuator saturation	7	1.000	0.109	1.555	0.006	0.579	0.039	5.000	0.836
adaptive impedance control	combined extreme stress	0	1.000	0.117	1.640	0.004	0.531	0.120	5.000	0.793
adaptive impedance control	combined extreme stress	1	0.833	0.126	1.725	0.008	0.523	0.134	7.000	0.800
adaptive impedance control	combined extreme stress	2	1.000	0.122	1.679	0.008	0.536	0.137	5.833	0.802
adaptive impedance control	combined extreme stress	3	1.000	0.126	1.562	0.011	0.537	0.141	7.333	0.804
adaptive impedance control	combined extreme stress	4	1.000	0.117	1.540	0.000	0.531	0.128	6.667	0.804
adaptive impedance control	combined extreme stress	5	1.000	0.127	1.547	0.000	0.529	0.139	8.000	0.804
adaptive impedance control	combined extreme stress	6	1.000	0.113	1.590	0.004	0.530	0.144	5.000	0.800
adaptive impedance control	combined extreme stress	7	1.000	0.112	1.574	0.000	0.527	0.126	5.667	0.798
adaptive impedance control	combined stress	0	0.667	0.128	1.873	0.028	0.551	0.051	5.000	0.798
adaptive impedance control	combined stress	1	1.000	0.125	1.783	0.015	0.539	0.081	5.000	0.795
adaptive impedance control	combined stress	2	1.000	0.123	1.743	0.015	0.544	0.103	5.000	0.798
adaptive impedance control	combined stress	3	1.000	0.122	1.652	0.009	0.555	0.075	5.000	0.796
adaptive impedance control	combined stress	4	0.833	0.123	1.770	0.026	0.530	0.060	5.000	0.793
adaptive impedance control	combined stress	5	1.000	0.113	1.623	0.006	0.546	0.098	5.000	0.794
adaptive impedance control	combined stress	6	1.000	0.118	1.661	0.006	0.543	0.100	5.333	0.801
adaptive impedance control	combined stress	7	1.000	0.116	1.610	0.002	0.534	0.088	6.333	0.791
adaptive impedance control	contact transition	0	1.000	0.112	1.440	0.000	0.581	0.019	5.000	0.829
adaptive impedance control	contact transition	1	1.000	0.106	1.489	0.002	0.574	0.026	5.000	0.829
adaptive impedance control	contact transition	2	1.000	0.108	1.445	0.000	0.573	0.034	5.000	0.830
adaptive impedance control	contact transition	3	0.833	0.105	1.430	0.000	0.577	0.041	5.000	0.828
adaptive impedance control	contact transition	4	1.000	0.112	1.486	0.000	0.573	0.032	5.000	0.827
adaptive impedance control	contact transition	5	1.000	0.109	1.476	0.000	0.573	0.028	5.000	0.829
adaptive impedance control	contact transition	6	1.000	0.107	1.389	0.000	0.572	0.036	5.000	0.830
adaptive impedance control	contact transition	7	1.000	0.113	1.451	0.000	0.569	0.034	5.000	0.822
adaptive impedance control	delayed mode switch	0	0.333	0.125	2.140	0.045	0.564	0.028	5.000	0.831
adaptive impedance control	delayed mode switch	1	0.500	0.130	1.987	0.032	0.559	0.041	7.500	0.830
adaptive impedance control	delayed mode switch	2	0.667	0.122	2.027	0.037	0.566	0.043	5.000	0.830
adaptive impedance control	delayed mode switch	3	0.667	0.132	1.915	0.022	0.566	0.037	9.667	0.833
adaptive impedance control	delayed mode switch	4	0.333	0.127	2.183	0.034	0.556	0.032	6.833	0.833
adaptive impedance control	delayed mode switch	5	0.833	0.116	1.808	0.032	0.548	0.051	6.667	0.826
adaptive impedance control	delayed mode switch	6	0.500	0.132	2.082	0.037	0.556	0.052	6.333	0.833
adaptive impedance control	delayed mode switch	7	0.500	0.116	1.994	0.028	0.563	0.037	6.167	0.832
adaptive impedance control	friction slip shift	0	0.833	0.118	1.269	0.000	0.576	0.013	5.000	0.826
adaptive impedance control	friction slip shift	1	1.000	0.123	1.260	0.000	0.573	0.019	5.000	0.823
adaptive impedance control	friction slip shift	2	1.000	0.124	1.270	0.000	0.567	0.022	5.167	0.829
adaptive impedance control	friction slip shift	3	1.000	0.125	1.271	0.000	0.573	0.017	5.000	0.831

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
adaptive impedance control	friction slip shift	4	1.000	0.112	1.253	0.000	0.569	0.021	5.000	0.820
adaptive impedance control	friction slip shift	5	1.000	0.121	1.252	0.000	0.575	0.009	5.000	0.824
adaptive impedance control	friction slip shift	6	1.000	0.112	1.221	0.000	0.567	0.013	5.000	0.827
adaptive impedance control	friction slip shift	7	1.000	0.114	1.293	0.000	0.571	0.019	5.000	0.822
adaptive impedance control	nominal surface tracking	0	0.667	0.112	1.285	0.000	0.599	0.013	5.000	0.855
adaptive impedance control	nominal surface tracking	1	1.000	0.116	1.241	0.000	0.592	0.013	5.000	0.855
adaptive impedance control	nominal surface tracking	2	0.500	0.111	1.235	0.000	0.594	0.011	5.000	0.855
adaptive impedance control	nominal surface tracking	3	0.500	0.114	1.248	0.000	0.601	0.013	5.000	0.855
adaptive impedance control	nominal surface tracking	4	0.833	0.107	1.251	0.000	0.588	0.015	5.000	0.854
adaptive impedance control	nominal surface tracking	5	0.833	0.116	1.270	0.000	0.592	0.013	5.000	0.854
adaptive impedance control	nominal surface tracking	6	0.833	0.111	1.253	0.000	0.590	0.011	5.000	0.854
adaptive impedance control	nominal surface tracking	7	0.667	0.108	1.242	0.000	0.596	0.015	5.000	0.856
adaptive impedance control	sensor noise burst	0	0.667	0.107	1.352	0.000	0.594	0.082	5.000	0.847
adaptive impedance control	sensor noise burst	1	1.000	0.105	1.437	0.000	0.584	0.082	5.000	0.843
adaptive impedance control	sensor noise burst	2	0.667	0.113	1.387	0.000	0.590	0.082	5.167	0.842
adaptive impedance control	sensor noise burst	3	0.833	0.110	1.398	0.000	0.592	0.075	5.000	0.845
adaptive impedance control	sensor noise burst	4	1.000	0.110	1.477	0.007	0.588	0.058	5.000	0.842
adaptive impedance control	sensor noise burst	5	1.000	0.111	1.434	0.002	0.586	0.067	5.000	0.845
adaptive impedance control	sensor noise burst	6	0.833	0.113	1.430	0.000	0.584	0.079	5.000	0.845
adaptive impedance control	sensor noise burst	7	1.000	0.103	1.527	0.002	0.588	0.041	5.000	0.844
adaptive impedance control	stick slip cycle	0	0.833	0.097	1.219	0.000	0.575	0.024	5.000	0.822
adaptive impedance control	stick slip cycle	1	1.000	0.103	1.234	0.000	0.566	0.039	5.000	0.821
adaptive impedance control	stick slip cycle	2	1.000	0.101	1.268	0.000	0.564	0.032	5.000	0.816
adaptive impedance control	stick slip cycle	3	1.000	0.101	1.208	0.000	0.569	0.026	5.000	0.815
adaptive impedance control	stick slip cycle	4	1.000	0.096	1.223	0.000	0.567	0.032	5.000	0.825
adaptive impedance control	stick slip cycle	5	1.000	0.103	1.258	0.000	0.571	0.024	5.000	0.819
adaptive impedance control	stick slip cycle	6	1.000	0.102	1.273	0.000	0.561	0.043	5.000	0.815
adaptive impedance control	stick slip cycle	7	1.000	0.099	1.249	0.000	0.562	0.021	5.000	0.819
adaptive impedance control	stiffness shift	0	0.333	0.169	1.860	0.028	0.601	0.013	5.000	0.852
adaptive impedance control	stiffness shift	1	0.667	0.169	2.006	0.034	0.591	0.021	5.000	0.850
adaptive impedance control	stiffness shift	2	0.333	0.173	1.969	0.034	0.594	0.013	5.167	0.850
adaptive impedance control	stiffness shift	3	0.333	0.165	1.944	0.034	0.597	0.013	5.000	0.848
adaptive impedance control	stiffness shift	4	0.667	0.158	1.965	0.037	0.591	0.013	6.000	0.851
adaptive impedance control	stiffness shift	5	0.500	0.173	1.917	0.030	0.590	0.013	5.000	0.849
adaptive impedance control	stiffness shift	6	0.667	0.179	1.958	0.032	0.596	0.011	5.000	0.850
adaptive impedance control	stiffness shift	7	0.667	0.164	1.853	0.030	0.596	0.015	5.667	0.850
adaptive impedance control	surface discontinuity	0	1.000	0.099	1.283	0.000	0.571	0.032	5.000	0.824
adaptive impedance control	surface discontinuity	1	1.000	0.104	1.311	0.000	0.561	0.039	6.167	0.822
adaptive impedance control	surface discontinuity	2	1.000	0.101	1.296	0.000	0.569	0.043	5.000	0.826
adaptive impedance control	surface discontinuity	3	1.000	0.103	1.286	0.000	0.573	0.041	5.000	0.825
adaptive impedance control	surface discontinuity	4	1.000	0.102	1.278	0.000	0.566	0.030	6.000	0.824
adaptive impedance control	surface discontinuity	5	1.000	0.105	1.346	0.000	0.566	0.036	5.000	0.825
adaptive impedance control	surface discontinuity	6	1.000	0.099	1.300	0.000	0.569	0.049	6.333	0.828

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
adaptive impedance control	surface discontinuity	7	1.000	0.098	1.265	0.000	0.563	0.041	5.000	0.822
adaptive impedance control	target force jump	0	0.667	0.085	1.289	0.000	0.590	0.024	5.000	0.838
adaptive impedance control	target force jump	1	1.000	0.081	1.305	0.000	0.580	0.034	5.000	0.834
adaptive impedance control	target force jump	2	0.667	0.085	1.370	0.000	0.585	0.038	5.000	0.837
adaptive impedance control	target force jump	3	0.667	0.088	1.276	0.000	0.589	0.036	5.000	0.836
adaptive impedance control	target force jump	4	0.667	0.080	1.274	0.000	0.586	0.025	5.000	0.839
adaptive impedance control	target force jump	5	0.833	0.083	1.303	0.000	0.585	0.025	5.000	0.835
adaptive impedance control	target force jump	6	0.833	0.084	1.336	0.000	0.582	0.042	5.000	0.838
adaptive impedance control	target force jump	7	0.833	0.084	1.348	0.000	0.584	0.030	5.000	0.835
admittance switching control	actuator saturation	0	1.000	0.118	1.725	0.007	0.562	0.039	5.000	0.805
admittance switching control	actuator saturation	1	1.000	0.113	1.623	0.002	0.566	0.043	5.000	0.809
admittance switching control	actuator saturation	2	1.000	0.114	1.617	0.002	0.537	0.036	5.167	0.813
admittance switching control	actuator saturation	3	1.000	0.123	1.642	0.004	0.526	0.045	5.000	0.808
admittance switching control	actuator saturation	4	1.000	0.115	1.583	0.002	0.556	0.039	6.167	0.809
admittance switching control	actuator saturation	5	1.000	0.112	1.648	0.002	0.564	0.045	5.000	0.810
admittance switching control	actuator saturation	6	1.000	0.120	1.659	0.006	0.528	0.030	5.000	0.808
admittance switching control	actuator saturation	7	1.000	0.114	1.625	0.002	0.560	0.041	5.000	0.811
admittance switching control	combined extreme stress	0	1.000	0.113	1.640	0.006	0.519	0.148	5.000	0.779
admittance switching control	combined extreme stress	1	0.833	0.116	1.669	0.007	0.513	0.150	5.167	0.779
admittance switching control	combined extreme stress	2	1.000	0.114	1.606	0.002	0.507	0.133	5.000	0.785
admittance switching control	combined extreme stress	3	1.000	0.115	1.603	0.002	0.489	0.161	5.000	0.783
admittance switching control	combined extreme stress	4	0.833	0.117	1.563	0.000	0.476	0.144	6.000	0.779
admittance switching control	combined extreme stress	5	1.000	0.117	1.525	0.000	0.504	0.140	5.333	0.782
admittance switching control	combined extreme stress	6	1.000	0.110	1.715	0.006	0.506	0.114	5.167	0.782
admittance switching control	combined extreme stress	7	1.000	0.112	1.570	0.000	0.507	0.142	5.000	0.782
admittance switching control	combined stress	0	0.833	0.126	1.808	0.019	0.515	0.082	5.000	0.784
admittance switching control	combined stress	1	1.000	0.125	1.716	0.007	0.522	0.107	5.000	0.785
admittance switching control	combined stress	2	0.833	0.125	1.757	0.017	0.521	0.075	5.000	0.784
admittance switching control	combined stress	3	1.000	0.122	1.655	0.004	0.507	0.084	5.000	0.786
admittance switching control	combined stress	4	0.833	0.124	1.766	0.011	0.522	0.081	5.000	0.781
admittance switching control	combined stress	5	1.000	0.112	1.650	0.006	0.547	0.084	5.000	0.790
admittance switching control	combined stress	6	1.000	0.115	1.684	0.006	0.530	0.075	5.000	0.789
admittance switching control	combined stress	7	1.000	0.116	1.694	0.007	0.532	0.077	5.667	0.783
admittance switching control	contact transition	0	1.000	0.116	1.781	0.011	0.564	0.036	5.000	0.812
admittance switching control	contact transition	1	0.833	0.111	1.925	0.013	0.564	0.034	5.000	0.812
admittance switching control	contact transition	2	1.000	0.110	1.846	0.011	0.541	0.036	5.000	0.813
admittance switching control	contact transition	3	1.000	0.111	1.880	0.011	0.552	0.028	5.000	0.813
admittance switching control	contact transition	4	1.000	0.115	1.818	0.011	0.558	0.043	5.000	0.812
admittance switching control	contact transition	5	1.000	0.116	1.814	0.011	0.566	0.030	5.000	0.814
admittance switching control	contact transition	6	1.000	0.110	1.848	0.011	0.551	0.032	5.333	0.814
admittance switching control	contact transition	7	1.000	0.121	1.827	0.011	0.545	0.030	5.000	0.809
admittance switching control	delayed mode switch	0	0.667	0.128	2.022	0.043	0.541	0.051	5.000	0.795
admittance switching control	delayed mode switch	1	0.500	0.133	1.931	0.030	0.509	0.047	7.167	0.790

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
admittance switching control	delayed mode switch	2	0.667	0.127	1.944	0.037	0.547	0.037	5.000	0.800
admittance switching control	delayed mode switch	3	0.667	0.133	1.863	0.015	0.502	0.047	8.333	0.794
admittance switching control	delayed mode switch	4	0.333	0.129	2.155	0.036	0.504	0.052	6.500	0.798
admittance switching control	delayed mode switch	5	1.000	0.116	1.810	0.026	0.526	0.054	6.667	0.794
admittance switching control	delayed mode switch	6	0.833	0.130	1.926	0.034	0.502	0.043	6.500	0.796
admittance switching control	delayed mode switch	7	0.667	0.120	1.986	0.030	0.541	0.041	6.000	0.799
admittance switching control	friction slip shift	0	1.000	0.117	1.462	0.000	0.552	0.019	5.000	0.812
admittance switching control	friction slip shift	1	1.000	0.120	1.469	0.000	0.556	0.030	5.000	0.811
admittance switching control	friction slip shift	2	1.000	0.126	1.503	0.000	0.549	0.019	5.000	0.813
admittance switching control	friction slip shift	3	1.000	0.130	1.562	0.000	0.554	0.019	5.000	0.814
admittance switching control	friction slip shift	4	1.000	0.116	1.548	0.000	0.556	0.017	5.000	0.811
admittance switching control	friction slip shift	5	1.000	0.117	1.463	0.000	0.564	0.024	5.000	0.811
admittance switching control	friction slip shift	6	1.000	0.113	1.567	0.002	0.552	0.015	5.000	0.813
admittance switching control	friction slip shift	7	1.000	0.116	1.537	0.002	0.556	0.024	5.000	0.811
admittance switching control	nominal surface tracking	0	1.000	0.112	1.610	0.000	0.571	0.013	5.000	0.827
admittance switching control	nominal surface tracking	1	1.000	0.116	1.598	0.000	0.573	0.015	5.000	0.827
admittance switching control	nominal surface tracking	2	1.000	0.111	1.590	0.002	0.560	0.021	5.000	0.827
admittance switching control	nominal surface tracking	3	1.000	0.115	1.606	0.002	0.564	0.015	5.000	0.827
admittance switching control	nominal surface tracking	4	1.000	0.109	1.637	0.004	0.567	0.015	5.000	0.827
admittance switching control	nominal surface tracking	5	1.000	0.117	1.600	0.002	0.575	0.015	5.000	0.827
admittance switching control	nominal surface tracking	6	1.000	0.109	1.562	0.000	0.562	0.013	5.000	0.827
admittance switching control	nominal surface tracking	7	1.000	0.109	1.630	0.004	0.569	0.009	5.000	0.828
admittance switching control	sensor noise burst	0	1.000	0.107	1.547	0.000	0.567	0.067	5.000	0.819
admittance switching control	sensor noise burst	1	1.000	0.112	1.546	0.000	0.566	0.088	5.000	0.817
admittance switching control	sensor noise burst	2	1.000	0.114	1.520	0.000	0.556	0.073	5.000	0.818
admittance switching control	sensor noise burst	3	1.000	0.112	1.559	0.002	0.560	0.075	5.000	0.819
admittance switching control	sensor noise burst	4	1.000	0.116	1.551	0.002	0.547	0.071	5.000	0.818
admittance switching control	sensor noise burst	5	1.000	0.112	1.504	0.000	0.573	0.077	5.000	0.818
admittance switching control	sensor noise burst	6	1.000	0.115	1.502	0.000	0.550	0.065	5.000	0.819
admittance switching control	sensor noise burst	7	1.000	0.106	1.510	0.000	0.567	0.062	5.000	0.816
admittance switching control	stick slip cycle	0	1.000	0.102	1.496	0.000	0.552	0.034	5.000	0.808
admittance switching control	stick slip cycle	1	1.000	0.104	1.466	0.000	0.552	0.039	5.000	0.806
admittance switching control	stick slip cycle	2	1.000	0.104	1.469	0.000	0.539	0.026	5.000	0.804
admittance switching control	stick slip cycle	3	1.000	0.107	1.548	0.002	0.547	0.028	5.000	0.804
admittance switching control	stick slip cycle	4	1.000	0.105	1.521	0.002	0.551	0.024	5.000	0.808
admittance switching control	stick slip cycle	5	1.000	0.109	1.448	0.000	0.554	0.039	5.000	0.806
admittance switching control	stick slip cycle	6	1.000	0.106	1.507	0.000	0.534	0.030	5.000	0.804
admittance switching control	stick slip cycle	7	1.000	0.104	1.467	0.000	0.549	0.036	5.000	0.805
admittance switching control	stiffness shift	0	1.000	0.176	1.827	0.024	0.573	0.019	5.500	0.816
admittance switching control	stiffness shift	1	0.667	0.178	1.957	0.032	0.562	0.019	5.000	0.815
admittance switching control	stiffness shift	2	0.833	0.185	1.883	0.030	0.556	0.021	5.833	0.815
admittance switching control	stiffness shift	3	1.000	0.177	1.867	0.032	0.547	0.021	5.333	0.813
admittance switching control	stiffness shift	4	1.000	0.171	1.929	0.036	0.552	0.024	6.500	0.816

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
admittance switching control	stiffness shift	5	0.667	0.185	1.865	0.030	0.575	0.019	5.000	0.815
admittance switching control	stiffness shift	6	0.833	0.190	1.901	0.032	0.558	0.024	5.000	0.816
admittance switching control	stiffness shift	7	1.000	0.174	1.825	0.022	0.551	0.021	6.500	0.816
admittance switching control	surface discontinuity	0	1.000	0.094	1.564	0.002	0.552	0.047	5.000	0.810
admittance switching control	surface discontinuity	1	1.000	0.100	1.725	0.006	0.554	0.041	5.000	0.809
admittance switching control	surface discontinuity	2	1.000	0.098	1.657	0.004	0.547	0.037	5.000	0.812
admittance switching control	surface discontinuity	3	1.000	0.100	1.601	0.002	0.552	0.051	5.000	0.812
admittance switching control	surface discontinuity	4	1.000	0.100	1.719	0.006	0.554	0.041	5.000	0.811
admittance switching control	surface discontinuity	5	1.000	0.104	1.598	0.002	0.562	0.041	5.000	0.812
admittance switching control	surface discontinuity	6	1.000	0.094	1.581	0.002	0.547	0.039	5.000	0.812
admittance switching control	surface discontinuity	7	1.000	0.098	1.697	0.007	0.549	0.043	5.000	0.810
admittance switching control	target force jump	0	1.000	0.096	1.775	0.007	0.560	0.030	5.000	0.814
admittance switching control	target force jump	1	0.833	0.095	1.824	0.009	0.560	0.036	5.167	0.810
admittance switching control	target force jump	2	1.000	0.100	1.736	0.004	0.556	0.036	5.000	0.811
admittance switching control	target force jump	3	1.000	0.098	1.833	0.009	0.556	0.032	5.000	0.814
admittance switching control	target force jump	4	0.833	0.093	1.827	0.011	0.560	0.034	5.167	0.814
admittance switching control	target force jump	5	1.000	0.092	1.754	0.009	0.566	0.037	5.000	0.811
admittance switching control	target force jump	6	1.000	0.097	1.846	0.009	0.547	0.034	5.000	0.812
admittance switching control	target force jump	7	0.833	0.098	1.882	0.009	0.566	0.032	5.000	0.812
conformal safety gain	actuator saturation	0	0.833	0.165	1.606	0.004	0.588	0.035	5.000	0.810
conformal safety gain	actuator saturation	1	0.833	0.161	1.488	0.000	0.588	0.035	5.000	0.805
conformal safety gain	actuator saturation	2	0.333	0.156	1.408	0.000	0.602	0.029	5.167	0.811
conformal safety gain	actuator saturation	3	0.500	0.168	1.613	0.002	0.599	0.035	5.000	0.807
conformal safety gain	actuator saturation	4	0.500	0.162	1.473	0.000	0.595	0.025	5.667	0.810
conformal safety gain	actuator saturation	5	0.833	0.157	1.458	0.000	0.592	0.027	5.000	0.810
conformal safety gain	actuator saturation	6	0.833	0.166	1.530	0.002	0.596	0.035	5.000	0.808
conformal safety gain	actuator saturation	7	0.833	0.160	1.477	0.000	0.596	0.023	5.000	0.807
conformal safety gain	combined extreme stress	0	1.000	0.167	1.544	0.002	0.546	0.082	5.000	0.772
conformal safety gain	combined extreme stress	1	1.000	0.160	1.416	0.002	0.541	0.109	5.167	0.777
conformal safety gain	combined extreme stress	2	1.000	0.163	1.373	0.000	0.569	0.102	5.667	0.780
conformal safety gain	combined extreme stress	3	0.833	0.165	1.437	0.000	0.556	0.137	5.000	0.781
conformal safety gain	combined extreme stress	4	0.833	0.158	1.367	0.000	0.556	0.119	5.167	0.782
conformal safety gain	combined extreme stress	5	1.000	0.163	1.393	0.000	0.545	0.108	5.500	0.782
conformal safety gain	combined extreme stress	6	1.000	0.163	1.540	0.004	0.559	0.098	5.000	0.778
conformal safety gain	combined extreme stress	7	1.000	0.160	1.404	0.000	0.551	0.115	5.167	0.776
conformal safety gain	combined stress	0	1.000	0.179	1.770	0.019	0.554	0.045	5.000	0.773
conformal safety gain	combined stress	1	1.000	0.177	1.634	0.006	0.543	0.054	5.000	0.769
conformal safety gain	combined stress	2	1.000	0.177	1.614	0.006	0.562	0.072	5.167	0.772
conformal safety gain	combined stress	3	1.000	0.177	1.608	0.002	0.568	0.082	5.167	0.771
conformal safety gain	combined stress	4	1.000	0.173	1.624	0.008	0.549	0.064	5.000	0.767
conformal safety gain	combined stress	5	1.000	0.163	1.486	0.002	0.559	0.055	5.000	0.770
conformal safety gain	combined stress	6	1.000	0.165	1.506	0.002	0.555	0.068	5.000	0.775
conformal safety gain	combined stress	7	0.833	0.163	1.448	0.000	0.551	0.056	5.500	0.765

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
conformal safety gain	contact transition	0	0.667	0.167	1.385	0.000	0.588	0.035	5.000	0.803
conformal safety gain	contact transition	1	0.833	0.157	1.384	0.000	0.591	0.023	5.000	0.803
conformal safety gain	contact transition	2	0.500	0.160	1.373	0.000	0.595	0.025	5.000	0.803
conformal safety gain	contact transition	3	0.667	0.156	1.314	0.000	0.591	0.029	5.000	0.802
conformal safety gain	contact transition	4	0.833	0.167	1.342	0.000	0.587	0.012	5.000	0.800
conformal safety gain	contact transition	5	0.833	0.163	1.345	0.000	0.588	0.029	5.000	0.803
conformal safety gain	contact transition	6	0.833	0.160	1.292	0.000	0.589	0.023	5.167	0.803
conformal safety gain	contact transition	7	0.667	0.165	1.352	0.000	0.589	0.027	5.000	0.797
conformal safety gain	delayed mode switch	0	0.500	0.176	2.020	0.039	0.576	0.027	5.000	0.803
conformal safety gain	delayed mode switch	1	0.667	0.181	1.882	0.027	0.568	0.025	7.167	0.803
conformal safety gain	delayed mode switch	2	0.667	0.175	1.922	0.031	0.588	0.027	5.000	0.803
conformal safety gain	delayed mode switch	3	0.833	0.179	1.791	0.016	0.577	0.023	7.833	0.805
conformal safety gain	delayed mode switch	4	0.333	0.175	2.093	0.033	0.577	0.019	6.000	0.805
conformal safety gain	delayed mode switch	5	1.000	0.163	1.753	0.023	0.557	0.027	6.667	0.800
conformal safety gain	delayed mode switch	6	0.833	0.183	1.928	0.031	0.578	0.025	6.667	0.806
conformal safety gain	delayed mode switch	7	0.667	0.166	1.897	0.027	0.581	0.031	5.500	0.804
conformal safety gain	friction slip shift	0	0.833	0.187	1.118	0.000	0.581	0.015	5.000	0.800
conformal safety gain	friction slip shift	1	1.000	0.192	1.148	0.000	0.579	0.014	5.000	0.797
conformal safety gain	friction slip shift	2	0.833	0.197	1.154	0.000	0.583	0.004	5.000	0.802
conformal safety gain	friction slip shift	3	0.833	0.194	1.119	0.000	0.587	0.017	5.000	0.804
conformal safety gain	friction slip shift	4	0.667	0.176	1.107	0.000	0.587	0.004	5.000	0.795
conformal safety gain	friction slip shift	5	1.000	0.185	1.104	0.000	0.576	0.010	5.000	0.798
conformal safety gain	friction slip shift	6	0.833	0.177	1.073	0.000	0.583	0.004	5.000	0.802
conformal safety gain	friction slip shift	7	0.667	0.172	1.084	0.000	0.578	0.010	5.000	0.796
conformal safety gain	nominal surface tracking	0	0.500	0.181	1.059	0.000	0.599	0.006	5.000	0.828
conformal safety gain	nominal surface tracking	1	0.500	0.183	1.062	0.000	0.599	0.002	5.000	0.827
conformal safety gain	nominal surface tracking	2	0.167	0.179	1.074	0.000	0.612	0.002	5.000	0.827
conformal safety gain	nominal surface tracking	3	0.167	0.182	1.093	0.000	0.613	0.004	5.000	0.828
conformal safety gain	nominal surface tracking	4	0.167	0.176	1.071	0.000	0.610	0.004	5.000	0.826
conformal safety gain	nominal surface tracking	5	0.333	0.182	1.061	0.000	0.602	0.002	5.000	0.826
conformal safety gain	nominal surface tracking	6	0.333	0.177	1.061	0.000	0.605	0.004	5.000	0.825
conformal safety gain	nominal surface tracking	7	0.167	0.181	1.068	0.000	0.613	0.004	5.000	0.829
conformal safety gain	sensor noise burst	0	0.833	0.158	1.331	0.000	0.589	0.069	5.000	0.818
conformal safety gain	sensor noise burst	1	0.833	0.162	1.331	0.000	0.587	0.071	5.000	0.814
conformal safety gain	sensor noise burst	2	0.500	0.165	1.385	0.000	0.600	0.071	5.000	0.813
conformal safety gain	sensor noise burst	3	0.667	0.165	1.431	0.000	0.595	0.071	5.000	0.817
conformal safety gain	sensor noise burst	4	0.667	0.169	1.393	0.002	0.597	0.052	5.167	0.814
conformal safety gain	sensor noise burst	5	1.000	0.163	1.321	0.000	0.590	0.077	5.000	0.817
conformal safety gain	sensor noise burst	6	0.667	0.167	1.354	0.000	0.600	0.060	5.000	0.816
conformal safety gain	sensor noise burst	7	0.500	0.161	1.391	0.002	0.602	0.072	5.000	0.816
conformal safety gain	stick slip cycle	0	1.000	0.155	1.077	0.000	0.583	0.012	5.000	0.796
conformal safety gain	stick slip cycle	1	1.000	0.163	1.156	0.000	0.572	0.021	5.000	0.796
conformal safety gain	stick slip cycle	2	1.000	0.158	1.144	0.000	0.577	0.021	5.000	0.791

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
conformal safety gain	stick slip cycle	3	1.000	0.163	1.104	0.000	0.573	0.021	5.000	0.789
conformal safety gain	stick slip cycle	4	0.833	0.152	1.094	0.000	0.580	0.027	5.000	0.799
conformal safety gain	stick slip cycle	5	1.000	0.165	1.145	0.000	0.576	0.015	5.000	0.793
conformal safety gain	stick slip cycle	6	0.833	0.160	1.172	0.000	0.580	0.025	5.000	0.789
conformal safety gain	stick slip cycle	7	0.667	0.159	1.118	0.000	0.580	0.025	5.000	0.794
conformal safety gain	stiffness shift	0	0.167	0.241	1.751	0.014	0.607	0.014	5.000	0.821
conformal safety gain	stiffness shift	1	0.000	0.241	1.863	0.025	0.599	0.012	5.000	0.819
conformal safety gain	stiffness shift	2	0.000	0.249	1.833	0.023	0.617	0.010	5.000	0.819
conformal safety gain	stiffness shift	3	0.333	0.237	1.829	0.025	0.597	0.013	5.167	0.818
conformal safety gain	stiffness shift	4	0.000	0.232	1.843	0.025	0.613	0.014	6.167	0.820
conformal safety gain	stiffness shift	5	0.167	0.248	1.781	0.021	0.596	0.010	5.000	0.818
conformal safety gain	stiffness shift	6	0.000	0.256	1.820	0.023	0.607	0.012	5.000	0.820
conformal safety gain	stiffness shift	7	0.000	0.241	1.787	0.017	0.608	0.017	5.667	0.820
conformal safety gain	surface discontinuity	0	1.000	0.148	1.135	0.000	0.574	0.031	5.000	0.799
conformal safety gain	surface discontinuity	1	0.833	0.148	1.151	0.000	0.578	0.029	5.000	0.797
conformal safety gain	surface discontinuity	2	1.000	0.146	1.120	0.000	0.584	0.029	5.000	0.800
conformal safety gain	surface discontinuity	3	0.833	0.152	1.173	0.000	0.584	0.023	5.000	0.798
conformal safety gain	surface discontinuity	4	0.833	0.151	1.110	0.000	0.588	0.014	5.000	0.797
conformal safety gain	surface discontinuity	5	0.833	0.153	1.190	0.000	0.578	0.027	5.000	0.800
conformal safety gain	surface discontinuity	6	1.000	0.147	1.159	0.000	0.586	0.029	5.000	0.802
conformal safety gain	surface discontinuity	7	0.667	0.148	1.144	0.000	0.587	0.033	5.000	0.797
conformal safety gain	target force jump	0	0.500	0.134	1.194	0.000	0.597	0.029	5.000	0.810
conformal safety gain	target force jump	1	0.667	0.124	1.138	0.000	0.596	0.018	5.000	0.806
conformal safety gain	target force jump	2	0.000	0.133	1.237	0.000	0.613	0.025	5.000	0.808
conformal safety gain	target force jump	3	0.500	0.138	1.201	0.000	0.604	0.033	5.000	0.808
conformal safety gain	target force jump	4	0.167	0.127	1.187	0.000	0.611	0.021	5.000	0.811
conformal safety gain	target force jump	5	0.500	0.125	1.197	0.000	0.601	0.026	5.000	0.807
conformal safety gain	target force jump	6	0.000	0.127	1.173	0.000	0.605	0.020	5.000	0.810
conformal safety gain	target force jump	7	0.333	0.129	1.189	0.000	0.605	0.020	5.000	0.806
ensemble uncertainty gain	actuator saturation	0	0.333	0.171	1.638	0.006	0.603	0.029	5.000	0.813
ensemble uncertainty gain	actuator saturation	1	0.833	0.164	1.522	0.004	0.599	0.033	5.000	0.810
ensemble uncertainty gain	actuator saturation	2	0.167	0.162	1.434	0.000	0.605	0.031	5.833	0.815
ensemble uncertainty gain	actuator saturation	3	0.500	0.176	1.584	0.002	0.594	0.029	5.000	0.812
ensemble uncertainty gain	actuator saturation	4	0.833	0.172	1.480	0.000	0.596	0.027	6.667	0.814
ensemble uncertainty gain	actuator saturation	5	0.667	0.161	1.478	0.000	0.598	0.025	5.000	0.814
ensemble uncertainty gain	actuator saturation	6	0.167	0.170	1.567	0.006	0.605	0.025	5.000	0.812
ensemble uncertainty gain	actuator saturation	7	0.667	0.168	1.498	0.002	0.598	0.023	5.000	0.812
ensemble uncertainty gain	combined extreme stress	0	1.000	0.172	1.575	0.004	0.561	0.081	5.167	0.774
ensemble uncertainty gain	combined extreme stress	1	1.000	0.172	1.608	0.006	0.552	0.113	5.833	0.779
ensemble uncertainty gain	combined extreme stress	2	1.000	0.170	1.549	0.000	0.570	0.093	5.000	0.783
ensemble uncertainty gain	combined extreme stress	3	0.833	0.175	1.477	0.002	0.551	0.085	5.667	0.784
ensemble uncertainty gain	combined extreme stress	4	1.000	0.173	1.560	0.004	0.556	0.105	6.833	0.783
ensemble uncertainty gain	combined extreme stress	5	1.000	0.180	1.674	0.008	0.554	0.124	8.000	0.784

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
ensemble uncertainty gain	combined extreme stress	6	0.833	0.164	1.656	0.006	0.566	0.104	5.000	0.782
ensemble uncertainty gain	combined extreme stress	7	1.000	0.174	1.604	0.004	0.555	0.104	5.833	0.779
ensemble uncertainty gain	combined stress	0	0.667	0.190	1.868	0.039	0.573	0.067	5.000	0.777
ensemble uncertainty gain	combined stress	1	0.833	0.185	1.712	0.016	0.565	0.070	5.000	0.773
ensemble uncertainty gain	combined stress	2	1.000	0.179	1.692	0.018	0.574	0.071	5.000	0.776
ensemble uncertainty gain	combined stress	3	1.000	0.185	1.669	0.006	0.571	0.063	5.000	0.775
ensemble uncertainty gain	combined stress	4	1.000	0.185	1.748	0.025	0.551	0.063	5.000	0.772
ensemble uncertainty gain	combined stress	5	0.833	0.170	1.507	0.002	0.571	0.069	6.000	0.774
ensemble uncertainty gain	combined stress	6	1.000	0.175	1.592	0.000	0.564	0.078	5.167	0.780
ensemble uncertainty gain	combined stress	7	1.000	0.179	1.660	0.006	0.555	0.059	7.000	0.770
ensemble uncertainty gain	contact transition	0	0.500	0.170	1.410	0.000	0.599	0.025	5.000	0.808
ensemble uncertainty gain	contact transition	1	0.333	0.159	1.396	0.000	0.606	0.025	5.000	0.807
ensemble uncertainty gain	contact transition	2	0.333	0.165	1.382	0.000	0.598	0.023	5.000	0.808
ensemble uncertainty gain	contact transition	3	0.500	0.156	1.332	0.000	0.595	0.018	5.000	0.807
ensemble uncertainty gain	contact transition	4	0.833	0.173	1.366	0.000	0.588	0.018	5.000	0.805
ensemble uncertainty gain	contact transition	5	0.333	0.167	1.390	0.000	0.601	0.031	5.000	0.808
ensemble uncertainty gain	contact transition	6	0.333	0.160	1.314	0.000	0.597	0.021	5.833	0.808
ensemble uncertainty gain	contact transition	7	0.500	0.171	1.397	0.000	0.596	0.027	5.000	0.802
ensemble uncertainty gain	delayed mode switch	0	0.167	0.183	2.082	0.049	0.593	0.029	5.000	0.806
ensemble uncertainty gain	delayed mode switch	1	0.500	0.195	1.918	0.039	0.581	0.031	8.833	0.806
ensemble uncertainty gain	delayed mode switch	2	0.333	0.184	1.926	0.043	0.591	0.020	5.000	0.807
ensemble uncertainty gain	delayed mode switch	3	0.667	0.205	1.878	0.020	0.574	0.025	12.167	0.810
ensemble uncertainty gain	delayed mode switch	4	0.333	0.198	2.135	0.041	0.573	0.026	8.667	0.809
ensemble uncertainty gain	delayed mode switch	5	0.833	0.175	1.816	0.033	0.571	0.045	7.333	0.803
ensemble uncertainty gain	delayed mode switch	6	0.500	0.199	1.948	0.041	0.587	0.023	7.333	0.809
ensemble uncertainty gain	delayed mode switch	7	0.667	0.185	1.981	0.037	0.582	0.029	7.667	0.808
ensemble uncertainty gain	friction slip shift	0	0.333	0.189	1.144	0.000	0.596	0.016	5.000	0.805
ensemble uncertainty gain	friction slip shift	1	0.667	0.189	1.151	0.000	0.593	0.014	5.000	0.802
ensemble uncertainty gain	friction slip shift	2	0.833	0.198	1.182	0.000	0.586	0.004	5.000	0.809
ensemble uncertainty gain	friction slip shift	3	0.667	0.198	1.153	0.000	0.586	0.012	5.000	0.809
ensemble uncertainty gain	friction slip shift	4	0.833	0.178	1.112	0.000	0.585	0.014	5.000	0.801
ensemble uncertainty gain	friction slip shift	5	0.833	0.187	1.137	0.000	0.585	0.006	5.000	0.803
ensemble uncertainty gain	friction slip shift	6	0.667	0.174	1.103	0.000	0.591	0.006	5.000	0.807
ensemble uncertainty gain	friction slip shift	7	0.833	0.171	1.105	0.000	0.583	0.016	5.000	0.802
ensemble uncertainty gain	nominal surface tracking	0	0.167	0.177	1.067	0.000	0.618	0.004	5.000	0.831
ensemble uncertainty gain	nominal surface tracking	1	0.333	0.177	1.080	0.000	0.614	0.006	5.000	0.830
ensemble uncertainty gain	nominal surface tracking	2	0.167	0.178	1.093	0.000	0.618	0.002	5.000	0.832
ensemble uncertainty gain	nominal surface tracking	3	0.333	0.187	1.118	0.000	0.610	0.006	5.000	0.832
ensemble uncertainty gain	nominal surface tracking	4	0.167	0.175	1.081	0.000	0.608	0.006	5.000	0.831
ensemble uncertainty gain	nominal surface tracking	5	0.333	0.180	1.087	0.000	0.613	0.006	5.000	0.830
ensemble uncertainty gain	nominal surface tracking	6	0.000	0.169	1.080	0.000	0.614	0.000	5.000	0.829
ensemble uncertainty gain	nominal surface tracking	7	0.167	0.181	1.096	0.000	0.615	0.002	5.000	0.833
ensemble uncertainty gain	sensor noise burst	0	0.333	0.157	1.342	0.000	0.609	0.045	5.000	0.822

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
ensemble uncertainty gain	sensor noise burst	1	0.333	0.158	1.367	0.000	0.605	0.077	5.000	0.818
ensemble uncertainty gain	sensor noise burst	2	0.167	0.168	1.335	0.000	0.602	0.077	5.000	0.818
ensemble uncertainty gain	sensor noise burst	3	0.500	0.165	1.386	0.000	0.602	0.060	5.000	0.821
ensemble uncertainty gain	sensor noise burst	4	0.833	0.167	1.393	0.000	0.596	0.052	5.167	0.818
ensemble uncertainty gain	sensor noise burst	5	0.500	0.163	1.366	0.000	0.600	0.066	5.000	0.821
ensemble uncertainty gain	sensor noise burst	6	0.333	0.165	1.410	0.000	0.609	0.066	5.000	0.820
ensemble uncertainty gain	sensor noise burst	7	0.500	0.165	1.450	0.000	0.605	0.056	5.000	0.821
ensemble uncertainty gain	stick slip cycle	0	0.667	0.153	1.093	0.000	0.593	0.021	5.000	0.800
ensemble uncertainty gain	stick slip cycle	1	0.667	0.161	1.169	0.000	0.591	0.029	5.000	0.800
ensemble uncertainty gain	stick slip cycle	2	1.000	0.159	1.153	0.000	0.579	0.033	5.000	0.797
ensemble uncertainty gain	stick slip cycle	3	1.000	0.162	1.109	0.000	0.573	0.021	5.000	0.794
ensemble uncertainty gain	stick slip cycle	4	0.833	0.154	1.100	0.000	0.583	0.018	5.000	0.804
ensemble uncertainty gain	stick slip cycle	5	0.833	0.171	1.179	0.000	0.589	0.027	5.000	0.798
ensemble uncertainty gain	stick slip cycle	6	0.833	0.159	1.187	0.000	0.585	0.019	5.000	0.795
ensemble uncertainty gain	stick slip cycle	7	1.000	0.160	1.159	0.000	0.579	0.021	5.000	0.799
ensemble uncertainty gain	stiffness shift	0	0.167	0.261	1.772	0.025	0.618	0.010	5.667	0.825
ensemble uncertainty gain	stiffness shift	1	0.000	0.255	1.868	0.037	0.619	0.010	5.000	0.822
ensemble uncertainty gain	stiffness shift	2	0.000	0.267	1.876	0.033	0.617	0.010	6.667	0.823
ensemble uncertainty gain	stiffness shift	3	0.000	0.259	1.836	0.027	0.605	0.014	6.000	0.822
ensemble uncertainty gain	stiffness shift	4	0.000	0.250	1.855	0.031	0.615	0.016	6.833	0.825
ensemble uncertainty gain	stiffness shift	5	0.000	0.265	1.807	0.033	0.612	0.015	5.167	0.822
ensemble uncertainty gain	stiffness shift	6	0.000	0.272	1.862	0.029	0.616	0.012	5.000	0.824
ensemble uncertainty gain	stiffness shift	7	0.000	0.260	1.812	0.027	0.616	0.016	7.000	0.824
ensemble uncertainty gain	surface discontinuity	0	0.333	0.153	1.254	0.000	0.592	0.033	5.000	0.803
ensemble uncertainty gain	surface discontinuity	1	0.667	0.153	1.297	0.000	0.593	0.031	5.667	0.801
ensemble uncertainty gain	surface discontinuity	2	0.833	0.152	1.254	0.000	0.591	0.029	5.000	0.805
ensemble uncertainty gain	surface discontinuity	3	0.667	0.159	1.282	0.000	0.584	0.025	5.000	0.803
ensemble uncertainty gain	surface discontinuity	4	0.667	0.161	1.219	0.000	0.586	0.021	5.000	0.802
ensemble uncertainty gain	surface discontinuity	5	0.500	0.161	1.315	0.000	0.593	0.033	5.000	0.803
ensemble uncertainty gain	surface discontinuity	6	0.500	0.152	1.300	0.000	0.594	0.027	5.167	0.807
ensemble uncertainty gain	surface discontinuity	7	0.500	0.155	1.239	0.000	0.594	0.045	5.000	0.802
ensemble uncertainty gain	target force jump	0	0.333	0.133	1.208	0.000	0.613	0.016	5.000	0.814
ensemble uncertainty gain	target force jump	1	0.167	0.122	1.173	0.000	0.608	0.025	5.000	0.811
ensemble uncertainty gain	target force jump	2	0.000	0.135	1.284	0.000	0.614	0.025	5.000	0.813
ensemble uncertainty gain	target force jump	3	0.333	0.138	1.201	0.000	0.603	0.028	5.000	0.813
ensemble uncertainty gain	target force jump	4	0.000	0.128	1.211	0.000	0.610	0.022	5.000	0.816
ensemble uncertainty gain	target force jump	5	0.333	0.126	1.210	0.000	0.608	0.026	5.000	0.811
ensemble uncertainty gain	target force jump	6	0.167	0.122	1.201	0.000	0.608	0.018	5.000	0.814
ensemble uncertainty gain	target force jump	7	0.333	0.130	1.210	0.000	0.608	0.029	5.000	0.811
fixed impedance	actuator saturation	0	1.000	0.196	1.677	0.011	0.564	0.041	22.333	0.814
fixed impedance	actuator saturation	1	1.000	0.206	1.751	0.019	0.563	0.038	26.500	0.812
fixed impedance	actuator saturation	2	1.000	0.184	1.728	0.017	0.569	0.034	6.500	0.815
fixed impedance	actuator saturation	3	1.000	0.206	1.708	0.011	0.570	0.042	12.000	0.812

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
fixed impedance	actuator saturation	4	0.833	0.200	1.752	0.026	0.561	0.047	5.667	0.813
fixed impedance	actuator saturation	5	1.000	0.185	1.729	0.017	0.556	0.038	6.833	0.814
fixed impedance	actuator saturation	6	1.000	0.202	1.709	0.015	0.573	0.036	6.667	0.812
fixed impedance	actuator saturation	7	1.000	0.204	1.751	0.026	0.561	0.041	10.000	0.813
fixed impedance	combined extreme stress	0	0.000	0.252	1.735	0.017	0.506	0.080	96.000	0.770
fixed impedance	combined extreme stress	1	0.333	0.238	1.691	0.008	0.498	0.087	96.000	0.776
fixed impedance	combined extreme stress	2	0.333	0.239	1.710	0.017	0.508	0.091	96.000	0.778
fixed impedance	combined extreme stress	3	0.333	0.247	1.697	0.009	0.509	0.089	80.833	0.779
fixed impedance	combined extreme stress	4	0.667	0.224	1.712	0.011	0.502	0.089	96.000	0.779
fixed impedance	combined extreme stress	5	0.333	0.242	1.679	0.006	0.498	0.106	96.000	0.779
fixed impedance	combined extreme stress	6	0.667	0.228	1.736	0.011	0.515	0.076	96.000	0.777
fixed impedance	combined extreme stress	7	0.500	0.234	1.743	0.017	0.500	0.078	96.000	0.775
fixed impedance	combined stress	0	0.167	0.259	1.633	0.002	0.528	0.044	96.000	0.781
fixed impedance	combined stress	1	0.167	0.246	1.634	0.002	0.525	0.053	96.000	0.782
fixed impedance	combined stress	2	0.333	0.238	1.640	0.006	0.536	0.059	96.000	0.784
fixed impedance	combined stress	3	0.000	0.260	1.708	0.011	0.534	0.062	96.000	0.782
fixed impedance	combined stress	4	0.333	0.250	1.688	0.009	0.517	0.070	96.000	0.779
fixed impedance	combined stress	5	0.167	0.259	1.752	0.015	0.528	0.078	96.000	0.779
fixed impedance	combined stress	6	0.500	0.236	1.741	0.021	0.536	0.055	82.667	0.785
fixed impedance	combined stress	7	0.333	0.239	1.718	0.011	0.521	0.055	96.000	0.778
fixed impedance	contact transition	0	1.000	0.191	1.817	0.032	0.568	0.036	5.000	0.812
fixed impedance	contact transition	1	1.000	0.197	1.913	0.047	0.562	0.040	7.500	0.812
fixed impedance	contact transition	2	1.000	0.190	1.844	0.036	0.569	0.045	5.000	0.813
fixed impedance	contact transition	3	1.000	0.191	1.932	0.049	0.568	0.036	5.000	0.812
fixed impedance	contact transition	4	0.833	0.197	1.867	0.042	0.561	0.032	5.000	0.812
fixed impedance	contact transition	5	1.000	0.194	1.868	0.042	0.562	0.044	7.500	0.813
fixed impedance	contact transition	6	1.000	0.194	1.912	0.049	0.573	0.038	5.500	0.813
fixed impedance	contact transition	7	0.667	0.208	1.860	0.038	0.561	0.040	6.500	0.809
fixed impedance	delayed mode switch	0	0.667	0.219	1.898	0.045	0.549	0.047	67.833	0.809
fixed impedance	delayed mode switch	1	0.667	0.222	1.878	0.034	0.544	0.042	67.333	0.809
fixed impedance	delayed mode switch	2	0.833	0.210	1.813	0.030	0.557	0.047	96.000	0.811
fixed impedance	delayed mode switch	3	1.000	0.214	1.751	0.015	0.549	0.040	96.000	0.811
fixed impedance	delayed mode switch	4	0.667	0.219	1.887	0.034	0.545	0.030	96.000	0.811
fixed impedance	delayed mode switch	5	1.000	0.200	1.838	0.049	0.536	0.045	42.000	0.805
fixed impedance	delayed mode switch	6	0.833	0.218	1.799	0.030	0.557	0.040	36.833	0.812
fixed impedance	delayed mode switch	7	0.667	0.208	1.882	0.038	0.545	0.038	80.833	0.810
fixed impedance	friction slip shift	0	0.833	0.191	1.809	0.038	0.566	0.023	5.000	0.811
fixed impedance	friction slip shift	1	1.000	0.187	1.771	0.028	0.564	0.025	5.000	0.811
fixed impedance	friction slip shift	2	1.000	0.197	1.778	0.032	0.568	0.019	5.000	0.813
fixed impedance	friction slip shift	3	0.667	0.201	1.875	0.044	0.570	0.023	5.000	0.814
fixed impedance	friction slip shift	4	0.833	0.188	1.902	0.049	0.565	0.023	5.000	0.809
fixed impedance	friction slip shift	5	0.833	0.187	1.788	0.034	0.562	0.021	5.000	0.811
fixed impedance	friction slip shift	6	0.667	0.188	1.928	0.051	0.573	0.023	5.000	0.812
fixed impedance	friction slip shift	7	0.667	0.197	1.925	0.049	0.561	0.021	5.000	0.809
fixed impedance	nominal surface tracking	0	1.000	0.175	1.828	0.041	0.578	0.011	5.000	0.827
fixed impedance	nominal surface tracking	1	1.000	0.184	1.849	0.041	0.576	0.013	5.000	0.827
fixed impedance	nominal surface tracking	2	0.667	0.172	1.783	0.024	0.583	0.011	5.000	0.827
fixed impedance	nominal surface tracking	3	1.000	0.180	1.810	0.032	0.582	0.011	5.000	0.827
fixed impedance	nominal surface tracking	4	1.000	0.175	1.845	0.036	0.581	0.013	5.000	0.827
fixed impedance	nominal surface tracking	5	0.833	0.180	1.834	0.042	0.576	0.011	5.000	0.827
fixed impedance	nominal surface tracking	6	1.000	0.184	1.895	0.047	0.584	0.015	5.000	0.826
fixed impedance	nominal surface tracking	7	1.000	0.174	1.820	0.038	0.575	0.011	5.000	0.828
fixed impedance	sensor noise burst	0	1.000	0.169	1.811	0.036	0.573	0.066	5.000	0.821
fixed impedance	sensor noise burst	1	0.833	0.171	1.776	0.021	0.564	0.068	5.000	0.819
fixed impedance	sensor noise burst	2	1.000	0.184	1.758	0.021	0.569	0.073	5.333	0.818
fixed impedance	sensor noise burst	3	1.000	0.187	1.817	0.034	0.572	0.070	5.000	0.819
fixed impedance	sensor noise burst	4	1.000	0.186	1.765	0.028	0.565	0.055	5.000	0.818
fixed impedance	sensor noise burst	5	1.000	0.177	1.736	0.023	0.562	0.068	5.333	0.820
fixed impedance	sensor noise burst	6	1.000	0.182	1.751	0.026	0.574	0.070	5.000	0.819
fixed impedance	sensor noise burst	7	1.000	0.177	1.762	0.032	0.562	0.070	5.000	0.819

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
fixed impedance	stick slip cycle	0	1.000	0.160	1.878	0.047	0.561	0.032	5.000	0.806
fixed impedance	stick slip cycle	1	1.000	0.166	1.799	0.034	0.555	0.045	5.000	0.805
fixed impedance	stick slip cycle	2	1.000	0.158	1.790	0.030	0.554	0.043	5.000	0.803
fixed impedance	stick slip cycle	3	1.000	0.169	1.893	0.042	0.562	0.036	5.000	0.802
fixed impedance	stick slip cycle	4	1.000	0.169	1.956	0.055	0.555	0.045	5.000	0.807
fixed impedance	stick slip cycle	5	1.000	0.166	1.779	0.030	0.554	0.030	5.000	0.805
fixed impedance	stick slip cycle	6	1.000	0.174	1.855	0.038	0.566	0.038	5.000	0.801
fixed impedance	stick slip cycle	7	1.000	0.160	1.813	0.038	0.550	0.036	5.000	0.804
fixed impedance	stiffness shift	0	0.000	0.301	1.733	0.019	0.580	0.023	54.000	0.824
fixed impedance	stiffness shift	1	0.000	0.295	1.761	0.019	0.575	0.019	66.333	0.822
fixed impedance	stiffness shift	2	0.000	0.303	1.755	0.013	0.577	0.028	35.333	0.823
fixed impedance	stiffness shift	3	0.000	0.306	1.765	0.040	0.577	0.025	69.333	0.822
fixed impedance	stiffness shift	4	0.000	0.289	1.775	0.028	0.576	0.023	67.333	0.823
fixed impedance	stiffness shift	5	0.000	0.306	1.766	0.026	0.572	0.021	81.333	0.822
fixed impedance	stiffness shift	6	0.000	0.311	1.758	0.021	0.584	0.025	65.667	0.823
fixed impedance	stiffness shift	7	0.000	0.283	1.704	0.013	0.577	0.017	40.500	0.823
fixed impedance	surface discontinuity	0	1.000	0.209	1.737	0.017	0.555	0.038	41.667	0.806
fixed impedance	surface discontinuity	1	0.833	0.213	1.756	0.025	0.555	0.032	54.333	0.806
fixed impedance	surface discontinuity	2	0.833	0.213	1.782	0.027	0.564	0.028	25.833	0.808
fixed impedance	surface discontinuity	3	1.000	0.216	1.729	0.013	0.561	0.034	42.667	0.807
fixed impedance	surface discontinuity	4	0.833	0.227	1.829	0.036	0.557	0.036	42.833	0.807
fixed impedance	surface discontinuity	5	1.000	0.208	1.688	0.008	0.555	0.038	42.167	0.809
fixed impedance	surface discontinuity	6	1.000	0.202	1.729	0.015	0.564	0.034	24.500	0.810
fixed impedance	surface discontinuity	7	1.000	0.199	1.757	0.019	0.551	0.030	15.333	0.808
fixed impedance	target force jump	0	1.000	0.142	1.879	0.042	0.572	0.040	5.000	0.817
fixed impedance	target force jump	1	0.833	0.140	1.967	0.062	0.562	0.045	5.000	0.816
fixed impedance	target force jump	2	1.000	0.148	1.863	0.040	0.572	0.030	5.000	0.816
fixed impedance	target force jump	3	1.000	0.147	1.924	0.049	0.572	0.047	5.000	0.817
fixed impedance	target force jump	4	0.833	0.142	1.968	0.051	0.568	0.042	5.000	0.818
fixed impedance	target force jump	5	0.833	0.143	1.923	0.049	0.566	0.040	5.000	0.815
fixed impedance	target force jump	6	0.833	0.138	1.970	0.055	0.570	0.040	5.000	0.818
fixed impedance	target force jump	7	0.833	0.144	1.965	0.051	0.564	0.047	5.000	0.816
gain scheduled impedance	actuator saturation	0	0.167	0.130	1.652	0.008	0.609	0.025	5.000	0.853
gain scheduled impedance	actuator saturation	1	0.167	0.124	1.543	0.004	0.608	0.040	5.000	0.852
gain scheduled impedance	actuator saturation	2	0.000	0.121	1.420	0.000	0.617	0.028	5.000	0.855
gain scheduled impedance	actuator saturation	3	0.000	0.132	1.605	0.002	0.614	0.032	5.000	0.851
gain scheduled impedance	actuator saturation	4	0.167	0.123	1.487	0.000	0.610	0.023	6.000	0.854
gain scheduled impedance	actuator saturation	5	0.333	0.121	1.508	0.000	0.602	0.027	5.000	0.854
gain scheduled impedance	actuator saturation	6	0.167	0.126	1.586	0.006	0.605	0.027	5.000	0.851
gain scheduled impedance	actuator saturation	7	0.167	0.122	1.518	0.004	0.614	0.025	5.000	0.853
gain scheduled impedance	combined extreme stress	0	1.000	0.130	1.532	0.002	0.572	0.105	5.167	0.815
gain scheduled impedance	combined extreme stress	1	0.833	0.140	1.644	0.008	0.563	0.150	6.500	0.820
gain scheduled impedance	combined extreme stress	2	1.000	0.130	1.511	0.002	0.589	0.106	5.000	0.824
gain scheduled impedance	combined extreme stress	3	1.000	0.138	1.478	0.002	0.578	0.107	5.000	0.825
gain scheduled impedance	combined extreme stress	4	1.000	0.132	1.539	0.000	0.576	0.123	6.667	0.825
gain scheduled impedance	combined extreme stress	5	1.000	0.133	1.464	0.002	0.567	0.125	6.333	0.825
gain scheduled impedance	combined extreme stress	6	1.000	0.124	1.492	0.004	0.574	0.133	5.000	0.822
gain scheduled impedance	combined extreme stress	7	0.833	0.125	1.490	0.002	0.576	0.094	5.000	0.819
gain scheduled impedance	combined stress	0	0.500	0.147	1.826	0.025	0.588	0.063	5.000	0.819
gain scheduled impedance	combined stress	1	1.000	0.146	1.770	0.015	0.582	0.080	5.000	0.817
gain scheduled impedance	combined stress	2	0.833	0.138	1.647	0.011	0.599	0.061	5.000	0.820
gain scheduled impedance	combined stress	3	0.667	0.141	1.675	0.004	0.595	0.090	5.000	0.818
gain scheduled impedance	combined stress	4	1.000	0.141	1.658	0.006	0.575	0.077	5.000	0.815
gain scheduled impedance	combined stress	5	1.000	0.129	1.502	0.002	0.580	0.079	5.000	0.817
gain scheduled impedance	combined stress	6	0.833	0.131	1.571	0.008	0.579	0.111	5.000	0.822

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
gain scheduled impedance	combined stress	7	1.000	0.130	1.479	0.002	0.586	0.056	6.167	0.813
gain scheduled impedance	contact transition	0	0.167	0.126	1.406	0.000	0.616	0.023	5.000	0.847
gain scheduled impedance	contact transition	1	0.167	0.121	1.395	0.000	0.609	0.027	5.000	0.847
gain scheduled impedance	contact transition	2	0.000	0.120	1.402	0.000	0.618	0.017	5.000	0.848
gain scheduled impedance	contact transition	3	0.000	0.117	1.352	0.000	0.618	0.034	5.000	0.847
gain scheduled impedance	contact transition	4	0.167	0.122	1.367	0.000	0.606	0.019	5.000	0.846
gain scheduled impedance	contact transition	5	0.333	0.123	1.389	0.000	0.609	0.027	5.000	0.847
gain scheduled impedance	contact transition	6	0.167	0.117	1.285	0.000	0.605	0.017	5.167	0.848
gain scheduled impedance	contact transition	7	0.000	0.127	1.409	0.000	0.615	0.027	5.000	0.841
gain scheduled impedance	delayed mode switch	0	0.167	0.143	2.082	0.042	0.603	0.042	5.000	0.848
gain scheduled impedance	delayed mode switch	1	0.500	0.151	1.956	0.027	0.594	0.040	7.667	0.847
gain scheduled impedance	delayed mode switch	2	0.000	0.143	1.954	0.034	0.612	0.036	5.000	0.848
gain scheduled impedance	delayed mode switch	3	0.500	0.152	1.845	0.017	0.602	0.036	9.500	0.850
gain scheduled impedance	delayed mode switch	4	0.167	0.144	2.115	0.033	0.594	0.029	6.500	0.849
gain scheduled impedance	delayed mode switch	5	0.667	0.132	1.767	0.023	0.585	0.040	6.833	0.844
gain scheduled impedance	delayed mode switch	6	0.500	0.150	1.937	0.031	0.595	0.017	6.667	0.850
gain scheduled impedance	delayed mode switch	7	0.500	0.134	1.926	0.027	0.606	0.021	5.667	0.849
gain scheduled impedance	friction slip shift	0	0.500	0.132	1.145	0.000	0.602	0.009	5.000	0.844
gain scheduled impedance	friction slip shift	1	0.500	0.134	1.158	0.000	0.599	0.009	5.000	0.842
gain scheduled impedance	friction slip shift	2	0.167	0.138	1.163	0.000	0.608	0.006	5.167	0.847
gain scheduled impedance	friction slip shift	3	0.167	0.139	1.153	0.000	0.610	0.017	5.000	0.849
gain scheduled impedance	friction slip shift	4	0.000	0.123	1.124	0.000	0.606	0.011	5.000	0.840
gain scheduled impedance	friction slip shift	5	0.500	0.127	1.133	0.000	0.601	0.013	5.000	0.843
gain scheduled impedance	friction slip shift	6	0.333	0.122	1.092	0.000	0.599	0.004	5.000	0.846
gain scheduled impedance	friction slip shift	7	0.000	0.115	1.093	0.000	0.612	0.008	5.000	0.841
gain scheduled impedance	nominal surface tracking	0	0.000	0.120	1.068	0.000	0.620	0.004	5.000	0.870
gain scheduled impedance	nominal surface tracking	1	0.000	0.122	1.087	0.000	0.617	0.000	5.000	0.871
gain scheduled impedance	nominal surface tracking	2	0.000	0.123	1.084	0.000	0.628	0.002	5.000	0.870
gain scheduled impedance	nominal surface tracking	3	0.000	0.128	1.111	0.000	0.624	0.006	5.000	0.870
gain scheduled impedance	nominal surface tracking	4	0.000	0.123	1.089	0.000	0.621	0.002	5.000	0.869
gain scheduled impedance	nominal surface tracking	5	0.000	0.125	1.093	0.000	0.612	0.002	5.000	0.869
gain scheduled impedance	nominal surface tracking	6	0.000	0.116	1.068	0.000	0.614	0.002	5.000	0.869
gain scheduled impedance	nominal surface tracking	7	0.000	0.122	1.082	0.000	0.631	0.004	5.000	0.871
gain scheduled impedance	sensor noise burst	0	0.000	0.116	1.357	0.000	0.620	0.061	5.000	0.863
gain scheduled impedance	sensor noise burst	1	0.167	0.119	1.339	0.000	0.609	0.053	5.000	0.859
gain scheduled impedance	sensor noise burst	2	0.000	0.121	1.322	0.000	0.622	0.055	5.000	0.859
gain scheduled impedance	sensor noise burst	3	0.000	0.124	1.448	0.000	0.617	0.057	5.000	0.861
gain scheduled impedance	sensor noise burst	4	0.000	0.122	1.415	0.002	0.616	0.055	5.000	0.859
gain scheduled impedance	sensor noise burst	5	0.000	0.121	1.363	0.000	0.610	0.062	5.333	0.861
gain scheduled impedance	sensor noise burst	6	0.167	0.122	1.333	0.000	0.610	0.068	5.000	0.861
gain scheduled impedance	sensor noise burst	7	0.000	0.116	1.405	0.000	0.619	0.059	5.000	0.861
gain scheduled impedance	stick slip cycle	0	0.167	0.111	1.123	0.000	0.606	0.023	5.000	0.841
gain scheduled impedance	stick slip cycle	1	0.833	0.119	1.185	0.000	0.592	0.032	5.000	0.840

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
gain scheduled impedance	stick slip cycle	2	0.333	0.114	1.155	0.000	0.606	0.023	5.000	0.835
gain scheduled impedance	stick slip cycle	3	0.667	0.115	1.110	0.000	0.602	0.030	5.000	0.834
gain scheduled impedance	stick slip cycle	4	0.667	0.107	1.111	0.000	0.601	0.023	5.000	0.843
gain scheduled impedance	stick slip cycle	5	0.500	0.123	1.195	0.000	0.600	0.040	5.000	0.838
gain scheduled impedance	stick slip cycle	6	0.667	0.116	1.188	0.000	0.594	0.034	5.000	0.835
gain scheduled impedance	stick slip cycle	7	0.500	0.114	1.139	0.000	0.600	0.021	5.000	0.838
gain scheduled impedance	stiffness shift	0	0.000	0.196	1.806	0.019	0.619	0.008	5.333	0.867
gain scheduled impedance	stiffness shift	1	0.000	0.194	1.914	0.032	0.616	0.011	5.000	0.865
gain scheduled impedance	stiffness shift	2	0.000	0.203	1.876	0.029	0.631	0.006	5.500	0.865
gain scheduled impedance	stiffness shift	3	0.000	0.193	1.858	0.027	0.621	0.009	5.500	0.864
gain scheduled impedance	stiffness shift	4	0.000	0.188	1.894	0.027	0.626	0.013	6.333	0.866
gain scheduled impedance	stiffness shift	5	0.000	0.199	1.843	0.028	0.610	0.006	5.167	0.865
gain scheduled impedance	stiffness shift	6	0.167	0.206	1.854	0.025	0.612	0.008	5.000	0.866
gain scheduled impedance	stiffness shift	7	0.000	0.192	1.816	0.023	0.624	0.011	6.167	0.866
gain scheduled impedance	surface discontinuity	0	0.500	0.111	1.165	0.000	0.605	0.029	5.000	0.844
gain scheduled impedance	surface discontinuity	1	0.667	0.115	1.202	0.000	0.596	0.034	5.000	0.842
gain scheduled impedance	surface discontinuity	2	0.000	0.110	1.156	0.000	0.616	0.027	5.000	0.845
gain scheduled impedance	surface discontinuity	3	0.167	0.114	1.194	0.000	0.608	0.034	5.000	0.844
gain scheduled impedance	surface discontinuity	4	0.333	0.114	1.165	0.000	0.608	0.023	5.000	0.843
gain scheduled impedance	surface discontinuity	5	0.667	0.119	1.230	0.000	0.602	0.031	5.000	0.845
gain scheduled impedance	surface discontinuity	6	0.333	0.109	1.190	0.000	0.603	0.021	5.000	0.847
gain scheduled impedance	surface discontinuity	7	0.333	0.112	1.160	0.000	0.608	0.023	5.000	0.842
gain scheduled impedance	target force jump	0	0.000	0.104	1.242	0.000	0.629	0.027	5.000	0.855
gain scheduled impedance	target force jump	1	0.167	0.096	1.177	0.000	0.617	0.021	5.000	0.852
gain scheduled impedance	target force jump	2	0.000	0.104	1.278	0.000	0.628	0.029	5.000	0.854
gain scheduled impedance	target force jump	3	0.000	0.107	1.201	0.000	0.626	0.027	5.000	0.854
gain scheduled impedance	target force jump	4	0.000	0.098	1.194	0.000	0.622	0.023	5.000	0.857
gain scheduled impedance	target force jump	5	0.167	0.096	1.221	0.000	0.618	0.035	5.000	0.853
gain scheduled impedance	target force jump	6	0.000	0.096	1.210	0.000	0.618	0.019	5.000	0.856
gain scheduled impedance	target force jump	7	0.000	0.100	1.227	0.000	0.629	0.015	5.000	0.852
hist gradient gain regressor	actuator saturation	0	0.500	0.166	1.633	0.006	0.605	0.031	5.000	0.816
hist gradient gain regressor	actuator saturation	1	0.167	0.162	1.528	0.002	0.616	0.029	5.000	0.814
hist gradient gain regressor	actuator saturation	2	0.500	0.158	1.404	0.000	0.602	0.031	5.667	0.818
hist gradient gain regressor	actuator saturation	3	0.500	0.170	1.558	0.000	0.600	0.035	5.000	0.814
hist gradient gain regressor	actuator saturation	4	0.667	0.167	1.472	0.000	0.597	0.033	6.667	0.817
hist gradient gain regressor	actuator saturation	5	0.833	0.158	1.470	0.000	0.595	0.031	5.000	0.818
hist gradient gain regressor	actuator saturation	6	0.000	0.171	1.571	0.004	0.614	0.033	5.000	0.814
hist gradient gain regressor	actuator saturation	7	0.667	0.161	1.496	0.000	0.591	0.023	5.000	0.816
hist gradient gain regressor	combined extreme stress	0	1.000	0.170	1.706	0.012	0.548	0.114	5.167	0.776
hist gradient gain regressor	combined extreme stress	1	0.667	0.182	1.681	0.014	0.564	0.106	8.667	0.783
hist gradient gain regressor	combined extreme stress	2	0.833	0.171	1.731	0.004	0.559	0.122	6.667	0.784
hist gradient gain regressor	combined extreme stress	3	0.833	0.180	1.557	0.006	0.562	0.105	7.000	0.787
hist gradient gain regressor	combined extreme stress	4	1.000	0.168	1.686	0.002	0.559	0.131	7.167	0.788

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
hist gradient gain regressor	combined extreme stress	5	0.833	0.188	1.705	0.006	0.546	0.113	9.333	0.787
hist gradient gain regressor	combined extreme stress	6	0.833	0.167	1.745	0.012	0.578	0.104	5.000	0.783
hist gradient gain regressor	combined extreme stress	7	1.000	0.171	1.650	0.006	0.550	0.095	6.500	0.782
hist gradient gain regressor	combined stress	0	1.000	0.188	1.775	0.037	0.562	0.058	5.167	0.778
hist gradient gain regressor	combined stress	1	0.833	0.184	1.724	0.029	0.572	0.061	5.000	0.775
hist gradient gain regressor	combined stress	2	0.833	0.180	1.678	0.022	0.558	0.068	5.000	0.779
hist gradient gain regressor	combined stress	3	1.000	0.176	1.670	0.008	0.564	0.078	5.000	0.776
hist gradient gain regressor	combined stress	4	0.833	0.183	1.816	0.027	0.548	0.062	5.167	0.774
hist gradient gain regressor	combined stress	5	1.000	0.165	1.605	0.004	0.559	0.086	6.500	0.774
hist gradient gain regressor	combined stress	6	0.833	0.179	1.585	0.010	0.576	0.072	6.333	0.782
hist gradient gain regressor	combined stress	7	1.000	0.177	1.558	0.002	0.541	0.051	7.833	0.772
hist gradient gain regressor	contact transition	0	0.500	0.163	1.376	0.000	0.600	0.025	5.000	0.810
hist gradient gain regressor	contact transition	1	0.167	0.154	1.383	0.000	0.616	0.023	5.000	0.809
hist gradient gain regressor	contact transition	2	0.500	0.158	1.360	0.000	0.598	0.019	5.000	0.811
hist gradient gain regressor	contact transition	3	0.667	0.150	1.332	0.000	0.597	0.037	5.000	0.810
hist gradient gain regressor	contact transition	4	0.667	0.163	1.333	0.000	0.598	0.017	5.000	0.808
hist gradient gain regressor	contact transition	5	0.667	0.159	1.351	0.000	0.597	0.029	5.000	0.810
hist gradient gain regressor	contact transition	6	0.333	0.160	1.306	0.000	0.609	0.018	6.000	0.811
hist gradient gain regressor	contact transition	7	0.833	0.163	1.391	0.000	0.580	0.023	5.000	0.804
hist gradient gain regressor	delayed mode switch	0	0.500	0.182	2.041	0.049	0.588	0.033	5.000	0.813
hist gradient gain regressor	delayed mode switch	1	0.333	0.198	1.897	0.039	0.597	0.039	8.667	0.810
hist gradient gain regressor	delayed mode switch	2	0.333	0.180	1.916	0.039	0.590	0.033	5.333	0.812
hist gradient gain regressor	delayed mode switch	3	0.500	0.203	1.886	0.016	0.581	0.027	13.167	0.814
hist gradient gain regressor	delayed mode switch	4	0.333	0.193	2.096	0.037	0.581	0.023	9.667	0.814
hist gradient gain regressor	delayed mode switch	5	0.833	0.172	1.738	0.025	0.572	0.045	7.500	0.808
hist gradient gain regressor	delayed mode switch	6	0.500	0.205	1.924	0.043	0.597	0.029	7.667	0.815
hist gradient gain regressor	delayed mode switch	7	0.667	0.179	1.913	0.027	0.579	0.025	8.333	0.813
hist gradient gain regressor	friction slip shift	0	0.833	0.181	1.121	0.000	0.588	0.008	5.000	0.809
hist gradient gain regressor	friction slip shift	1	0.667	0.183	1.125	0.000	0.601	0.008	5.000	0.805
hist gradient gain regressor	friction slip shift	2	0.500	0.191	1.165	0.000	0.592	0.008	5.000	0.812
hist gradient gain regressor	friction slip shift	3	0.333	0.190	1.125	0.000	0.595	0.017	5.000	0.815
hist gradient gain regressor	friction slip shift	4	0.833	0.168	1.100	0.000	0.589	0.017	5.000	0.803
hist gradient gain regressor	friction slip shift	5	0.833	0.177	1.108	0.000	0.592	0.006	5.000	0.807
hist gradient gain regressor	friction slip shift	6	0.500	0.170	1.083	0.000	0.601	0.008	5.000	0.811
hist gradient gain regressor	friction slip shift	7	1.000	0.158	1.096	0.000	0.574	0.008	5.000	0.806
hist gradient gain regressor	nominal surface tracking	0	0.000	0.171	1.055	0.000	0.618	0.008	5.000	0.837
hist gradient gain regressor	nominal surface tracking	1	0.000	0.174	1.067	0.000	0.624	0.008	5.000	0.838
hist gradient gain regressor	nominal surface tracking	2	0.167	0.172	1.078	0.000	0.621	0.012	5.000	0.837
hist gradient gain regressor	nominal surface tracking	3	0.000	0.177	1.100	0.000	0.618	0.004	5.000	0.837
hist gradient gain regressor	nominal surface tracking	4	0.000	0.167	1.078	0.000	0.622	0.002	5.000	0.837
hist gradient gain regressor	nominal surface tracking	5	0.167	0.176	1.075	0.000	0.615	0.002	5.000	0.836
hist gradient gain regressor	nominal surface tracking	6	0.000	0.168	1.051	0.000	0.627	0.008	5.000	0.835
hist gradient gain regressor	nominal surface tracking	7	0.167	0.171	1.083	0.000	0.617	0.006	5.000	0.839

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
hist gradient gain regressor	sensor noise burst	0	0.167	0.149	1.290	0.000	0.609	0.077	5.000	0.828
hist gradient gain regressor	sensor noise burst	1	0.167	0.155	1.332	0.000	0.614	0.073	5.000	0.824
hist gradient gain regressor	sensor noise burst	2	0.333	0.159	1.333	0.000	0.605	0.062	5.000	0.824
hist gradient gain regressor	sensor noise burst	3	0.333	0.160	1.363	0.000	0.607	0.056	5.000	0.827
hist gradient gain regressor	sensor noise burst	4	0.833	0.163	1.361	0.000	0.603	0.073	5.000	0.824
hist gradient gain regressor	sensor noise burst	5	0.167	0.154	1.357	0.000	0.612	0.067	5.000	0.827
hist gradient gain regressor	sensor noise burst	6	0.000	0.166	1.413	0.000	0.619	0.058	5.000	0.827
hist gradient gain regressor	sensor noise burst	7	0.500	0.153	1.387	0.000	0.599	0.056	5.000	0.827
hist gradient gain regressor	stick slip cycle	0	0.833	0.145	1.066	0.000	0.591	0.021	5.000	0.805
hist gradient gain regressor	stick slip cycle	1	0.500	0.158	1.139	0.000	0.596	0.016	5.000	0.805
hist gradient gain regressor	stick slip cycle	2	0.833	0.152	1.137	0.000	0.578	0.019	5.000	0.800
hist gradient gain regressor	stick slip cycle	3	1.000	0.153	1.079	0.000	0.579	0.027	5.167	0.799
hist gradient gain regressor	stick slip cycle	4	0.667	0.144	1.084	0.000	0.587	0.019	5.000	0.810
hist gradient gain regressor	stick slip cycle	5	1.000	0.163	1.182	0.000	0.586	0.037	5.000	0.803
hist gradient gain regressor	stick slip cycle	6	0.333	0.156	1.191	0.000	0.598	0.023	5.000	0.799
hist gradient gain regressor	stick slip cycle	7	1.000	0.151	1.147	0.000	0.571	0.021	5.000	0.805
hist gradient gain regressor	stiffness shift	0	0.000	0.257	1.766	0.014	0.618	0.017	5.500	0.832
hist gradient gain regressor	stiffness shift	1	0.000	0.256	1.862	0.031	0.623	0.014	5.000	0.829
hist gradient gain regressor	stiffness shift	2	0.000	0.268	1.849	0.029	0.617	0.010	6.833	0.830
hist gradient gain regressor	stiffness shift	3	0.000	0.258	1.810	0.025	0.613	0.013	5.833	0.828
hist gradient gain regressor	stiffness shift	4	0.000	0.246	1.826	0.029	0.620	0.017	7.000	0.832
hist gradient gain regressor	stiffness shift	5	0.167	0.267	1.801	0.031	0.612	0.010	5.667	0.830
hist gradient gain regressor	stiffness shift	6	0.000	0.277	1.831	0.027	0.632	0.016	5.000	0.830
hist gradient gain regressor	stiffness shift	7	0.000	0.257	1.768	0.025	0.606	0.010	7.333	0.830
hist gradient gain regressor	surface discontinuity	0	1.000	0.150	1.265	0.000	0.590	0.035	5.000	0.806
hist gradient gain regressor	surface discontinuity	1	0.500	0.156	1.297	0.000	0.600	0.031	5.833	0.803
hist gradient gain regressor	surface discontinuity	2	0.667	0.150	1.263	0.000	0.589	0.031	5.000	0.808
hist gradient gain regressor	surface discontinuity	3	0.667	0.156	1.296	0.000	0.592	0.025	5.000	0.807
hist gradient gain regressor	surface discontinuity	4	0.833	0.156	1.264	0.000	0.593	0.029	5.000	0.805
hist gradient gain regressor	surface discontinuity	5	0.500	0.161	1.343	0.000	0.594	0.029	5.000	0.806
hist gradient gain regressor	surface discontinuity	6	0.167	0.155	1.318	0.000	0.614	0.027	5.333	0.809
hist gradient gain regressor	surface discontinuity	7	0.833	0.151	1.271	0.000	0.578	0.023	5.000	0.804
hist gradient gain regressor	target force jump	0	0.000	0.128	1.202	0.000	0.616	0.023	5.000	0.818
hist gradient gain regressor	target force jump	1	0.000	0.117	1.130	0.000	0.625	0.024	5.000	0.815
hist gradient gain regressor	target force jump	2	0.333	0.128	1.253	0.000	0.615	0.021	5.000	0.817
hist gradient gain regressor	target force jump	3	0.167	0.131	1.186	0.000	0.616	0.037	5.000	0.817
hist gradient gain regressor	target force jump	4	0.000	0.121	1.223	0.000	0.618	0.033	5.000	0.820
hist gradient gain regressor	target force jump	5	0.000	0.118	1.235	0.000	0.617	0.029	5.000	0.816
hist gradient gain regressor	target force jump	6	0.000	0.123	1.179	0.000	0.631	0.024	5.000	0.818
hist gradient gain regressor	target force jump	7	0.333	0.124	1.199	0.000	0.606	0.024	5.000	0.816
impedance token policy v4	actuator saturation	0	0.167	0.162	1.445	0.000	0.491	0.019	5.000	0.735
impedance token policy v4	actuator saturation	1	0.167	0.162	1.401	0.000	0.491	0.025	5.000	0.732
impedance token policy v4	actuator saturation	2	0.167	0.154	1.301	0.000	0.464	0.013	5.000	0.732

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
impedance token policy v4	actuator saturation	3	0.000	0.166	1.473	0.000	0.454	0.015	5.000	0.728
impedance token policy v4	actuator saturation	4	0.167	0.161	1.323	0.000	0.484	0.027	5.167	0.733
impedance token policy v4	actuator saturation	5	0.333	0.153	1.344	0.000	0.469	0.019	5.000	0.735
impedance token policy v4	actuator saturation	6	0.000	0.167	1.448	0.000	0.450	0.021	5.000	0.730
impedance token policy v4	actuator saturation	7	0.000	0.160	1.344	0.000	0.478	0.025	5.000	0.733
impedance token policy v4	combined extreme stress	0	0.000	0.140	1.441	0.000	0.376	0.099	5.000	0.718
impedance token policy v4	combined extreme stress	1	0.000	0.144	1.324	0.002	0.420	0.101	5.000	0.718
impedance token policy v4	combined extreme stress	2	0.000	0.142	1.353	0.000	0.407	0.097	5.000	0.723
impedance token policy v4	combined extreme stress	3	0.000	0.150	1.334	0.000	0.409	0.076	5.000	0.723
impedance token policy v4	combined extreme stress	4	0.000	0.141	1.315	0.000	0.441	0.095	5.000	0.725
impedance token policy v4	combined extreme stress	5	0.000	0.147	1.337	0.000	0.428	0.103	5.167	0.717
impedance token policy v4	combined extreme stress	6	0.000	0.140	1.398	0.000	0.395	0.083	5.000	0.718
impedance token policy v4	combined extreme stress	7	0.000	0.139	1.392	0.000	0.399	0.110	5.000	0.709
impedance token policy v4	combined stress	0	0.000	0.156	1.572	0.006	0.381	0.054	5.000	0.721
impedance token policy v4	combined stress	1	0.000	0.155	1.452	0.004	0.371	0.075	5.000	0.717
impedance token policy v4	combined stress	2	0.000	0.151	1.507	0.000	0.377	0.070	5.000	0.723
impedance token policy v4	combined stress	3	0.000	0.155	1.472	0.000	0.368	0.054	5.000	0.721
impedance token policy v4	combined stress	4	0.000	0.152	1.520	0.002	0.376	0.054	5.000	0.717
impedance token policy v4	combined stress	5	0.000	0.139	1.349	0.000	0.347	0.075	5.000	0.721
impedance token policy v4	combined stress	6	0.000	0.145	1.388	0.000	0.369	0.050	5.167	0.721
impedance token policy v4	combined stress	7	0.000	0.147	1.359	0.000	0.353	0.070	5.167	0.712
impedance token policy v4	contact transition	0	0.000	0.162	1.259	0.000	0.447	0.025	5.000	0.725
impedance token policy v4	contact transition	1	0.000	0.156	1.260	0.000	0.470	0.023	5.000	0.723
impedance token policy v4	contact transition	2	0.000	0.160	1.255	0.000	0.446	0.017	5.167	0.726
impedance token policy v4	contact transition	3	0.000	0.154	1.188	0.000	0.450	0.023	5.167	0.726
impedance token policy v4	contact transition	4	0.000	0.163	1.247	0.000	0.428	0.017	5.167	0.722
impedance token policy v4	contact transition	5	0.000	0.162	1.272	0.000	0.432	0.017	5.000	0.723
impedance token policy v4	contact transition	6	0.000	0.152	1.138	0.000	0.447	0.033	5.000	0.727
impedance token policy v4	contact transition	7	0.000	0.165	1.298	0.000	0.419	0.019	5.167	0.721
impedance token policy v4	delayed mode switch	0	0.500	0.156	1.754	0.014	0.513	0.041	5.000	0.739
impedance token policy v4	delayed mode switch	1	0.667	0.163	1.636	0.010	0.502	0.033	6.667	0.741
impedance token policy v4	delayed mode switch	2	0.667	0.154	1.650	0.008	0.486	0.027	5.000	0.742
impedance token policy v4	delayed mode switch	3	0.333	0.168	1.610	0.006	0.501	0.021	7.167	0.736
impedance token policy v4	delayed mode switch	4	0.167	0.163	1.906	0.017	0.502	0.021	5.667	0.738
impedance token policy v4	delayed mode switch	5	0.167	0.147	1.561	0.006	0.480	0.025	6.500	0.735
impedance token policy v4	delayed mode switch	6	0.500	0.167	1.758	0.015	0.510	0.029	6.000	0.741
impedance token policy v4	delayed mode switch	7	0.500	0.149	1.666	0.008	0.518	0.037	5.333	0.739
impedance token policy v4	friction slip shift	0	0.000	0.189	1.052	0.000	0.449	0.004	5.000	0.728
impedance token policy v4	friction slip shift	1	0.167	0.195	1.057	0.000	0.441	0.006	5.000	0.724
impedance token policy v4	friction slip shift	2	0.000	0.196	1.066	0.000	0.443	0.006	5.000	0.728
impedance token policy v4	friction slip shift	3	0.167	0.195	1.081	0.000	0.478	0.010	5.000	0.734
impedance token policy v4	friction slip shift	4	0.000	0.180	1.067	0.000	0.418	0.004	5.000	0.725
impedance token policy v4	friction slip shift	5	0.167	0.187	1.061	0.000	0.433	0.011	5.000	0.728

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
impedance token policy v4	friction slip shift	6	0.167	0.186	1.070	0.000	0.457	0.012	5.000	0.731
impedance token policy v4	friction slip shift	7	0.000	0.177	1.079	0.000	0.445	0.015	5.167	0.727
impedance token policy v4	nominal surface tracking	0	1.000	0.188	1.037	0.000	0.566	0.006	5.000	0.763
impedance token policy v4	nominal surface tracking	1	1.000	0.193	1.048	0.000	0.556	0.006	5.000	0.756
impedance token policy v4	nominal surface tracking	2	0.833	0.190	1.036	0.000	0.549	0.004	5.000	0.754
impedance token policy v4	nominal surface tracking	3	0.833	0.191	1.023	0.000	0.548	0.004	5.000	0.753
impedance token policy v4	nominal surface tracking	4	1.000	0.186	1.063	0.000	0.559	0.010	5.000	0.756
impedance token policy v4	nominal surface tracking	5	1.000	0.187	1.020	0.000	0.550	0.004	5.000	0.761
impedance token policy v4	nominal surface tracking	6	0.833	0.185	1.038	0.000	0.544	0.004	5.000	0.757
impedance token policy v4	nominal surface tracking	7	1.000	0.186	1.050	0.000	0.562	0.000	5.000	0.759
impedance token policy v4	sensor noise burst	0	0.167	0.165	1.208	0.000	0.522	0.048	5.000	0.740
impedance token policy v4	sensor noise burst	1	0.333	0.165	1.248	0.000	0.519	0.054	5.167	0.739
impedance token policy v4	sensor noise burst	2	0.333	0.164	1.229	0.000	0.485	0.059	5.333	0.740
impedance token policy v4	sensor noise burst	3	0.167	0.168	1.292	0.000	0.500	0.059	5.000	0.735
impedance token policy v4	sensor noise burst	4	0.333	0.171	1.273	0.000	0.500	0.044	5.000	0.736
impedance token policy v4	sensor noise burst	5	0.333	0.166	1.280	0.000	0.498	0.059	5.333	0.738
impedance token policy v4	sensor noise burst	6	0.167	0.167	1.269	0.000	0.513	0.027	5.000	0.737
impedance token policy v4	sensor noise burst	7	0.167	0.168	1.295	0.000	0.509	0.046	5.000	0.739
impedance token policy v4	stick slip cycle	0	0.000	0.164	1.089	0.000	0.444	0.023	5.167	0.729
impedance token policy v4	stick slip cycle	1	0.000	0.171	1.063	0.000	0.419	0.010	5.167	0.722
impedance token policy v4	stick slip cycle	2	0.167	0.163	1.071	0.000	0.400	0.011	5.167	0.726
impedance token policy v4	stick slip cycle	3	0.000	0.168	1.068	0.000	0.389	0.013	5.000	0.724
impedance token policy v4	stick slip cycle	4	0.167	0.160	1.055	0.000	0.461	0.013	5.167	0.731
impedance token policy v4	stick slip cycle	5	0.000	0.170	1.110	0.000	0.403	0.012	5.000	0.726
impedance token policy v4	stick slip cycle	6	0.000	0.163	1.107	0.000	0.387	0.015	5.000	0.722
impedance token policy v4	stick slip cycle	7	0.000	0.168	1.079	0.000	0.439	0.017	5.167	0.723
impedance token policy v4	stiffness shift	0	0.000	0.224	1.563	0.000	0.508	0.011	5.000	0.740
impedance token policy v4	stiffness shift	1	0.333	0.216	1.695	0.010	0.511	0.017	5.000	0.752
impedance token policy v4	stiffness shift	2	0.500	0.218	1.668	0.008	0.516	0.012	5.000	0.763
impedance token policy v4	stiffness shift	3	0.500	0.214	1.655	0.008	0.510	0.017	5.000	0.751
impedance token policy v4	stiffness shift	4	0.833	0.200	1.661	0.008	0.523	0.015	6.000	0.765
impedance token policy v4	stiffness shift	5	0.500	0.213	1.623	0.011	0.521	0.017	5.000	0.760
impedance token policy v4	stiffness shift	6	0.167	0.225	1.683	0.010	0.490	0.015	5.000	0.760
impedance token policy v4	stiffness shift	7	0.500	0.214	1.600	0.000	0.521	0.015	5.333	0.751
impedance token policy v4	surface discontinuity	0	0.000	0.150	1.109	0.000	0.447	0.017	5.000	0.725
impedance token policy v4	surface discontinuity	1	0.000	0.148	1.104	0.000	0.425	0.019	5.000	0.721
impedance token policy v4	surface discontinuity	2	0.000	0.152	1.087	0.000	0.424	0.013	5.000	0.725
impedance token policy v4	surface discontinuity	3	0.000	0.153	1.105	0.000	0.439	0.015	5.000	0.724
impedance token policy v4	surface discontinuity	4	0.000	0.149	1.081	0.000	0.433	0.017	5.167	0.722
impedance token policy v4	surface discontinuity	5	0.000	0.154	1.108	0.000	0.440	0.021	5.000	0.723
impedance token policy v4	surface discontinuity	6	0.000	0.148	1.078	0.000	0.426	0.012	5.000	0.726
impedance token policy v4	surface discontinuity	7	0.000	0.150	1.071	0.000	0.455	0.019	5.000	0.722
impedance token policy v4	target force jump	0	0.000	0.146	1.116	0.000	0.498	0.021	5.167	0.733

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
impedance token policy v4	target force jump	1	0.000	0.134	1.084	0.000	0.488	0.015	5.333	0.729
impedance token policy v4	target force jump	2	0.167	0.140	1.167	0.000	0.477	0.017	5.167	0.736
impedance token policy v4	target force jump	3	0.000	0.148	1.142	0.000	0.483	0.010	5.167	0.734
impedance token policy v4	target force jump	4	0.333	0.137	1.094	0.000	0.502	0.014	5.167	0.736
impedance token policy v4	target force jump	5	0.000	0.137	1.081	0.000	0.469	0.017	5.167	0.731
impedance token policy v4	target force jump	6	0.000	0.134	1.105	0.000	0.502	0.010	5.000	0.734
impedance token policy v4	target force jump	7	0.000	0.141	1.115	0.000	0.486	0.025	5.000	0.729
impedance token policy v5	actuator saturation	0	0.000	0.189	1.410	0.000	0.261	0.013	7.000	0.686
impedance token policy v5	actuator saturation	1	0.000	0.193	1.326	0.000	0.229	0.017	5.167	0.678
impedance token policy v5	actuator saturation	2	0.000	0.185	1.211	0.000	0.293	0.015	7.167	0.690
impedance token policy v5	actuator saturation	3	0.000	0.194	1.349	0.000	0.243	0.013	7.000	0.680
impedance token policy v5	actuator saturation	4	0.000	0.189	1.237	0.000	0.284	0.017	6.167	0.690
impedance token policy v5	actuator saturation	5	0.000	0.186	1.254	0.000	0.257	0.013	6.167	0.689
impedance token policy v5	actuator saturation	6	0.000	0.192	1.341	0.000	0.280	0.019	6.500	0.685
impedance token policy v5	actuator saturation	7	0.000	0.192	1.250	0.000	0.227	0.012	6.667	0.677
impedance token policy v5	combined extreme stress	0	0.000	0.172	1.392	0.000	0.171	0.079	6.000	0.663
impedance token policy v5	combined extreme stress	1	0.000	0.166	1.343	0.002	0.252	0.078	6.000	0.676
impedance token policy v5	combined extreme stress	2	0.000	0.162	1.297	0.000	0.233	0.076	6.167	0.676
impedance token policy v5	combined extreme stress	3	0.000	0.169	1.275	0.000	0.248	0.070	5.667	0.677
impedance token policy v5	combined extreme stress	4	0.000	0.163	1.270	0.000	0.256	0.066	5.833	0.680
impedance token policy v5	combined extreme stress	5	0.000	0.165	1.300	0.000	0.245	0.056	5.667	0.677
impedance token policy v5	combined extreme stress	6	0.000	0.162	1.320	0.000	0.239	0.064	7.333	0.674
impedance token policy v5	combined extreme stress	7	0.000	0.164	1.302	0.000	0.255	0.052	5.500	0.673
impedance token policy v5	combined stress	0	0.000	0.183	1.419	0.000	0.121	0.037	6.000	0.664
impedance token policy v5	combined stress	1	0.000	0.187	1.386	0.000	0.105	0.035	5.667	0.657
impedance token policy v5	combined stress	2	0.000	0.183	1.392	0.000	0.140	0.043	5.333	0.659
impedance token policy v5	combined stress	3	0.000	0.180	1.311	0.000	0.118	0.050	6.000	0.652
impedance token policy v5	combined stress	4	0.000	0.188	1.410	0.002	0.097	0.052	6.167	0.652
impedance token policy v5	combined stress	5	0.000	0.175	1.279	0.000	0.056	0.043	5.500	0.645
impedance token policy v5	combined stress	6	0.000	0.176	1.286	0.000	0.172	0.039	5.500	0.662
impedance token policy v5	combined stress	7	0.000	0.171	1.270	0.000	0.101	0.035	6.167	0.654
impedance token policy v5	contact transition	0	0.000	0.198	1.181	0.000	0.248	0.012	6.500	0.678
impedance token policy v5	contact transition	1	0.000	0.193	1.164	0.000	0.293	0.012	5.500	0.679
impedance token policy v5	contact transition	2	0.000	0.198	1.158	0.000	0.256	0.013	6.000	0.680
impedance token policy v5	contact transition	3	0.000	0.194	1.117	0.000	0.253	0.010	5.667	0.673
impedance token policy v5	contact transition	4	0.000	0.205	1.172	0.000	0.227	0.012	8.333	0.670
impedance token policy v5	contact transition	5	0.000	0.199	1.199	0.000	0.287	0.015	5.667	0.680
impedance token policy v5	contact transition	6	0.000	0.194	1.107	0.000	0.293	0.010	5.000	0.680
impedance token policy v5	contact transition	7	0.000	0.197	1.217	0.000	0.220	0.008	7.500	0.668
impedance token policy v5	delayed mode switch	0	0.000	0.195	1.708	0.012	0.278	0.021	5.500	0.685
impedance token policy v5	delayed mode switch	1	0.000	0.192	1.637	0.010	0.230	0.015	5.833	0.683
impedance token policy v5	delayed mode switch	2	0.000	0.192	1.620	0.008	0.277	0.017	6.667	0.682
impedance token policy v5	delayed mode switch	3	0.000	0.184	1.564	0.008	0.271	0.014	6.500	0.688

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
impedance token policy v5	delayed mode switch	4	0.000	0.187	1.759	0.015	0.258	0.017	5.500	0.684
impedance token policy v5	delayed mode switch	5	0.000	0.180	1.454	0.002	0.273	0.017	7.667	0.682
impedance token policy v5	delayed mode switch	6	0.000	0.193	1.585	0.002	0.271	0.017	8.500	0.686
impedance token policy v5	delayed mode switch	7	0.000	0.177	1.645	0.012	0.290	0.014	5.167	0.686
impedance token policy v5	friction slip shift	0	0.000	0.241	0.980	0.000	0.264	0.000	6.333	0.675
impedance token policy v5	friction slip shift	1	0.000	0.245	0.978	0.000	0.290	0.000	5.167	0.680
impedance token policy v5	friction slip shift	2	0.000	0.247	0.987	0.000	0.310	0.004	5.000	0.689
impedance token policy v5	friction slip shift	3	0.000	0.245	0.989	0.000	0.346	0.004	5.333	0.687
impedance token policy v5	friction slip shift	4	0.000	0.237	0.958	0.000	0.289	0.000	5.333	0.670
impedance token policy v5	friction slip shift	5	0.000	0.242	0.983	0.000	0.308	0.000	5.000	0.681
impedance token policy v5	friction slip shift	6	0.000	0.246	0.937	0.000	0.296	0.000	5.000	0.680
impedance token policy v5	friction slip shift	7	0.000	0.234	0.986	0.000	0.290	0.002	5.500	0.672
impedance token policy v5	nominal surface tracking	0	0.000	0.251	0.961	0.000	0.351	0.004	5.333	0.703
impedance token policy v5	nominal surface tracking	1	0.000	0.252	0.942	0.000	0.398	0.000	5.000	0.704
impedance token policy v5	nominal surface tracking	2	0.000	0.244	0.947	0.000	0.388	0.002	5.000	0.705
impedance token policy v5	nominal surface tracking	3	0.000	0.247	0.924	0.000	0.398	0.000	5.167	0.709
impedance token policy v5	nominal surface tracking	4	0.000	0.245	0.938	0.000	0.372	0.000	5.167	0.699
impedance token policy v5	nominal surface tracking	5	0.000	0.250	0.951	0.000	0.382	0.000	5.333	0.706
impedance token policy v5	nominal surface tracking	6	0.000	0.250	0.923	0.000	0.362	0.000	6.000	0.705
impedance token policy v5	nominal surface tracking	7	0.000	0.247	0.921	0.000	0.374	0.000	6.167	0.705
impedance token policy v5	sensor noise burst	0	0.000	0.205	1.141	0.000	0.308	0.029	6.333	0.692
impedance token policy v5	sensor noise burst	1	0.000	0.202	1.164	0.000	0.294	0.046	8.167	0.689
impedance token policy v5	sensor noise burst	2	0.000	0.202	1.181	0.000	0.297	0.038	5.667	0.691
impedance token policy v5	sensor noise burst	3	0.000	0.200	1.273	0.000	0.328	0.037	7.167	0.698
impedance token policy v5	sensor noise burst	4	0.000	0.208	1.242	0.000	0.272	0.040	5.500	0.684
impedance token policy v5	sensor noise burst	5	0.000	0.201	1.187	0.000	0.316	0.048	8.000	0.696
impedance token policy v5	sensor noise burst	6	0.000	0.199	1.150	0.000	0.335	0.033	5.000	0.698
impedance token policy v5	sensor noise burst	7	0.000	0.200	1.172	0.000	0.293	0.034	6.333	0.689
impedance token policy v5	stick slip cycle	0	0.000	0.218	0.958	0.000	0.244	0.000	7.167	0.667
impedance token policy v5	stick slip cycle	1	0.000	0.216	0.994	0.000	0.274	0.000	5.167	0.675
impedance token policy v5	stick slip cycle	2	0.000	0.216	0.979	0.000	0.180	0.002	8.500	0.668
impedance token policy v5	stick slip cycle	3	0.000	0.222	0.950	0.000	0.179	0.000	7.833	0.665
impedance token policy v5	stick slip cycle	4	0.000	0.217	0.958	0.000	0.291	0.000	6.000	0.671
impedance token policy v5	stick slip cycle	5	0.000	0.218	1.038	0.000	0.243	0.006	5.833	0.670
impedance token policy v5	stick slip cycle	6	0.000	0.213	1.000	0.000	0.198	0.004	5.667	0.661
impedance token policy v5	stick slip cycle	7	0.000	0.219	0.981	0.000	0.192	0.000	5.833	0.666
impedance token policy v5	stiffness shift	0	0.000	0.249	1.510	0.000	0.343	0.011	5.000	0.710
impedance token policy v5	stiffness shift	1	0.000	0.249	1.612	0.006	0.282	0.015	5.167	0.700
impedance token policy v5	stiffness shift	2	0.000	0.257	1.578	0.002	0.347	0.012	5.167	0.710
impedance token policy v5	stiffness shift	3	0.000	0.243	1.556	0.000	0.351	0.013	5.000	0.708
impedance token policy v5	stiffness shift	4	0.000	0.247	1.574	0.002	0.350	0.012	6.000	0.702
impedance token policy v5	stiffness shift	5	0.000	0.248	1.537	0.004	0.265	0.017	5.000	0.696
impedance token policy v5	stiffness shift	6	0.000	0.257	1.586	0.004	0.328	0.015	5.000	0.710

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
impedance token policy v5	stiffness shift	7	0.000	0.248	1.523	0.000	0.341	0.013	5.333	0.704
impedance token policy v5	surface discontinuity	0	0.000	0.193	1.014	0.000	0.196	0.002	6.667	0.673
impedance token policy v5	surface discontinuity	1	0.000	0.184	1.031	0.000	0.257	0.004	6.667	0.679
impedance token policy v5	surface discontinuity	2	0.000	0.191	0.991	0.000	0.274	0.000	5.500	0.684
impedance token policy v5	surface discontinuity	3	0.000	0.189	1.016	0.000	0.221	0.000	6.500	0.675
impedance token policy v5	surface discontinuity	4	0.000	0.196	0.987	0.000	0.263	0.000	6.667	0.677
impedance token policy v5	surface discontinuity	5	0.000	0.186	1.035	0.000	0.236	0.006	6.333	0.681
impedance token policy v5	surface discontinuity	6	0.000	0.188	1.016	0.000	0.270	0.004	5.500	0.679
impedance token policy v5	surface discontinuity	7	0.000	0.189	1.020	0.000	0.246	0.002	7.333	0.676
impedance token policy v5	target force jump	0	0.000	0.186	1.034	0.000	0.319	0.002	8.500	0.682
impedance token policy v5	target force jump	1	0.000	0.175	0.995	0.000	0.310	0.004	9.000	0.675
impedance token policy v5	target force jump	2	0.000	0.179	1.072	0.000	0.305	0.008	7.333	0.681
impedance token policy v5	target force jump	3	0.000	0.189	1.022	0.000	0.320	0.002	8.667	0.678
impedance token policy v5	target force jump	4	0.000	0.183	1.025	0.000	0.327	0.002	10.667	0.681
impedance token policy v5	target force jump	5	0.000	0.179	1.034	0.000	0.306	0.004	8.667	0.673
impedance token policy v5	target force jump	6	0.000	0.176	1.021	0.000	0.349	0.004	8.833	0.686
impedance token policy v5	target force jump	7	0.000	0.182	1.057	0.000	0.313	0.004	9.000	0.673
learned gain regressor	actuator saturation	0	0.167	0.171	1.707	0.015	0.615	0.019	5.000	0.831
learned gain regressor	actuator saturation	1	0.000	0.167	1.596	0.008	0.621	0.033	5.000	0.829
learned gain regressor	actuator saturation	2	0.000	0.167	1.468	0.000	0.618	0.027	5.833	0.832
learned gain regressor	actuator saturation	3	0.000	0.170	1.662	0.008	0.615	0.029	5.000	0.829
learned gain regressor	actuator saturation	4	0.333	0.170	1.552	0.002	0.611	0.023	6.667	0.832
learned gain regressor	actuator saturation	5	0.500	0.162	1.558	0.010	0.608	0.023	5.000	0.833
learned gain regressor	actuator saturation	6	0.167	0.183	1.594	0.002	0.619	0.019	5.000	0.828
learned gain regressor	actuator saturation	7	0.333	0.163	1.564	0.008	0.609	0.031	5.000	0.831
learned gain regressor	combined extreme stress	0	1.000	0.159	1.616	0.002	0.561	0.079	5.000	0.796
learned gain regressor	combined extreme stress	1	0.833	0.165	1.633	0.008	0.570	0.106	5.500	0.803
learned gain regressor	combined extreme stress	2	0.500	0.160	1.680	0.004	0.585	0.093	5.000	0.803
learned gain regressor	combined extreme stress	3	0.833	0.165	1.434	0.006	0.574	0.110	5.500	0.806
learned gain regressor	combined extreme stress	4	0.667	0.163	1.501	0.000	0.574	0.106	6.833	0.806
learned gain regressor	combined extreme stress	5	0.833	0.172	1.697	0.012	0.566	0.097	7.333	0.806
learned gain regressor	combined extreme stress	6	0.833	0.169	1.637	0.008	0.582	0.077	5.000	0.801
learned gain regressor	combined extreme stress	7	1.000	0.169	1.556	0.006	0.574	0.113	6.167	0.801
learned gain regressor	combined stress	0	0.667	0.193	1.865	0.042	0.575	0.040	5.167	0.796
learned gain regressor	combined stress	1	0.667	0.186	1.743	0.027	0.576	0.062	5.167	0.792
learned gain regressor	combined stress	2	0.833	0.178	1.635	0.013	0.581	0.054	5.000	0.796
learned gain regressor	combined stress	3	0.833	0.187	1.693	0.008	0.578	0.054	5.333	0.794
learned gain regressor	combined stress	4	0.833	0.184	1.787	0.029	0.568	0.054	5.000	0.791
learned gain regressor	combined stress	5	1.000	0.169	1.593	0.004	0.575	0.048	5.167	0.794
learned gain regressor	combined stress	6	1.000	0.174	1.675	0.012	0.579	0.071	5.167	0.798
learned gain regressor	combined stress	7	0.833	0.177	1.742	0.017	0.568	0.058	6.667	0.787
learned gain regressor	contact transition	0	0.500	0.158	1.435	0.000	0.601	0.019	5.000	0.824
learned gain regressor	contact transition	1	0.333	0.149	1.435	0.000	0.615	0.019	5.000	0.823

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
learned gain regressor	contact transition	2	0.167	0.161	1.438	0.000	0.610	0.021	5.000	0.824
learned gain regressor	contact transition	3	0.500	0.148	1.392	0.000	0.607	0.019	5.000	0.823
learned gain regressor	contact transition	4	0.333	0.167	1.418	0.000	0.602	0.011	5.000	0.821
learned gain regressor	contact transition	5	0.500	0.152	1.454	0.000	0.608	0.029	5.000	0.824
learned gain regressor	contact transition	6	0.167	0.158	1.370	0.000	0.610	0.027	6.167	0.824
learned gain regressor	contact transition	7	0.500	0.159	1.408	0.000	0.603	0.023	5.000	0.816
learned gain regressor	delayed mode switch	0	0.167	0.170	2.105	0.050	0.594	0.033	5.000	0.827
learned gain regressor	delayed mode switch	1	0.333	0.192	2.005	0.050	0.593	0.038	9.000	0.825
learned gain regressor	delayed mode switch	2	0.333	0.171	2.033	0.048	0.607	0.027	5.000	0.826
learned gain regressor	delayed mode switch	3	0.333	0.190	1.971	0.031	0.591	0.017	11.500	0.828
learned gain regressor	delayed mode switch	4	0.167	0.172	2.206	0.044	0.590	0.025	7.167	0.829
learned gain regressor	delayed mode switch	5	0.333	0.161	1.905	0.038	0.585	0.025	7.333	0.822
learned gain regressor	delayed mode switch	6	0.333	0.188	2.053	0.048	0.600	0.027	7.333	0.829
learned gain regressor	delayed mode switch	7	0.333	0.176	2.025	0.041	0.595	0.027	7.500	0.828
learned gain regressor	friction slip shift	0	0.500	0.201	1.207	0.000	0.589	0.010	5.000	0.821
learned gain regressor	friction slip shift	1	0.333	0.202	1.209	0.000	0.602	0.004	5.000	0.816
learned gain regressor	friction slip shift	2	0.333	0.223	1.225	0.000	0.598	0.008	5.500	0.824
learned gain regressor	friction slip shift	3	0.333	0.196	1.220	0.000	0.600	0.006	5.000	0.826
learned gain regressor	friction slip shift	4	0.667	0.184	1.179	0.000	0.600	0.006	5.000	0.814
learned gain regressor	friction slip shift	5	0.667	0.203	1.202	0.000	0.597	0.006	5.167	0.818
learned gain regressor	friction slip shift	6	0.500	0.179	1.162	0.000	0.603	0.015	5.000	0.822
learned gain regressor	friction slip shift	7	0.833	0.173	1.158	0.000	0.590	0.002	5.000	0.817
learned gain regressor	nominal surface tracking	0	0.000	0.189	1.147	0.000	0.628	0.002	5.000	0.852
learned gain regressor	nominal surface tracking	1	0.000	0.201	1.139	0.000	0.637	0.006	5.000	0.852
learned gain regressor	nominal surface tracking	2	0.000	0.195	1.158	0.000	0.637	0.004	5.000	0.851
learned gain regressor	nominal surface tracking	3	0.000	0.201	1.182	0.000	0.631	0.002	5.000	0.852
learned gain regressor	nominal surface tracking	4	0.000	0.190	1.163	0.000	0.631	0.004	5.000	0.851
learned gain regressor	nominal surface tracking	5	0.000	0.184	1.154	0.000	0.632	0.006	5.000	0.851
learned gain regressor	nominal surface tracking	6	0.000	0.176	1.152	0.000	0.638	0.004	5.000	0.850
learned gain regressor	nominal surface tracking	7	0.000	0.176	1.161	0.000	0.626	0.004	5.000	0.853
learned gain regressor	sensor noise burst	0	0.167	0.162	1.401	0.000	0.616	0.053	5.000	0.843
learned gain regressor	sensor noise burst	1	0.000	0.174	1.409	0.000	0.621	0.050	5.000	0.838
learned gain regressor	sensor noise burst	2	0.000	0.184	1.452	0.000	0.620	0.057	5.000	0.838
learned gain regressor	sensor noise burst	3	0.000	0.176	1.440	0.000	0.625	0.065	5.000	0.841
learned gain regressor	sensor noise burst	4	0.167	0.182	1.501	0.004	0.620	0.050	5.000	0.838
learned gain regressor	sensor noise burst	5	0.000	0.182	1.453	0.000	0.620	0.059	5.000	0.841
learned gain regressor	sensor noise burst	6	0.000	0.192	1.410	0.000	0.623	0.069	5.333	0.840
learned gain regressor	sensor noise burst	7	0.000	0.174	1.499	0.000	0.619	0.063	5.000	0.840
learned gain regressor	stick slip cycle	0	0.833	0.146	1.179	0.000	0.588	0.021	5.000	0.816
learned gain regressor	stick slip cycle	1	0.667	0.161	1.215	0.000	0.598	0.033	5.000	0.815
learned gain regressor	stick slip cycle	2	0.833	0.171	1.216	0.000	0.592	0.015	5.000	0.809
learned gain regressor	stick slip cycle	3	0.833	0.165	1.198	0.000	0.589	0.008	5.000	0.808
learned gain regressor	stick slip cycle	4	0.833	0.148	1.173	0.000	0.589	0.025	5.000	0.821

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
learned gain regressor	stick slip cycle	5	0.833	0.164	1.229	0.000	0.589	0.019	5.000	0.813
learned gain regressor	stick slip cycle	6	0.833	0.163	1.226	0.000	0.591	0.023	5.000	0.809
learned gain regressor	stick slip cycle	7	1.000	0.159	1.208	0.000	0.585	0.023	5.000	0.814
learned gain regressor	stiffness shift	0	0.000	0.252	1.866	0.027	0.632	0.002	5.500	0.849
learned gain regressor	stiffness shift	1	0.000	0.253	1.944	0.042	0.635	0.010	5.000	0.846
learned gain regressor	stiffness shift	2	0.000	0.258	1.915	0.034	0.636	0.006	5.667	0.846
learned gain regressor	stiffness shift	3	0.000	0.251	1.924	0.044	0.628	0.006	6.000	0.845
learned gain regressor	stiffness shift	4	0.000	0.232	1.912	0.042	0.638	0.012	6.667	0.847
learned gain regressor	stiffness shift	5	0.000	0.262	1.915	0.036	0.627	0.004	5.000	0.846
learned gain regressor	stiffness shift	6	0.000	0.265	1.932	0.034	0.638	0.008	5.000	0.846
learned gain regressor	stiffness shift	7	0.000	0.239	1.871	0.034	0.630	0.010	6.833	0.847
learned gain regressor	surface discontinuity	0	0.833	0.144	1.295	0.000	0.594	0.029	5.167	0.821
learned gain regressor	surface discontinuity	1	0.667	0.141	1.302	0.000	0.599	0.031	5.833	0.818
learned gain regressor	surface discontinuity	2	0.333	0.135	1.294	0.000	0.605	0.027	5.000	0.822
learned gain regressor	surface discontinuity	3	0.667	0.144	1.318	0.000	0.604	0.031	5.000	0.821
learned gain regressor	surface discontinuity	4	0.500	0.138	1.207	0.000	0.602	0.029	5.000	0.820
learned gain regressor	surface discontinuity	5	0.333	0.144	1.360	0.000	0.600	0.023	5.000	0.822
learned gain regressor	surface discontinuity	6	0.333	0.144	1.335	0.000	0.610	0.027	5.500	0.823
learned gain regressor	surface discontinuity	7	0.833	0.139	1.222	0.000	0.599	0.039	5.000	0.818
learned gain regressor	target force jump	0	0.167	0.131	1.254	0.000	0.616	0.016	5.000	0.832
learned gain regressor	target force jump	1	0.167	0.120	1.190	0.000	0.622	0.023	5.000	0.828
learned gain regressor	target force jump	2	0.000	0.131	1.272	0.000	0.624	0.027	5.000	0.831
learned gain regressor	target force jump	3	0.167	0.135	1.209	0.000	0.622	0.021	5.000	0.830
learned gain regressor	target force jump	4	0.167	0.127	1.215	0.000	0.620	0.021	5.000	0.834
learned gain regressor	target force jump	5	0.000	0.122	1.255	0.000	0.625	0.016	5.000	0.829
learned gain regressor	target force jump	6	0.000	0.122	1.234	0.000	0.624	0.029	5.000	0.832
learned gain regressor	target force jump	7	0.000	0.129	1.264	0.000	0.621	0.025	5.000	0.829
oracle impedance	actuator saturation	0	0.500	0.103	1.699	0.009	0.603	0.030	5.000	0.831
oracle impedance	actuator saturation	1	0.500	0.103	1.578	0.004	0.603	0.023	5.000	0.828
oracle impedance	actuator saturation	2	0.667	0.098	1.472	0.002	0.596	0.036	5.000	0.833
oracle impedance	actuator saturation	3	0.667	0.105	1.699	0.006	0.600	0.028	5.000	0.828
oracle impedance	actuator saturation	4	0.500	0.101	1.534	0.002	0.599	0.021	5.333	0.831
oracle impedance	actuator saturation	5	0.500	0.099	1.532	0.004	0.601	0.041	5.000	0.831
oracle impedance	actuator saturation	6	0.833	0.102	1.677	0.011	0.592	0.028	5.000	0.827
oracle impedance	actuator saturation	7	0.167	0.103	1.537	0.004	0.612	0.036	5.000	0.829
oracle impedance	combined extreme stress	0	1.000	0.106	1.528	0.004	0.450	0.160	5.000	0.776
oracle impedance	combined extreme stress	1	1.000	0.103	1.530	0.004	0.457	0.143	5.000	0.785
oracle impedance	combined extreme stress	2	1.000	0.100	1.427	0.000	0.516	0.164	5.000	0.786
oracle impedance	combined extreme stress	3	0.833	0.109	1.524	0.002	0.498	0.126	5.000	0.789
oracle impedance	combined extreme stress	4	1.000	0.099	1.422	0.000	0.502	0.123	5.167	0.788
oracle impedance	combined extreme stress	5	0.667	0.106	1.453	0.000	0.489	0.179	5.000	0.789
oracle impedance	combined extreme stress	6	1.000	0.103	1.634	0.006	0.513	0.132	5.000	0.784
oracle impedance	combined extreme stress	7	1.000	0.100	1.467	0.000	0.478	0.139	5.000	0.778

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
oracle impedance	combined stress	0	0.667	0.114	1.845	0.015	0.546	0.088	5.000	0.778
oracle impedance	combined stress	1	0.833	0.112	1.759	0.013	0.502	0.081	5.000	0.776
oracle impedance	combined stress	2	0.833	0.110	1.726	0.011	0.518	0.102	5.000	0.779
oracle impedance	combined stress	3	1.000	0.110	1.709	0.008	0.555	0.096	5.167	0.776
oracle impedance	combined stress	4	0.833	0.109	1.695	0.011	0.477	0.077	5.000	0.773
oracle impedance	combined stress	5	1.000	0.103	1.498	0.000	0.521	0.089	5.000	0.775
oracle impedance	combined stress	6	1.000	0.104	1.540	0.002	0.516	0.093	5.000	0.783
oracle impedance	combined stress	7	0.833	0.098	1.554	0.006	0.485	0.085	5.000	0.771
oracle impedance	contact transition	0	0.333	0.108	1.437	0.000	0.597	0.030	5.000	0.819
oracle impedance	contact transition	1	0.333	0.100	1.453	0.002	0.595	0.017	5.000	0.819
oracle impedance	contact transition	2	0.500	0.104	1.437	0.000	0.588	0.040	5.000	0.821
oracle impedance	contact transition	3	0.500	0.103	1.406	0.000	0.595	0.030	5.000	0.819
oracle impedance	contact transition	4	0.500	0.103	1.408	0.000	0.592	0.021	5.000	0.818
oracle impedance	contact transition	5	0.667	0.106	1.440	0.000	0.595	0.036	5.000	0.820
oracle impedance	contact transition	6	0.667	0.101	1.351	0.000	0.591	0.032	5.000	0.821
oracle impedance	contact transition	7	0.333	0.105	1.473	0.000	0.603	0.026	5.000	0.810
oracle impedance	delayed mode switch	0	0.500	0.115	2.125	0.030	0.573	0.042	5.000	0.819
oracle impedance	delayed mode switch	1	0.500	0.115	1.997	0.021	0.560	0.049	5.833	0.819
oracle impedance	delayed mode switch	2	0.667	0.114	1.995	0.028	0.569	0.034	5.000	0.819
oracle impedance	delayed mode switch	3	0.667	0.107	1.911	0.015	0.561	0.028	5.833	0.824
oracle impedance	delayed mode switch	4	0.333	0.105	2.160	0.024	0.553	0.028	5.333	0.823
oracle impedance	delayed mode switch	5	0.667	0.105	1.798	0.021	0.554	0.064	6.167	0.813
oracle impedance	delayed mode switch	6	0.667	0.112	2.004	0.025	0.560	0.036	5.833	0.824
oracle impedance	delayed mode switch	7	0.500	0.101	1.988	0.017	0.575	0.028	5.000	0.819
oracle impedance	friction slip shift	0	1.000	0.119	1.168	0.000	0.588	0.017	5.000	0.810
oracle impedance	friction slip shift	1	0.500	0.122	1.201	0.000	0.591	0.007	5.000	0.809
oracle impedance	friction slip shift	2	1.000	0.123	1.204	0.000	0.576	0.019	5.000	0.814
oracle impedance	friction slip shift	3	0.667	0.123	1.194	0.000	0.588	0.015	5.000	0.817
oracle impedance	friction slip shift	4	0.500	0.111	1.149	0.000	0.594	0.009	5.000	0.804
oracle impedance	friction slip shift	5	0.667	0.118	1.162	0.000	0.589	0.015	5.000	0.808
oracle impedance	friction slip shift	6	1.000	0.114	1.128	0.000	0.585	0.011	5.000	0.812
oracle impedance	friction slip shift	7	0.333	0.113	1.125	0.000	0.598	0.015	5.000	0.805
oracle impedance	nominal surface tracking	0	0.000	0.109	1.105	0.000	0.630	0.006	5.000	0.855
oracle impedance	nominal surface tracking	1	0.167	0.114	1.127	0.000	0.619	0.006	5.000	0.856
oracle impedance	nominal surface tracking	2	0.167	0.107	1.122	0.000	0.612	0.000	5.000	0.855
oracle impedance	nominal surface tracking	3	0.000	0.110	1.146	0.000	0.620	0.002	5.000	0.856
oracle impedance	nominal surface tracking	4	0.000	0.104	1.128	0.000	0.621	0.002	5.000	0.854
oracle impedance	nominal surface tracking	5	0.167	0.113	1.121	0.000	0.625	0.004	5.000	0.854
oracle impedance	nominal surface tracking	6	0.000	0.110	1.102	0.000	0.619	0.000	5.000	0.854
oracle impedance	nominal surface tracking	7	0.000	0.106	1.116	0.000	0.633	0.000	5.000	0.855
oracle impedance	sensor noise burst	0	0.000	0.104	1.330	0.000	0.619	0.060	5.000	0.844
oracle impedance	sensor noise burst	1	0.667	0.104	1.411	0.000	0.603	0.064	5.000	0.839
oracle impedance	sensor noise burst	2	0.500	0.107	1.351	0.000	0.602	0.079	5.000	0.839
oracle impedance	sensor noise burst	3	0.333	0.107	1.454	0.000	0.609	0.070	5.000	0.842
oracle impedance	sensor noise burst	4	0.333	0.105	1.394	0.004	0.608	0.049	5.000	0.839
oracle impedance	sensor noise burst	5	0.500	0.105	1.420	0.000	0.605	0.068	5.000	0.843
oracle impedance	sensor noise burst	6	0.500	0.106	1.387	0.000	0.601	0.062	5.000	0.842
oracle impedance	sensor noise burst	7	0.000	0.105	1.481	0.002	0.615	0.064	5.000	0.841
oracle impedance	stick slip cycle	0	0.500	0.101	1.135	0.000	0.597	0.030	5.000	0.807
oracle impedance	stick slip cycle	1	0.667	0.105	1.224	0.000	0.579	0.025	5.000	0.806
oracle impedance	stick slip cycle	2	1.000	0.102	1.190	0.000	0.572	0.028	5.000	0.800
oracle impedance	stick slip cycle	3	0.833	0.104	1.165	0.000	0.581	0.023	5.000	0.797
oracle impedance	stick slip cycle	4	0.833	0.101	1.128	0.000	0.589	0.017	5.000	0.810
oracle impedance	stick slip cycle	5	0.667	0.107	1.217	0.000	0.588	0.032	5.000	0.802
oracle impedance	stick slip cycle	6	1.000	0.102	1.223	0.000	0.573	0.025	5.000	0.798
oracle impedance	stick slip cycle	7	1.000	0.100	1.203	0.000	0.583	0.023	5.000	0.802
oracle impedance	stiffness shift	0	0.000	0.149	1.840	0.019	0.620	0.011	5.000	0.852
oracle impedance	stiffness shift	1	0.167	0.148	2.012	0.023	0.610	0.011	5.000	0.850
oracle impedance	stiffness shift	2	0.167	0.151	1.969	0.021	0.607	0.011	5.000	0.850
oracle impedance	stiffness shift	3	0.167	0.148	1.947	0.022	0.612	0.009	5.000	0.849
oracle impedance	stiffness shift	4	0.000	0.139	1.960	0.024	0.618	0.013	5.667	0.850
oracle impedance	stiffness shift	5	0.167	0.155	1.914	0.019	0.614	0.011	5.000	0.849
oracle impedance	stiffness shift	6	0.167	0.155	1.945	0.021	0.607	0.011	5.000	0.852
oracle impedance	stiffness shift	7	0.000	0.142	1.880	0.022	0.620	0.009	5.333	0.850
oracle impedance	surface discontinuity	0	0.833	0.089	1.157	0.000	0.583	0.034	5.000	0.813
oracle impedance	surface discontinuity	1	0.833	0.086	1.149	0.000	0.578	0.040	5.000	0.813
oracle impedance	surface discontinuity	2	1.000	0.090	1.137	0.000	0.577	0.043	5.000	0.817
oracle impedance	surface discontinuity	3	0.833	0.088	1.161	0.000	0.590	0.028	5.000	0.816

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
oracle impedance	surface discontinuity	4	0.667	0.089	1.137	0.000	0.589	0.027	5.000	0.814
oracle impedance	surface discontinuity	5	0.833	0.092	1.206	0.000	0.589	0.043	5.000	0.817
oracle impedance	surface discontinuity	6	1.000	0.085	1.164	0.000	0.586	0.030	5.000	0.822
oracle impedance	surface discontinuity	7	0.833	0.086	1.137	0.000	0.583	0.030	5.000	0.810
oracle impedance	target force jump	0	0.000	0.089	1.261	0.000	0.613	0.019	5.000	0.830
oracle impedance	target force jump	1	0.333	0.086	1.200	0.000	0.605	0.025	5.000	0.827
oracle impedance	target force jump	2	0.500	0.090	1.343	0.000	0.602	0.023	5.000	0.832
oracle impedance	target force jump	3	0.333	0.091	1.262	0.000	0.608	0.027	5.000	0.829
oracle impedance	target force jump	4	0.333	0.086	1.240	0.000	0.615	0.015	5.000	0.834
oracle impedance	target force jump	5	0.333	0.087	1.243	0.000	0.607	0.042	5.000	0.829
oracle impedance	target force jump	6	0.667	0.084	1.243	0.000	0.599	0.017	5.000	0.833
oracle impedance	target force jump	7	0.000	0.088	1.254	0.000	0.617	0.019	5.000	0.827
random forest gain regressor	actuator saturation	0	1.000	0.172	1.637	0.006	0.560	0.035	5.000	0.795
random forest gain regressor	actuator saturation	1	1.000	0.168	1.543	0.002	0.553	0.029	5.000	0.789
random forest gain regressor	actuator saturation	2	0.833	0.165	1.412	0.000	0.578	0.029	5.333	0.797
random forest gain regressor	actuator saturation	3	1.000	0.175	1.577	0.000	0.571	0.031	5.000	0.794
random forest gain regressor	actuator saturation	4	0.833	0.170	1.457	0.000	0.576	0.037	6.500	0.795
random forest gain regressor	actuator saturation	5	0.833	0.164	1.513	0.002	0.573	0.023	5.000	0.795
random forest gain regressor	actuator saturation	6	0.667	0.174	1.543	0.002	0.583	0.021	5.000	0.796
random forest gain regressor	actuator saturation	7	1.000	0.167	1.510	0.002	0.564	0.027	5.000	0.793
random forest gain regressor	combined extreme stress	0	1.000	0.173	1.734	0.018	0.519	0.137	5.000	0.758
random forest gain regressor	combined extreme stress	1	0.833	0.179	1.715	0.010	0.517	0.100	9.667	0.760
random forest gain regressor	combined extreme stress	2	1.000	0.171	1.629	0.004	0.542	0.102	6.167	0.766
random forest gain regressor	combined extreme stress	3	0.833	0.181	1.549	0.004	0.526	0.111	7.167	0.765
random forest gain regressor	combined extreme stress	4	1.000	0.170	1.571	0.004	0.533	0.103	7.500	0.766
random forest gain regressor	combined extreme stress	5	1.000	0.184	1.719	0.012	0.524	0.114	9.000	0.765
random forest gain regressor	combined extreme stress	6	0.833	0.167	1.788	0.018	0.544	0.088	5.000	0.765
random forest gain regressor	combined extreme stress	7	1.000	0.176	1.643	0.004	0.531	0.081	6.333	0.761
random forest gain regressor	combined stress	0	0.833	0.193	1.857	0.033	0.533	0.074	5.500	0.763
random forest gain regressor	combined stress	1	0.833	0.185	1.752	0.025	0.535	0.084	5.167	0.760
random forest gain regressor	combined stress	2	1.000	0.181	1.731	0.020	0.544	0.067	5.000	0.762
random forest gain regressor	combined stress	3	1.000	0.177	1.664	0.008	0.550	0.068	5.000	0.762
random forest gain regressor	combined stress	4	0.833	0.185	1.756	0.020	0.533	0.059	5.333	0.757
random forest gain regressor	combined stress	5	1.000	0.170	1.601	0.008	0.552	0.068	6.333	0.762
random forest gain regressor	combined stress	6	1.000	0.180	1.609	0.012	0.540	0.074	5.667	0.763
random forest gain regressor	combined stress	7	1.000	0.182	1.623	0.004	0.531	0.047	8.333	0.756
random forest gain regressor	contact transition	0	1.000	0.171	1.420	0.000	0.563	0.023	5.000	0.793
random forest gain regressor	contact transition	1	0.833	0.160	1.410	0.000	0.570	0.023	5.000	0.793
random forest gain regressor	contact transition	2	1.000	0.165	1.410	0.000	0.572	0.018	5.000	0.793
random forest gain regressor	contact transition	3	1.000	0.155	1.343	0.000	0.565	0.029	5.000	0.790
random forest gain regressor	contact transition	4	0.833	0.171	1.332	0.000	0.576	0.014	5.000	0.791
random forest gain regressor	contact transition	5	0.833	0.165	1.370	0.000	0.569	0.029	5.000	0.793
random forest gain regressor	contact transition	6	0.833	0.161	1.329	0.000	0.574	0.027	5.833	0.792
random forest gain regressor	contact transition	7	0.833	0.169	1.375	0.000	0.566	0.031	5.000	0.789
random forest gain regressor	delayed mode switch	0	0.500	0.184	2.069	0.049	0.540	0.033	5.000	0.789
random forest gain regressor	delayed mode switch	1	0.500	0.199	1.903	0.041	0.548	0.027	9.500	0.790
random forest gain regressor	delayed mode switch	2	0.667	0.187	1.963	0.041	0.566	0.020	5.333	0.790

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
random forest gain regressor	delayed mode switch	3	0.667	0.205	1.863	0.019	0.555	0.010	13.000	0.792
random forest gain regressor	delayed mode switch	4	0.333	0.198	2.102	0.039	0.562	0.021	9.000	0.790
random forest gain regressor	delayed mode switch	5	0.833	0.175	1.772	0.029	0.546	0.024	7.833	0.786
random forest gain regressor	delayed mode switch	6	0.833	0.203	1.940	0.041	0.566	0.022	8.333	0.792
random forest gain regressor	delayed mode switch	7	0.500	0.187	2.007	0.037	0.559	0.022	8.333	0.790
random forest gain regressor	friction slip shift	0	0.833	0.188	1.139	0.000	0.552	0.019	5.000	0.791
random forest gain regressor	friction slip shift	1	1.000	0.193	1.163	0.000	0.552	0.006	5.000	0.790
random forest gain regressor	friction slip shift	2	1.000	0.199	1.183	0.000	0.564	0.006	5.000	0.792
random forest gain regressor	friction slip shift	3	0.833	0.194	1.128	0.000	0.560	0.012	5.000	0.790
random forest gain regressor	friction slip shift	4	1.000	0.177	1.120	0.000	0.565	0.006	5.000	0.787
random forest gain regressor	friction slip shift	5	1.000	0.188	1.137	0.000	0.556	0.010	5.000	0.789
random forest gain regressor	friction slip shift	6	1.000	0.172	1.076	0.000	0.572	0.012	5.000	0.791
random forest gain regressor	friction slip shift	7	0.833	0.168	1.103	0.000	0.559	0.014	5.000	0.787
random forest gain regressor	nominal surface tracking	0	1.000	0.177	1.069	0.000	0.568	0.006	5.000	0.809
random forest gain regressor	nominal surface tracking	1	0.833	0.178	1.066	0.000	0.573	0.002	5.000	0.806
random forest gain regressor	nominal surface tracking	2	0.833	0.180	1.085	0.000	0.583	0.002	5.000	0.811
random forest gain regressor	nominal surface tracking	3	0.833	0.183	1.116	0.000	0.581	0.008	5.000	0.811
random forest gain regressor	nominal surface tracking	4	0.833	0.177	1.081	0.000	0.585	0.004	5.000	0.807
random forest gain regressor	nominal surface tracking	5	1.000	0.180	1.080	0.000	0.573	0.004	5.000	0.808
random forest gain regressor	nominal surface tracking	6	0.833	0.172	1.071	0.000	0.581	0.004	5.000	0.806
random forest gain regressor	nominal surface tracking	7	0.667	0.180	1.084	0.000	0.589	0.004	5.000	0.811
random forest gain regressor	sensor noise burst	0	1.000	0.156	1.320	0.000	0.554	0.066	5.000	0.798
random forest gain regressor	sensor noise burst	1	1.000	0.163	1.396	0.000	0.563	0.070	5.000	0.796
random forest gain regressor	sensor noise burst	2	1.000	0.170	1.412	0.000	0.568	0.060	5.000	0.792
random forest gain regressor	sensor noise burst	3	1.000	0.164	1.356	0.000	0.565	0.064	5.000	0.796
random forest gain regressor	sensor noise burst	4	0.833	0.169	1.377	0.004	0.568	0.064	5.000	0.795
random forest gain regressor	sensor noise burst	5	1.000	0.164	1.390	0.000	0.567	0.058	5.000	0.796
random forest gain regressor	sensor noise burst	6	1.000	0.170	1.353	0.000	0.574	0.048	5.000	0.795
random forest gain regressor	sensor noise burst	7	1.000	0.161	1.381	0.000	0.564	0.062	5.000	0.797
random forest gain regressor	stick slip cycle	0	1.000	0.151	1.105	0.000	0.547	0.029	5.000	0.785
random forest gain regressor	stick slip cycle	1	1.000	0.162	1.162	0.000	0.550	0.025	5.000	0.784
random forest gain regressor	stick slip cycle	2	1.000	0.160	1.158	0.000	0.560	0.016	5.000	0.782
random forest gain regressor	stick slip cycle	3	1.000	0.158	1.101	0.000	0.550	0.021	5.000	0.781
random forest gain regressor	stick slip cycle	4	1.000	0.149	1.099	0.000	0.561	0.025	5.000	0.785
random forest gain regressor	stick slip cycle	5	1.000	0.166	1.177	0.000	0.561	0.020	5.000	0.784
random forest gain regressor	stick slip cycle	6	1.000	0.159	1.186	0.000	0.561	0.029	5.000	0.780
random forest gain regressor	stick slip cycle	7	1.000	0.156	1.124	0.000	0.551	0.025	5.000	0.783
random forest gain regressor	stiffness shift	0	0.167	0.269	1.760	0.025	0.562	0.012	6.167	0.799
random forest gain regressor	stiffness shift	1	0.000	0.268	1.872	0.037	0.563	0.014	5.000	0.797
random forest gain regressor	stiffness shift	2	0.000	0.282	1.850	0.037	0.581	0.008	7.667	0.798
random forest gain regressor	stiffness shift	3	0.000	0.266	1.831	0.029	0.564	0.013	6.167	0.795
random forest gain regressor	stiffness shift	4	0.167	0.254	1.829	0.033	0.580	0.014	7.000	0.798
random forest gain regressor	stiffness shift	5	0.167	0.277	1.830	0.031	0.567	0.010	5.333	0.795

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
random forest gain regressor	stiffness shift	6	0.000	0.284	1.846	0.033	0.581	0.012	5.333	0.797
random forest gain regressor	stiffness shift	7	0.000	0.267	1.802	0.027	0.573	0.010	7.500	0.801
random forest gain regressor	surface discontinuity	0	1.000	0.153	1.260	0.000	0.542	0.033	5.000	0.786
random forest gain regressor	surface discontinuity	1	1.000	0.158	1.315	0.000	0.552	0.023	5.667	0.786
random forest gain regressor	surface discontinuity	2	0.833	0.153	1.285	0.000	0.563	0.023	5.000	0.790
random forest gain regressor	surface discontinuity	3	1.000	0.155	1.285	0.000	0.557	0.031	5.000	0.786
random forest gain regressor	surface discontinuity	4	1.000	0.161	1.255	0.000	0.563	0.021	5.000	0.786
random forest gain regressor	surface discontinuity	5	1.000	0.163	1.327	0.000	0.566	0.025	5.000	0.788
random forest gain regressor	surface discontinuity	6	0.833	0.153	1.313	0.000	0.575	0.029	5.167	0.791
random forest gain regressor	surface discontinuity	7	0.833	0.155	1.266	0.000	0.569	0.029	5.000	0.791
random forest gain regressor	target force jump	0	1.000	0.132	1.222	0.000	0.565	0.025	5.000	0.797
random forest gain regressor	target force jump	1	1.000	0.123	1.178	0.000	0.569	0.024	5.000	0.793
random forest gain regressor	target force jump	2	1.000	0.134	1.269	0.000	0.576	0.018	5.000	0.793
random forest gain regressor	target force jump	3	0.833	0.135	1.202	0.000	0.572	0.029	5.000	0.795
random forest gain regressor	target force jump	4	0.833	0.125	1.214	0.000	0.579	0.018	5.000	0.797
random forest gain regressor	target force jump	5	0.833	0.121	1.239	0.000	0.577	0.031	5.000	0.793
random forest gain regressor	target force jump	6	0.833	0.123	1.211	0.000	0.580	0.029	5.000	0.796
random forest gain regressor	target force jump	7	0.833	0.129	1.222	0.000	0.578	0.020	5.000	0.794
risk averse impedance	actuator saturation	0	0.000	0.231	1.373	0.000	0.068	0.018	5.333	0.588
risk averse impedance	actuator saturation	1	0.000	0.235	1.306	0.000	0.036	0.018	5.500	0.586
risk averse impedance	actuator saturation	2	0.000	0.225	1.196	0.000	0.062	0.012	6.667	0.591
risk averse impedance	actuator saturation	3	0.000	0.234	1.337	0.000	0.046	0.012	5.667	0.585
risk averse impedance	actuator saturation	4	0.000	0.232	1.237	0.000	0.044	0.020	5.333	0.590
risk averse impedance	actuator saturation	5	0.000	0.229	1.239	0.000	0.066	0.014	6.167	0.589
risk averse impedance	actuator saturation	6	0.000	0.234	1.310	0.000	0.036	0.016	5.667	0.586
risk averse impedance	actuator saturation	7	0.000	0.233	1.258	0.000	0.038	0.014	5.167	0.589
risk averse impedance	combined extreme stress	0	0.000	0.208	1.244	0.000	0.000	0.064	5.167	0.550
risk averse impedance	combined extreme stress	1	0.000	0.203	1.178	0.000	0.002	0.059	5.167	0.558
risk averse impedance	combined extreme stress	2	0.000	0.199	1.227	0.000	0.000	0.071	6.000	0.560
risk averse impedance	combined extreme stress	3	0.000	0.208	1.199	0.000	0.010	0.066	5.333	0.561
risk averse impedance	combined extreme stress	4	0.000	0.194	1.174	0.000	0.000	0.071	5.667	0.563
risk averse impedance	combined extreme stress	5	0.000	0.204	1.176	0.000	0.012	0.044	5.500	0.561
risk averse impedance	combined extreme stress	6	0.000	0.202	1.310	0.000	0.000	0.055	6.167	0.557
risk averse impedance	combined extreme stress	7	0.000	0.201	1.233	0.000	0.000	0.065	5.333	0.557
risk averse impedance	combined stress	0	0.000	0.225	1.364	0.000	0.000	0.030	5.167	0.551
risk averse impedance	combined stress	1	0.000	0.226	1.315	0.000	0.000	0.034	5.167	0.548
risk averse impedance	combined stress	2	0.000	0.220	1.380	0.000	0.000	0.040	5.667	0.553
risk averse impedance	combined stress	3	0.000	0.221	1.359	0.000	0.000	0.050	5.167	0.548
risk averse impedance	combined stress	4	0.000	0.226	1.417	0.000	0.000	0.032	5.667	0.548
risk averse impedance	combined stress	5	0.000	0.214	1.182	0.000	0.000	0.046	5.500	0.547
risk averse impedance	combined stress	6	0.000	0.216	1.313	0.000	0.000	0.038	5.333	0.555
risk averse impedance	combined stress	7	0.000	0.209	1.192	0.000	0.000	0.036	5.667	0.546
risk averse impedance	contact transition	0	0.000	0.242	1.173	0.000	0.024	0.016	5.500	0.581

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
risk averse impedance	contact transition	1	0.000	0.234	1.193	0.000	0.004	0.012	5.500	0.580
risk averse impedance	contact transition	2	0.000	0.243	1.165	0.000	0.008	0.016	5.500	0.583
risk averse impedance	contact transition	3	0.000	0.235	1.131	0.000	0.012	0.012	5.667	0.581
risk averse impedance	contact transition	4	0.000	0.248	1.161	0.000	0.010	0.012	5.500	0.579
risk averse impedance	contact transition	5	0.000	0.239	1.170	0.000	0.008	0.012	5.667	0.580
risk averse impedance	contact transition	6	0.000	0.238	1.104	0.000	0.030	0.008	6.000	0.582
risk averse impedance	contact transition	7	0.000	0.240	1.195	0.000	0.000	0.014	5.667	0.574
risk averse impedance	delayed mode switch	0	0.000	0.233	1.719	0.012	0.018	0.022	6.833	0.584
risk averse impedance	delayed mode switch	1	0.000	0.231	1.593	0.010	0.052	0.022	6.833	0.583
risk averse impedance	delayed mode switch	2	0.000	0.233	1.643	0.004	0.014	0.022	5.000	0.584
risk averse impedance	delayed mode switch	3	0.000	0.226	1.579	0.006	0.065	0.018	6.000	0.586
risk averse impedance	delayed mode switch	4	0.000	0.226	1.759	0.016	0.054	0.020	5.000	0.587
risk averse impedance	delayed mode switch	5	0.000	0.216	1.451	0.000	0.064	0.014	7.500	0.580
risk averse impedance	delayed mode switch	6	0.000	0.235	1.607	0.004	0.073	0.022	7.333	0.587
risk averse impedance	delayed mode switch	7	0.000	0.217	1.625	0.008	0.048	0.018	5.167	0.586
risk averse impedance	friction slip shift	0	0.000	0.299	0.947	0.000	0.000	0.000	5.167	0.579
risk averse impedance	friction slip shift	1	0.000	0.305	0.979	0.000	0.000	0.000	5.000	0.575
risk averse impedance	friction slip shift	2	0.000	0.308	1.000	0.000	0.000	0.004	5.000	0.583
risk averse impedance	friction slip shift	3	0.000	0.304	0.980	0.000	0.012	0.002	5.000	0.584
risk averse impedance	friction slip shift	4	0.000	0.286	0.952	0.000	0.000	0.000	5.167	0.574
risk averse impedance	friction slip shift	5	0.000	0.300	0.964	0.000	0.000	0.002	5.000	0.577
risk averse impedance	friction slip shift	6	0.000	0.298	0.930	0.000	0.006	0.000	5.000	0.581
risk averse impedance	friction slip shift	7	0.000	0.290	0.934	0.000	0.000	0.000	5.167	0.576
risk averse impedance	nominal surface tracking	0	0.000	0.309	0.891	0.000	0.211	0.000	5.167	0.607
risk averse impedance	nominal surface tracking	1	0.000	0.312	0.911	0.000	0.207	0.000	5.167	0.608
risk averse impedance	nominal surface tracking	2	0.000	0.302	0.913	0.000	0.190	0.000	5.000	0.608
risk averse impedance	nominal surface tracking	3	0.000	0.304	0.936	0.000	0.205	0.000	5.000	0.608
risk averse impedance	nominal surface tracking	4	0.000	0.294	0.910	0.000	0.192	0.000	5.167	0.607
risk averse impedance	nominal surface tracking	5	0.000	0.307	0.914	0.000	0.207	0.000	5.000	0.606
risk averse impedance	nominal surface tracking	6	0.000	0.303	0.905	0.000	0.205	0.000	5.000	0.606
risk averse impedance	nominal surface tracking	7	0.000	0.304	0.927	0.000	0.200	0.000	5.000	0.609
risk averse impedance	sensor noise burst	0	0.000	0.245	1.105	0.000	0.155	0.028	6.000	0.599
risk averse impedance	sensor noise burst	1	0.000	0.246	1.154	0.000	0.117	0.040	6.000	0.595
risk averse impedance	sensor noise burst	2	0.000	0.249	1.196	0.000	0.103	0.042	5.667	0.596
risk averse impedance	sensor noise burst	3	0.000	0.246	1.225	0.000	0.137	0.038	5.167	0.598
risk averse impedance	sensor noise burst	4	0.000	0.250	1.223	0.000	0.109	0.028	5.000	0.596
risk averse impedance	sensor noise burst	5	0.000	0.246	1.154	0.000	0.143	0.038	5.667	0.597
risk averse impedance	sensor noise burst	6	0.000	0.250	1.218	0.000	0.129	0.038	5.500	0.598
risk averse impedance	sensor noise burst	7	0.000	0.248	1.244	0.000	0.123	0.040	5.500	0.598
risk averse impedance	stick slip cycle	0	0.000	0.264	0.924	0.000	0.000	0.000	5.833	0.575
risk averse impedance	stick slip cycle	1	0.000	0.265	0.994	0.000	0.000	0.004	6.000	0.575
risk averse impedance	stick slip cycle	2	0.000	0.260	0.975	0.000	0.000	0.002	5.500	0.571
risk averse impedance	stick slip cycle	3	0.000	0.268	0.947	0.000	0.000	0.000	5.500	0.568

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
risk averse impedance	stick slip cycle	4	0.000	0.256	0.938	0.000	0.008	0.000	7.333	0.580
risk averse impedance	stick slip cycle	5	0.000	0.268	0.996	0.000	0.000	0.004	5.333	0.572
risk averse impedance	stick slip cycle	6	0.000	0.256	1.009	0.000	0.000	0.004	5.833	0.569
risk averse impedance	stick slip cycle	7	0.000	0.263	0.953	0.000	0.000	0.000	6.000	0.574
risk averse impedance	stiffness shift	0	0.000	0.313	1.474	0.000	0.191	0.012	5.000	0.603
risk averse impedance	stiffness shift	1	0.000	0.308	1.585	0.006	0.179	0.014	5.167	0.601
risk averse impedance	stiffness shift	2	0.000	0.319	1.569	0.002	0.172	0.012	5.000	0.602
risk averse impedance	stiffness shift	3	0.000	0.309	1.542	0.000	0.172	0.014	5.000	0.599
risk averse impedance	stiffness shift	4	0.000	0.297	1.561	0.002	0.178	0.016	5.833	0.603
risk averse impedance	stiffness shift	5	0.000	0.316	1.523	0.002	0.179	0.014	5.000	0.600
risk averse impedance	stiffness shift	6	0.000	0.323	1.549	0.004	0.185	0.012	5.000	0.602
risk averse impedance	stiffness shift	7	0.000	0.308	1.523	0.000	0.180	0.012	5.333	0.602
risk averse impedance	surface discontinuity	0	0.000	0.233	0.974	0.000	0.000	0.002	5.833	0.578
risk averse impedance	surface discontinuity	1	0.000	0.229	0.992	0.000	0.000	0.002	5.333	0.576
risk averse impedance	surface discontinuity	2	0.000	0.234	0.980	0.000	0.000	0.002	5.667	0.581
risk averse impedance	surface discontinuity	3	0.000	0.232	1.011	0.000	0.012	0.000	5.833	0.578
risk averse impedance	surface discontinuity	4	0.000	0.236	0.971	0.000	0.008	0.000	5.333	0.577
risk averse impedance	surface discontinuity	5	0.000	0.234	0.971	0.000	0.000	0.004	5.167	0.578
risk averse impedance	surface discontinuity	6	0.000	0.229	0.997	0.000	0.016	0.002	5.500	0.581
risk averse impedance	surface discontinuity	7	0.000	0.233	0.988	0.000	0.000	0.004	6.000	0.577
risk averse impedance	target force jump	0	0.000	0.215	1.028	0.000	0.084	0.006	8.333	0.590
risk averse impedance	target force jump	1	0.000	0.203	0.990	0.000	0.067	0.002	9.500	0.586
risk averse impedance	target force jump	2	0.000	0.206	1.083	0.000	0.063	0.006	8.500	0.590
risk averse impedance	target force jump	3	0.000	0.220	1.031	0.000	0.063	0.004	8.500	0.588
risk averse impedance	target force jump	4	0.000	0.206	1.034	0.000	0.104	0.006	7.833	0.592
risk averse impedance	target force jump	5	0.000	0.206	1.048	0.000	0.067	0.006	7.667	0.587
risk averse impedance	target force jump	6	0.000	0.203	1.021	0.000	0.106	0.002	10.000	0.591
risk averse impedance	target force jump	7	0.000	0.209	1.040	0.000	0.065	0.006	10.000	0.587
robust mpc impedance	actuator saturation	0	0.000	0.220	1.469	0.000	0.413	0.022	5.000	0.673
robust mpc impedance	actuator saturation	1	0.000	0.215	1.397	0.000	0.384	0.024	5.167	0.673
robust mpc impedance	actuator saturation	2	0.000	0.209	1.276	0.000	0.396	0.018	5.167	0.675
robust mpc impedance	actuator saturation	3	0.000	0.221	1.435	0.000	0.376	0.016	5.000	0.673
robust mpc impedance	actuator saturation	4	0.000	0.214	1.337	0.000	0.391	0.022	5.333	0.674
robust mpc impedance	actuator saturation	5	0.000	0.215	1.356	0.000	0.396	0.024	5.000	0.675
robust mpc impedance	actuator saturation	6	0.000	0.219	1.407	0.000	0.383	0.018	5.000	0.672
robust mpc impedance	actuator saturation	7	0.000	0.217	1.352	0.000	0.380	0.018	5.000	0.674
robust mpc impedance	combined extreme stress	0	0.000	0.196	1.423	0.000	0.352	0.066	5.167	0.644
robust mpc impedance	combined extreme stress	1	0.000	0.192	1.427	0.004	0.327	0.072	5.333	0.648
robust mpc impedance	combined extreme stress	2	0.000	0.189	1.302	0.000	0.340	0.091	5.000	0.651
robust mpc impedance	combined extreme stress	3	0.000	0.195	1.299	0.000	0.340	0.076	5.167	0.650
robust mpc impedance	combined extreme stress	4	0.000	0.184	1.304	0.000	0.329	0.103	5.167	0.652
robust mpc impedance	combined extreme stress	5	0.000	0.190	1.250	0.000	0.351	0.081	5.000	0.649
robust mpc impedance	combined extreme stress	6	0.000	0.188	1.377	0.000	0.363	0.110	5.333	0.650

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
robust mpc impedance	combined extreme stress	7	0.000	0.186	1.301	0.000	0.347	0.095	5.500	0.649
robust mpc impedance	combined stress	0	0.000	0.217	1.536	0.006	0.328	0.039	5.167	0.649
robust mpc impedance	combined stress	1	0.000	0.212	1.395	0.000	0.304	0.049	5.000	0.650
robust mpc impedance	combined stress	2	0.000	0.211	1.449	0.000	0.321	0.035	5.000	0.652
robust mpc impedance	combined stress	3	0.000	0.209	1.444	0.000	0.312	0.059	5.167	0.651
robust mpc impedance	combined stress	4	0.000	0.209	1.456	0.002	0.266	0.057	5.167	0.649
robust mpc impedance	combined stress	5	0.000	0.199	1.328	0.000	0.305	0.056	5.167	0.649
robust mpc impedance	combined stress	6	0.000	0.203	1.425	0.002	0.342	0.066	5.000	0.652
robust mpc impedance	combined stress	7	0.000	0.195	1.309	0.000	0.259	0.049	5.000	0.648
robust mpc impedance	contact transition	0	0.000	0.228	1.274	0.000	0.403	0.018	5.000	0.672
robust mpc impedance	contact transition	1	0.000	0.215	1.292	0.000	0.398	0.016	5.000	0.672
robust mpc impedance	contact transition	2	0.000	0.224	1.244	0.000	0.403	0.024	5.000	0.673
robust mpc impedance	contact transition	3	0.000	0.215	1.214	0.000	0.394	0.012	5.167	0.672
robust mpc impedance	contact transition	4	0.000	0.224	1.230	0.000	0.391	0.010	5.000	0.671
robust mpc impedance	contact transition	5	0.000	0.224	1.277	0.000	0.404	0.014	5.000	0.673
robust mpc impedance	contact transition	6	0.000	0.218	1.188	0.000	0.410	0.014	5.167	0.673
robust mpc impedance	contact transition	7	0.000	0.221	1.261	0.000	0.377	0.018	5.000	0.670
robust mpc impedance	delayed mode switch	0	0.000	0.223	1.840	0.023	0.415	0.033	5.167	0.669
robust mpc impedance	delayed mode switch	1	0.000	0.223	1.698	0.016	0.389	0.021	6.667	0.669
robust mpc impedance	delayed mode switch	2	0.000	0.223	1.766	0.016	0.407	0.031	5.000	0.671
robust mpc impedance	delayed mode switch	3	0.000	0.223	1.653	0.010	0.385	0.020	7.167	0.671
robust mpc impedance	delayed mode switch	4	0.000	0.216	1.881	0.030	0.380	0.024	5.500	0.670
robust mpc impedance	delayed mode switch	5	0.000	0.205	1.572	0.010	0.387	0.018	6.833	0.667
robust mpc impedance	delayed mode switch	6	0.000	0.227	1.737	0.014	0.402	0.023	5.833	0.671
robust mpc impedance	delayed mode switch	7	0.000	0.209	1.765	0.014	0.397	0.022	5.000	0.670
robust mpc impedance	friction slip shift	0	0.000	0.270	1.024	0.000	0.400	0.006	5.000	0.669
robust mpc impedance	friction slip shift	1	0.000	0.273	1.033	0.000	0.366	0.006	5.000	0.669
robust mpc impedance	friction slip shift	2	0.000	0.278	1.064	0.000	0.383	0.002	5.000	0.671
robust mpc impedance	friction slip shift	3	0.000	0.274	1.046	0.000	0.396	0.006	5.000	0.672
robust mpc impedance	friction slip shift	4	0.000	0.256	1.022	0.000	0.360	0.002	5.000	0.668
robust mpc impedance	friction slip shift	5	0.000	0.266	1.029	0.000	0.373	0.006	5.000	0.669
robust mpc impedance	friction slip shift	6	0.000	0.268	1.007	0.000	0.402	0.000	5.000	0.670
robust mpc impedance	friction slip shift	7	0.000	0.258	1.009	0.000	0.368	0.004	5.000	0.668
robust mpc impedance	nominal surface tracking	0	0.000	0.276	0.966	0.000	0.441	0.000	5.000	0.685
robust mpc impedance	nominal surface tracking	1	0.000	0.277	0.976	0.000	0.427	0.000	5.000	0.686
robust mpc impedance	nominal surface tracking	2	0.000	0.269	0.986	0.000	0.428	0.000	5.000	0.685
robust mpc impedance	nominal surface tracking	3	0.000	0.271	1.011	0.000	0.423	0.002	5.000	0.686
robust mpc impedance	nominal surface tracking	4	0.000	0.261	0.990	0.000	0.421	0.000	5.000	0.685
robust mpc impedance	nominal surface tracking	5	0.000	0.276	0.982	0.000	0.433	0.004	5.000	0.685
robust mpc impedance	nominal surface tracking	6	0.000	0.271	0.976	0.000	0.432	0.000	5.000	0.685
robust mpc impedance	nominal surface tracking	7	0.000	0.271	0.985	0.000	0.430	0.000	5.000	0.686
robust mpc impedance	sensor noise burst	0	0.000	0.227	1.235	0.000	0.428	0.041	5.000	0.680
robust mpc impedance	sensor noise burst	1	0.000	0.224	1.231	0.000	0.402	0.046	5.167	0.678

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
robust mpc impedance	sensor noise burst	2	0.000	0.228	1.231	0.000	0.400	0.053	5.000	0.678
robust mpc impedance	sensor noise burst	3	0.000	0.226	1.276	0.000	0.403	0.055	5.000	0.680
robust mpc impedance	sensor noise burst	4	0.000	0.230	1.262	0.000	0.395	0.048	5.000	0.678
robust mpc impedance	sensor noise burst	5	0.000	0.227	1.241	0.000	0.403	0.061	5.000	0.679
robust mpc impedance	sensor noise burst	6	0.000	0.227	1.255	0.000	0.415	0.049	5.000	0.679
robust mpc impedance	sensor noise burst	7	0.000	0.225	1.266	0.000	0.396	0.042	5.167	0.679
robust mpc impedance	stick slip cycle	0	0.000	0.236	1.002	0.000	0.381	0.010	5.167	0.664
robust mpc impedance	stick slip cycle	1	0.000	0.234	1.043	0.000	0.371	0.004	5.000	0.664
robust mpc impedance	stick slip cycle	2	0.000	0.235	1.044	0.000	0.349	0.004	5.000	0.661
robust mpc impedance	stick slip cycle	3	0.000	0.238	1.018	0.000	0.340	0.002	5.333	0.661
robust mpc impedance	stick slip cycle	4	0.000	0.228	1.011	0.000	0.377	0.002	5.167	0.666
robust mpc impedance	stick slip cycle	5	0.000	0.242	1.065	0.000	0.359	0.010	5.167	0.663
robust mpc impedance	stick slip cycle	6	0.000	0.231	1.067	0.000	0.363	0.012	5.000	0.660
robust mpc impedance	stick slip cycle	7	0.000	0.235	1.028	0.000	0.360	0.008	5.000	0.663
robust mpc impedance	stiffness shift	0	0.000	0.307	1.597	0.004	0.432	0.012	5.000	0.682
robust mpc impedance	stiffness shift	1	0.000	0.298	1.681	0.012	0.409	0.016	5.000	0.682
robust mpc impedance	stiffness shift	2	0.000	0.313	1.667	0.008	0.418	0.012	5.167	0.682
robust mpc impedance	stiffness shift	3	0.000	0.299	1.656	0.008	0.411	0.014	5.333	0.681
robust mpc impedance	stiffness shift	4	0.000	0.289	1.673	0.010	0.428	0.018	6.167	0.683
robust mpc impedance	stiffness shift	5	0.000	0.313	1.626	0.014	0.403	0.022	5.000	0.682
robust mpc impedance	stiffness shift	6	0.000	0.317	1.655	0.008	0.418	0.014	5.000	0.682
robust mpc impedance	stiffness shift	7	0.000	0.299	1.613	0.008	0.420	0.014	5.833	0.683
robust mpc impedance	surface discontinuity	0	0.000	0.210	1.047	0.000	0.406	0.012	5.000	0.669
robust mpc impedance	surface discontinuity	1	0.000	0.205	1.071	0.000	0.386	0.016	5.167	0.669
robust mpc impedance	surface discontinuity	2	0.000	0.209	1.043	0.000	0.398	0.006	5.167	0.671
robust mpc impedance	surface discontinuity	3	0.000	0.209	1.075	0.000	0.382	0.008	5.000	0.671
robust mpc impedance	surface discontinuity	4	0.000	0.211	1.044	0.000	0.390	0.010	5.333	0.671
robust mpc impedance	surface discontinuity	5	0.000	0.214	1.088	0.000	0.402	0.014	5.000	0.671
robust mpc impedance	surface discontinuity	6	0.000	0.207	1.046	0.000	0.404	0.004	5.000	0.671
robust mpc impedance	surface discontinuity	7	0.000	0.208	1.035	0.000	0.394	0.008	5.000	0.671
robust mpc impedance	target force jump	0	0.000	0.196	1.112	0.000	0.449	0.012	6.000	0.675
robust mpc impedance	target force jump	1	0.000	0.183	1.056	0.000	0.420	0.004	6.167	0.675
robust mpc impedance	target force jump	2	0.000	0.188	1.149	0.000	0.425	0.015	6.667	0.676
robust mpc impedance	target force jump	3	0.000	0.198	1.135	0.000	0.416	0.006	5.667	0.676
robust mpc impedance	target force jump	4	0.000	0.186	1.100	0.000	0.433	0.006	6.167	0.677
robust mpc impedance	target force jump	5	0.000	0.185	1.121	0.000	0.435	0.019	6.167	0.674
robust mpc impedance	target force jump	6	0.000	0.184	1.104	0.000	0.439	0.008	7.000	0.677
robust mpc impedance	target force jump	7	0.000	0.187	1.088	0.000	0.423	0.006	6.333	0.675
token no memory ablation	actuator saturation	0	1.000	0.172	1.537	0.002	0.507	0.027	5.000	0.758
token no memory ablation	actuator saturation	1	1.000	0.169	1.443	0.000	0.505	0.023	5.000	0.755
token no memory ablation	actuator saturation	2	0.833	0.164	1.280	0.000	0.499	0.025	5.000	0.757
token no memory ablation	actuator saturation	3	1.000	0.171	1.472	0.000	0.514	0.025	5.000	0.757
token no memory ablation	actuator saturation	4	1.000	0.169	1.373	0.000	0.491	0.023	5.333	0.759

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
token no memory ablation	actuator saturation	5	1.000	0.167	1.414	0.000	0.516	0.031	5.000	0.758
token no memory ablation	actuator saturation	6	1.000	0.167	1.459	0.000	0.512	0.027	5.000	0.759
token no memory ablation	actuator saturation	7	1.000	0.168	1.379	0.000	0.510	0.023	5.000	0.758
token no memory ablation	combined extreme stress	0	0.000	0.157	1.482	0.000	0.441	0.108	5.000	0.719
token no memory ablation	combined extreme stress	1	0.167	0.155	1.412	0.002	0.461	0.104	5.000	0.726
token no memory ablation	combined extreme stress	2	0.000	0.156	1.412	0.000	0.458	0.091	5.000	0.726
token no memory ablation	combined extreme stress	3	0.333	0.159	1.391	0.000	0.483	0.107	5.000	0.731
token no memory ablation	combined extreme stress	4	0.333	0.155	1.363	0.000	0.452	0.124	5.167	0.731
token no memory ablation	combined extreme stress	5	0.333	0.158	1.422	0.000	0.465	0.118	5.000	0.729
token no memory ablation	combined extreme stress	6	0.000	0.152	1.519	0.004	0.470	0.108	5.000	0.728
token no memory ablation	combined extreme stress	7	0.000	0.158	1.475	0.004	0.461	0.100	5.000	0.724
token no memory ablation	combined stress	0	0.000	0.167	1.601	0.006	0.442	0.051	5.000	0.722
token no memory ablation	combined stress	1	0.167	0.173	1.535	0.006	0.461	0.057	5.000	0.721
token no memory ablation	combined stress	2	0.000	0.167	1.494	0.002	0.448	0.055	5.000	0.719
token no memory ablation	combined stress	3	0.000	0.167	1.519	0.002	0.462	0.067	5.000	0.723
token no memory ablation	combined stress	4	0.000	0.167	1.588	0.004	0.443	0.075	5.000	0.720
token no memory ablation	combined stress	5	0.000	0.157	1.444	0.000	0.468	0.069	5.000	0.720
token no memory ablation	combined stress	6	0.000	0.160	1.426	0.000	0.478	0.083	5.000	0.729
token no memory ablation	combined stress	7	0.000	0.162	1.407	0.000	0.457	0.061	5.833	0.717
token no memory ablation	contact transition	0	0.833	0.178	1.302	0.000	0.489	0.023	5.000	0.749
token no memory ablation	contact transition	1	0.833	0.168	1.280	0.000	0.489	0.016	5.000	0.746
token no memory ablation	contact transition	2	0.833	0.175	1.279	0.000	0.470	0.024	5.000	0.745
token no memory ablation	contact transition	3	1.000	0.169	1.264	0.000	0.499	0.022	5.000	0.749
token no memory ablation	contact transition	4	0.667	0.177	1.218	0.000	0.443	0.016	5.000	0.740
token no memory ablation	contact transition	5	0.833	0.177	1.294	0.000	0.497	0.022	5.000	0.748
token no memory ablation	contact transition	6	1.000	0.170	1.219	0.000	0.502	0.022	5.000	0.751
token no memory ablation	contact transition	7	0.667	0.174	1.259	0.000	0.461	0.023	5.000	0.737
token no memory ablation	delayed mode switch	0	0.667	0.177	1.924	0.027	0.487	0.027	5.000	0.749
token no memory ablation	delayed mode switch	1	1.000	0.179	1.774	0.018	0.492	0.029	6.667	0.749
token no memory ablation	delayed mode switch	2	0.833	0.180	1.815	0.016	0.483	0.025	5.000	0.745
token no memory ablation	delayed mode switch	3	0.833	0.178	1.750	0.010	0.509	0.020	7.667	0.757
token no memory ablation	delayed mode switch	4	0.500	0.174	1.985	0.023	0.473	0.033	5.667	0.752
token no memory ablation	delayed mode switch	5	0.500	0.167	1.648	0.016	0.485	0.031	6.333	0.745
token no memory ablation	delayed mode switch	6	0.833	0.180	1.799	0.016	0.500	0.025	6.167	0.755
token no memory ablation	delayed mode switch	7	0.500	0.169	1.814	0.012	0.489	0.029	5.167	0.750
token no memory ablation	friction slip shift	0	0.500	0.212	1.048	0.000	0.457	0.006	5.000	0.741
token no memory ablation	friction slip shift	1	0.667	0.209	1.092	0.000	0.502	0.006	5.000	0.746
token no memory ablation	friction slip shift	2	1.000	0.215	1.109	0.000	0.494	0.006	5.000	0.749
token no memory ablation	friction slip shift	3	0.833	0.211	1.064	0.000	0.504	0.010	5.000	0.753
token no memory ablation	friction slip shift	4	0.500	0.199	1.034	0.000	0.470	0.004	5.000	0.738
token no memory ablation	friction slip shift	5	0.833	0.208	1.053	0.000	0.486	0.014	5.000	0.744
token no memory ablation	friction slip shift	6	0.667	0.206	1.019	0.000	0.474	0.004	5.000	0.744
token no memory ablation	friction slip shift	7	0.500	0.200	1.018	0.000	0.480	0.004	5.000	0.739

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
token no memory ablation	nominal surface tracking	0	0.833	0.212	1.001	0.000	0.512	0.000	5.000	0.773
token no memory ablation	nominal surface tracking	1	0.833	0.217	1.003	0.000	0.507	0.004	5.000	0.770
token no memory ablation	nominal surface tracking	2	1.000	0.208	1.015	0.000	0.509	0.000	5.000	0.772
token no memory ablation	nominal surface tracking	3	1.000	0.208	1.027	0.000	0.519	0.002	5.000	0.773
token no memory ablation	nominal surface tracking	4	1.000	0.206	0.993	0.000	0.487	0.000	5.000	0.769
token no memory ablation	nominal surface tracking	5	0.833	0.215	1.020	0.000	0.524	0.002	5.000	0.772
token no memory ablation	nominal surface tracking	6	1.000	0.210	0.997	0.000	0.517	0.000	5.000	0.770
token no memory ablation	nominal surface tracking	7	1.000	0.212	1.025	0.000	0.515	0.000	5.000	0.773
token no memory ablation	sensor noise burst	0	1.000	0.175	1.213	0.000	0.509	0.066	5.000	0.765
token no memory ablation	sensor noise burst	1	1.000	0.173	1.276	0.000	0.526	0.058	5.000	0.766
token no memory ablation	sensor noise burst	2	1.000	0.178	1.319	0.000	0.515	0.060	5.167	0.763
token no memory ablation	sensor noise burst	3	1.000	0.176	1.391	0.000	0.535	0.043	5.000	0.768
token no memory ablation	sensor noise burst	4	1.000	0.177	1.277	0.000	0.484	0.050	5.000	0.762
token no memory ablation	sensor noise burst	5	1.000	0.176	1.322	0.000	0.524	0.054	5.000	0.767
token no memory ablation	sensor noise burst	6	1.000	0.173	1.275	0.000	0.516	0.060	5.000	0.766
token no memory ablation	sensor noise burst	7	1.000	0.176	1.329	0.000	0.512	0.062	5.000	0.765
token no memory ablation	stick slip cycle	0	0.667	0.178	1.021	0.000	0.481	0.008	5.000	0.742
token no memory ablation	stick slip cycle	1	0.500	0.182	1.110	0.000	0.491	0.018	5.000	0.743
token no memory ablation	stick slip cycle	2	0.333	0.185	1.071	0.000	0.446	0.014	5.000	0.731
token no memory ablation	stick slip cycle	3	0.167	0.180	1.035	0.000	0.492	0.008	5.000	0.736
token no memory ablation	stick slip cycle	4	0.667	0.176	1.023	0.000	0.477	0.012	5.000	0.745
token no memory ablation	stick slip cycle	5	0.333	0.187	1.085	0.000	0.475	0.016	5.000	0.738
token no memory ablation	stick slip cycle	6	0.500	0.175	1.121	0.000	0.496	0.010	5.000	0.740
token no memory ablation	stick slip cycle	7	0.667	0.180	1.062	0.000	0.482	0.016	5.000	0.741
token no memory ablation	stiffness shift	0	0.333	0.239	1.619	0.004	0.513	0.016	5.000	0.771
token no memory ablation	stiffness shift	1	0.333	0.232	1.750	0.012	0.526	0.016	5.000	0.771
token no memory ablation	stiffness shift	2	0.333	0.245	1.702	0.012	0.510	0.016	5.000	0.766
token no memory ablation	stiffness shift	3	0.500	0.231	1.684	0.006	0.528	0.014	5.000	0.769
token no memory ablation	stiffness shift	4	0.667	0.230	1.697	0.010	0.507	0.020	6.000	0.769
token no memory ablation	stiffness shift	5	0.333	0.244	1.694	0.016	0.521	0.017	5.000	0.769
token no memory ablation	stiffness shift	6	0.333	0.246	1.721	0.012	0.531	0.012	5.000	0.773
token no memory ablation	stiffness shift	7	0.167	0.236	1.659	0.010	0.515	0.018	5.500	0.770
token no memory ablation	surface discontinuity	0	1.000	0.158	1.109	0.000	0.484	0.016	5.000	0.747
token no memory ablation	surface discontinuity	1	1.000	0.154	1.122	0.000	0.499	0.016	5.000	0.748
token no memory ablation	surface discontinuity	2	0.833	0.162	1.088	0.000	0.478	0.010	5.000	0.745
token no memory ablation	surface discontinuity	3	1.000	0.158	1.124	0.000	0.512	0.016	5.000	0.751
token no memory ablation	surface discontinuity	4	0.833	0.163	1.078	0.000	0.472	0.016	5.000	0.745
token no memory ablation	surface discontinuity	5	0.833	0.165	1.154	0.000	0.500	0.024	5.000	0.749
token no memory ablation	surface discontinuity	6	1.000	0.156	1.131	0.000	0.499	0.022	5.000	0.753
token no memory ablation	surface discontinuity	7	0.833	0.162	1.106	0.000	0.488	0.022	5.000	0.746
token no memory ablation	target force jump	0	1.000	0.155	1.135	0.000	0.505	0.012	5.000	0.756
token no memory ablation	target force jump	1	0.667	0.146	1.065	0.000	0.477	0.004	5.167	0.745
token no memory ablation	target force jump	2	0.667	0.150	1.122	0.000	0.474	0.018	5.000	0.747

Method	Split	Seed	Succ.	Err.	Over.	Safe	Slip	Chat.	Lat.	Prog.
token no memory ablation	target force jump	3	1.000	0.156	1.110	0.000	0.512	0.018	5.000	0.754
token no memory ablation	target force jump	4	1.000	0.150	1.097	0.000	0.475	0.014	5.000	0.751
token no memory ablation	target force jump	5	1.000	0.148	1.109	0.000	0.490	0.018	5.167	0.749
token no memory ablation	target force jump	6	1.000	0.147	1.090	0.000	0.489	0.014	5.167	0.752
token no memory ablation	target force jump	7	0.500	0.152	1.125	0.000	0.480	0.010	5.167	0.746

E Full Paired Comparisons

Table 11: Full paired seed differences versus impedance-token policy v5.

Split	Comparison	dSucc.	CI	dErr.	dSafe	dSlip	dChat.	Wins
actuator saturation	adaptive impedance control	-0.958	0.053	-0.080	0.008	0.320	0.020	0
actuator saturation	admittance switching control	-1.000	0.000	-0.074	0.003	0.291	0.025	0
actuator saturation	conformal safety gain	-0.688	0.144	-0.028	0.001	0.335	0.015	0
actuator saturation	ensemble uncertainty gain	-0.521	0.190	-0.022	0.002	0.341	0.013	0
actuator saturation	fixed impedance	-0.979	0.041	0.008	0.018	0.306	0.025	0
actuator saturation	gain scheduled impedance	-0.146	0.074	-0.065	0.003	0.351	0.013	0
actuator saturation	hist gradient gain regressor	-0.479	0.190	-0.026	0.001	0.343	0.016	0
actuator saturation	impedance token policy v4	-0.125	0.082	-0.029	0.000	0.213	0.006	0
actuator saturation	learned gain regressor	-0.188	0.130	-0.021	0.006	0.355	0.010	0
actuator saturation	oracle impedance	-0.542	0.135	-0.088	0.005	0.341	0.015	0
actuator saturation	random forest gain regressor	-0.896	0.086	-0.021	0.002	0.311	0.014	0
actuator saturation	risk averse impedance	0.000	0.000	0.042	0.000	-0.210	0.000	0
actuator saturation	robust mpc impedance	0.000	0.000	0.026	0.000	0.131	0.005	0
actuator saturation	token no memory ablation	-0.979	0.041	-0.022	0.000	0.247	0.011	0
combined extreme stress	adaptive impedance control	-0.979	0.041	-0.045	0.004	0.293	0.066	0
combined extreme stress	admittance switching control	-0.958	0.053	-0.051	0.003	0.265	0.074	0
combined extreme stress	conformal safety gain	-0.958	0.053	-0.003	0.001	0.316	0.041	0
combined extreme stress	ensemble uncertainty gain	-0.958	0.053	0.007	0.004	0.321	0.033	0
combined extreme stress	fixed impedance	-0.396	0.150	0.072	0.012	0.267	0.019	0
combined extreme stress	gain scheduled impedance	-0.958	0.053	-0.034	0.002	0.337	0.050	0
combined extreme stress	hist gradient gain regressor	-0.875	0.082	0.009	0.007	0.321	0.044	0
combined extreme stress	impedance token policy v4	0.000	0.000	-0.023	0.000	0.172	0.028	0
combined extreme stress	learned gain regressor	-0.812	0.114	-0.000	0.005	0.336	0.030	0
combined extreme stress	oracle impedance	-0.938	0.086	-0.062	0.002	0.251	0.078	0
combined extreme stress	random forest gain regressor	-0.938	0.060	0.010	0.009	0.292	0.037	0
combined extreme stress	risk averse impedance	0.000	0.000	0.037	-0.000	-0.234	-0.006	0
combined extreme stress	robust mpc impedance	0.000	0.000	0.025	0.000	0.106	0.019	0
combined extreme stress	token no memory ablation	-0.146	0.114	-0.009	0.001	0.224	0.040	0
combined stress	adaptive impedance control	-0.938	0.086	-0.059	0.013	0.429	0.040	0
combined stress	admittance switching control	-0.938	0.060	-0.060	0.009	0.411	0.041	0
combined stress	conformal safety gain	-0.979	0.041	-0.008	0.005	0.441	0.020	0
combined stress	ensemble uncertainty gain	-0.917	0.087	0.001	0.014	0.452	0.026	0
combined stress	fixed impedance	-0.250	0.107	0.068	0.009	0.414	0.018	0
combined stress	gain scheduled impedance	-0.854	0.130	-0.043	0.009	0.472	0.035	0
combined stress	hist gradient gain regressor	-0.917	0.062	-0.001	0.017	0.446	0.025	0
combined stress	impedance token policy v4	0.000	0.000	-0.030	0.001	0.254	0.021	0
combined stress	learned gain regressor	-0.833	0.087	0.001	0.019	0.461	0.014	0
combined stress	oracle impedance	-0.875	0.082	-0.073	0.008	0.401	0.047	0
combined stress	random forest gain regressor	-0.938	0.060	0.001	0.016	0.426	0.026	0
combined stress	risk averse impedance	0.000	0.000	0.039	-0.000	-0.114	-0.003	0
combined stress	robust mpc impedance	0.000	0.000	0.027	0.001	0.191	0.010	0
combined stress	token no memory ablation	-0.021	0.041	-0.015	0.002	0.344	0.023	0
contact transition	adaptive impedance control	-0.979	0.041	-0.088	0.000	0.314	0.020	0
contact transition	admittance switching control	-0.979	0.041	-0.084	0.011	0.295	0.022	0
contact transition	conformal safety gain	-0.729	0.086	-0.036	0.000	0.330	0.014	0
contact transition	ensemble uncertainty gain	-0.458	0.120	-0.032	0.000	0.338	0.012	0
contact transition	fixed impedance	-0.938	0.086	-0.002	0.042	0.306	0.027	0
contact transition	gain scheduled impedance	-0.125	0.082	-0.076	0.000	0.352	0.013	0
contact transition	hist gradient gain regressor	-0.542	0.148	-0.039	0.000	0.340	0.013	0
contact transition	impedance token policy v4	0.000	0.000	-0.038	0.000	0.183	0.011	0
contact transition	learned gain regressor	-0.375	0.102	-0.041	0.000	0.347	0.010	0
contact transition	oracle impedance	-0.479	0.096	-0.094	0.000	0.335	0.018	0
contact transition	random forest gain regressor	-0.896	0.060	-0.033	0.000	0.310	0.013	0

Split	Comparison	dSucc.	CI	dErr.	dSafe	dSlip	dChat.	Wins
contact transition	risk averse impedance	0.000	0.000	0.042	0.000	-0.248	0.001	0
contact transition	robust mpc impedance	0.000	0.000	0.024	0.000	0.138	0.005	0
contact transition	token no memory ablation	-0.833	0.087	-0.024	0.000	0.221	0.009	0
delayed mode switch	adaptive impedance control	-0.542	0.120	-0.063	0.025	0.291	0.024	0
delayed mode switch	admittance switching control	-0.667	0.138	-0.060	0.023	0.253	0.030	0
delayed mode switch	conformal safety gain	-0.688	0.144	-0.013	0.020	0.307	0.009	0
delayed mode switch	ensemble uncertainty gain	-0.500	0.151	0.003	0.029	0.313	0.012	0
delayed mode switch	fixed impedance	-0.792	0.102	0.026	0.026	0.279	0.025	0
delayed mode switch	gain scheduled impedance	-0.375	0.160	-0.044	0.021	0.330	0.016	0
delayed mode switch	hist gradient gain regressor	-0.500	0.123	0.001	0.026	0.317	0.015	0
delayed mode switch	impedance token policy v4	-0.438	0.137	-0.029	0.002	0.233	0.013	0
delayed mode switch	learned gain regressor	-0.292	0.053	-0.010	0.035	0.326	0.011	0
delayed mode switch	oracle impedance	-0.562	0.086	-0.078	0.014	0.295	0.022	0
delayed mode switch	random forest gain regressor	-0.604	0.122	0.005	0.029	0.287	0.006	0
delayed mode switch	risk averse impedance	0.000	0.000	0.040	-0.001	-0.220	0.003	0
delayed mode switch	robust mpc impedance	0.000	0.000	0.031	0.008	0.127	0.007	0
delayed mode switch	token no memory ablation	-0.708	0.135	-0.012	0.009	0.221	0.011	0
friction slip shift	adaptive impedance control	-0.979	0.041	-0.124	0.000	0.272	0.015	0
friction slip shift	admittance switching control	-1.000	0.000	-0.123	0.000	0.256	0.020	0
friction slip shift	conformal safety gain	-0.833	0.087	-0.057	0.000	0.283	0.008	0
friction slip shift	ensemble uncertainty gain	-0.708	0.120	-0.057	0.000	0.289	0.009	0
friction slip shift	fixed impedance	-0.812	0.096	-0.050	0.041	0.267	0.021	0
friction slip shift	gain scheduled impedance	-0.271	0.150	-0.113	0.000	0.305	0.009	0
friction slip shift	hist gradient gain regressor	-0.688	0.157	-0.065	0.000	0.292	0.009	0
friction slip shift	impedance token policy v4	-0.083	0.062	-0.054	0.000	0.146	0.007	0
friction slip shift	learned gain regressor	-0.521	0.130	-0.047	0.000	0.298	0.006	0
friction slip shift	oracle impedance	-0.708	0.183	-0.124	0.000	0.290	0.012	0
friction slip shift	random forest gain regressor	-0.938	0.060	-0.057	0.000	0.261	0.009	0
friction slip shift	risk averse impedance	0.000	0.000	0.056	0.000	-0.297	-0.000	0
friction slip shift	robust mpc impedance	0.000	0.000	0.026	0.000	0.082	0.003	0
friction slip shift	token no memory ablation	-0.688	0.130	-0.035	0.000	0.184	0.005	0
nominal surface tracking	adaptive impedance control	-0.729	0.122	-0.136	0.000	0.216	0.012	0
nominal surface tracking	admittance switching control	-1.000	0.000	-0.136	0.002	0.189	0.014	0
nominal surface tracking	conformal safety gain	-0.292	0.102	-0.068	0.000	0.228	0.003	0
nominal surface tracking	ensemble uncertainty gain	-0.208	0.082	-0.070	0.000	0.236	0.003	0
nominal surface tracking	fixed impedance	-0.938	0.086	-0.070	0.038	0.201	0.012	0
nominal surface tracking	gain scheduled impedance	0.000	0.000	-0.126	0.000	0.243	0.002	0
nominal surface tracking	hist gradient gain regressor	-0.062	0.060	-0.076	0.000	0.242	0.005	0
nominal surface tracking	impedance token policy v4	-0.938	0.060	-0.060	0.000	0.176	0.004	0
nominal surface tracking	learned gain regressor	0.000	0.000	-0.059	0.000	0.254	0.003	0
nominal surface tracking	oracle impedance	-0.062	0.060	-0.139	0.000	0.244	0.002	0
nominal surface tracking	random forest gain regressor	-0.854	0.074	-0.070	0.000	0.201	0.003	0
nominal surface tracking	risk averse impedance	0.000	0.000	0.056	0.000	-0.176	-0.001	0
nominal surface tracking	robust mpc impedance	0.000	0.000	0.023	0.000	0.051	0.000	0
nominal surface tracking	token no memory ablation	-0.938	0.060	-0.037	0.000	0.133	0.000	0
sensor noise burst	adaptive impedance control	-0.875	0.102	-0.093	0.001	0.283	0.033	0
sensor noise burst	admittance switching control	-1.000	0.000	-0.090	0.000	0.255	0.034	0
sensor noise burst	conformal safety gain	-0.708	0.120	-0.038	0.000	0.290	0.030	0
sensor noise burst	ensemble uncertainty gain	-0.438	0.137	-0.039	0.000	0.298	0.024	0
sensor noise burst	fixed impedance	-0.979	0.041	-0.023	0.028	0.262	0.029	0
sensor noise burst	gain scheduled impedance	-0.042	0.053	-0.082	0.000	0.310	0.021	0
sensor noise burst	hist gradient gain regressor	-0.312	0.179	-0.045	0.000	0.303	0.027	0
sensor noise burst	impedance token policy v4	-0.250	0.062	-0.035	0.000	0.200	0.011	0
sensor noise burst	learned gain regressor	-0.042	0.053	-0.024	0.000	0.315	0.020	0
sensor noise burst	oracle impedance	-0.354	0.168	-0.097	0.001	0.302	0.026	0
sensor noise burst	random forest gain regressor	-0.979	0.041	-0.037	0.000	0.260	0.023	0
sensor noise burst	risk averse impedance	0.000	0.000	0.046	0.000	-0.178	-0.002	0
sensor noise burst	robust mpc impedance	0.000	0.000	0.025	0.000	0.100	0.011	0
sensor noise burst	token no memory ablation	-1.000	0.000	-0.026	0.000	0.210	0.019	0
stick slip cycle	adaptive impedance control	-0.979	0.041	-0.117	0.000	0.342	0.029	0
stick slip cycle	admittance switching control	-1.000	0.000	-0.112	0.000	0.322	0.031	0
stick slip cycle	conformal safety gain	-0.917	0.087	-0.058	0.000	0.352	0.020	0
stick slip cycle	ensemble uncertainty gain	-0.854	0.096	-0.057	0.000	0.359	0.022	0
stick slip cycle	fixed impedance	-1.000	0.000	-0.052	0.039	0.332	0.037	0
stick slip cycle	gain scheduled impedance	-0.542	0.148	-0.102	0.000	0.375	0.027	0
stick slip cycle	hist gradient gain regressor	-0.771	0.174	-0.064	0.000	0.361	0.022	0
stick slip cycle	impedance token policy v4	-0.042	0.053	-0.052	0.000	0.193	0.013	0
stick slip cycle	learned gain regressor	-0.833	0.062	-0.058	0.000	0.365	0.019	0
stick slip cycle	oracle impedance	-0.812	0.130	-0.115	0.000	0.358	0.024	0
stick slip cycle	random forest gain regressor	-1.000	0.000	-0.060	0.000	0.330	0.022	0
stick slip cycle	risk averse impedance	0.000	0.000	0.045	0.000	-0.224	0.000	0
stick slip cycle	robust mpc impedance	0.000	0.000	0.017	0.000	0.137	0.005	0
stick slip cycle	token no memory ablation	-0.479	0.130	-0.037	0.000	0.255	0.011	0
stiffness shift	adaptive impedance control	-0.521	0.114	-0.081	0.030	0.269	0.000	0
stiffness shift	admittance switching control	-0.875	0.102	-0.070	0.028	0.233	0.007	0
stiffness shift	conformal safety gain	-0.083	0.087	-0.007	0.020	0.280	-0.001	0

Split	Comparison	dSucc.	CI	dErr.	dSafe	dSlip	dChat.	Wins
stiffness shift	ensemble uncertainty gain	-0.021	0.041	0.011	0.028	0.289	-0.001	0
stiffness shift	fixed impedance	0.000	0.000	0.049	0.020	0.251	0.009	0
stiffness shift	gain scheduled impedance	-0.021	0.041	-0.053	0.024	0.294	-0.005	0
stiffness shift	hist gradient gain regressor	-0.021	0.041	0.011	0.024	0.292	-0.000	0
stiffness shift	impedance token policy v4	-0.417	0.175	-0.034	0.005	0.187	0.001	0
stiffness shift	learned gain regressor	0.000	0.000	0.002	0.035	0.307	-0.007	0
stiffness shift	oracle impedance	-0.104	0.060	-0.101	0.019	0.288	-0.003	0
stiffness shift	random forest gain regressor	-0.062	0.060	0.021	0.029	0.246	-0.002	0
stiffness shift	risk averse impedance	0.000	0.000	0.062	-0.000	-0.146	-0.000	0
stiffness shift	robust mpc impedance	0.000	0.000	0.055	0.007	0.092	0.002	0
stiffness shift	token no memory ablation	-0.375	0.102	-0.012	0.008	0.193	0.002	0
surface discontinuity	adaptive impedance control	-1.000	0.000	-0.088	0.000	0.322	0.037	0
surface discontinuity	admittance switching control	-1.000	0.000	-0.091	0.004	0.307	0.040	0
surface discontinuity	conformal safety gain	-0.875	0.082	-0.040	0.000	0.337	0.025	0
surface discontinuity	ensemble uncertainty gain	-0.583	0.107	-0.034	0.000	0.345	0.029	0
surface discontinuity	fixed impedance	-0.938	0.060	0.021	0.020	0.312	0.032	0
surface discontinuity	gain scheduled impedance	-0.375	0.160	-0.076	0.000	0.360	0.025	0
surface discontinuity	hist gradient gain regressor	-0.646	0.179	-0.035	0.000	0.348	0.027	0
surface discontinuity	impedance token policy v4	0.000	0.000	-0.039	0.000	0.191	0.015	0
surface discontinuity	learned gain regressor	-0.562	0.150	-0.048	0.000	0.356	0.027	0
surface discontinuity	oracle impedance	-0.854	0.074	-0.101	0.000	0.339	0.032	0
surface discontinuity	random forest gain regressor	-0.938	0.060	-0.033	0.000	0.316	0.025	0
surface discontinuity	risk averse impedance	0.000	0.000	0.043	0.000	-0.241	-0.000	0
surface discontinuity	robust mpc impedance	0.000	0.000	0.020	0.000	0.150	0.008	0
surface discontinuity	token no memory ablation	-0.917	0.062	-0.030	0.000	0.246	0.015	0
target force jump	adaptive impedance control	-0.771	0.086	-0.097	0.000	0.267	0.028	0
target force jump	admittance switching control	-0.938	0.060	-0.085	0.009	0.240	0.030	0
target force jump	conformal safety gain	-0.333	0.175	-0.052	0.000	0.285	0.020	0
target force jump	ensemble uncertainty gain	-0.208	0.102	-0.052	0.000	0.290	0.020	0
target force jump	fixed impedance	-0.896	0.060	-0.038	0.050	0.250	0.038	0
target force jump	gain scheduled impedance	-0.042	0.053	-0.081	0.000	0.305	0.021	0
target force jump	hist gradient gain regressor	-0.104	0.106	-0.057	0.000	0.299	0.023	0
target force jump	impedance token policy v4	-0.062	0.086	-0.042	0.000	0.169	0.013	0
target force jump	learned gain regressor	-0.083	0.062	-0.054	0.000	0.303	0.019	0
target force jump	oracle impedance	-0.312	0.157	-0.094	0.000	0.290	0.020	0
target force jump	random forest gain regressor	-0.896	0.060	-0.053	0.000	0.256	0.021	0
target force jump	risk averse impedance	0.000	0.000	0.027	0.000	-0.241	0.001	0
target force jump	robust mpc impedance	0.000	0.000	0.007	0.000	0.111	0.006	0
target force jump	token no memory ablation	-0.854	0.144	-0.031	0.000	0.169	0.010	0

F Full Ablation Metrics

Table 12: Full ablation metrics.

Method	Split	Succ.	Err.	Over.	Safe	Slip	Chat.
ablate no calibration guard	actuator saturation	0.250	0.145	1.425	0.000	0.292	0.029
ablate no calibration guard	combined extreme stress	0.094	0.135	1.409	0.001	0.239	0.103
ablate no calibration guard	combined stress	0.000	0.145	1.477	0.001	0.179	0.068
ablate no calibration guard	stick slip cycle	0.031	0.175	1.236	0.000	0.181	0.017
ablate no discrete tokens	actuator saturation	0.094	0.109	1.537	0.004	0.613	0.041
ablate no discrete tokens	combined extreme stress	0.938	0.119	1.550	0.002	0.574	0.136
ablate no discrete tokens	combined stress	0.688	0.121	1.641	0.007	0.588	0.086
ablate no discrete tokens	stick slip cycle	0.281	0.113	1.267	0.000	0.605	0.033
ablate no force update	actuator saturation	0.000	0.152	1.246	0.000	0.343	0.027
ablate no force update	combined extreme stress	0.000	0.219	1.560	0.001	0.303	0.044
ablate no force update	combined stress	0.000	0.200	1.422	0.000	0.209	0.036
ablate no force update	stick slip cycle	0.000	0.187	1.028	0.000	0.333	0.009
ablate no memory	actuator saturation	0.969	0.162	1.395	0.001	0.501	0.025
ablate no memory	combined extreme stress	0.125	0.157	1.421	0.000	0.459	0.114
ablate no memory	combined stress	0.000	0.169	1.469	0.002	0.434	0.060
ablate no memory	stick slip cycle	0.469	0.184	1.060	0.000	0.464	0.012
ablate no phase memory	actuator saturation	1.000	0.144	1.439	0.001	0.500	0.024
ablate no phase memory	combined extreme stress	0.219	0.145	1.451	0.000	0.433	0.100
ablate no phase memory	combined stress	0.094	0.155	1.512	0.002	0.406	0.061
ablate no phase memory	stick slip cycle	0.969	0.158	1.091	0.000	0.472	0.016
ablate no safety penalty	actuator saturation	0.000	0.185	1.298	0.000	0.264	0.017
ablate no safety penalty	combined extreme stress	0.000	0.163	1.311	0.000	0.234	0.070
ablate no safety penalty	combined stress	0.000	0.182	1.344	0.000	0.104	0.038
ablate no safety penalty	stick slip cycle	0.000	0.219	0.994	0.000	0.196	0.002
ablate no slip context	actuator saturation	0.000	0.172	1.323	0.000	0.256	0.020
ablate no slip context	combined extreme stress	0.000	0.157	1.360	0.000	0.217	0.084
ablate no slip context	combined stress	0.000	0.173	1.378	0.000	0.153	0.042
ablate no slip context	stick slip cycle	0.000	0.206	0.995	0.000	0.208	0.004
ablate no tail risk objective	actuator saturation	1.000	0.134	1.448	0.001	0.449	0.026
ablate no tail risk objective	combined extreme stress	0.625	0.135	1.434	0.001	0.413	0.116
ablate no tail risk objective	combined stress	0.312	0.146	1.503	0.001	0.361	0.066
ablate no tail risk objective	stick slip cycle	0.969	0.143	1.113	0.000	0.407	0.015

Method	Split	Succ.	Err.	Over.	Safe	Slip	Chat.
ablate no transition planner	actuator saturation	0.000	0.185	1.284	0.000	0.252	0.017
ablate no transition planner	combined extreme stress	0.000	0.163	1.323	0.000	0.238	0.071
ablate no transition planner	combined stress	0.000	0.181	1.355	0.000	0.104	0.035
ablate no transition planner	stick slip cycle	0.000	0.220	1.001	0.000	0.206	0.003
impedance token policy v4	actuator saturation	0.156	0.151	1.361	0.000	0.475	0.026
impedance token policy v4	combined extreme stress	0.000	0.140	1.414	0.001	0.417	0.107
impedance token policy v4	combined stress	0.000	0.152	1.450	0.001	0.371	0.065
impedance token policy v4	stick slip cycle	0.000	0.169	1.087	0.000	0.420	0.018
learned only token replacement	actuator saturation	0.688	0.157	1.533	0.004	0.593	0.026
learned only token replacement	combined extreme stress	0.969	0.169	1.664	0.012	0.552	0.117
learned only token replacement	combined stress	1.000	0.178	1.635	0.012	0.558	0.066
learned only token replacement	stick slip cycle	0.844	0.157	1.168	0.000	0.580	0.031
token full v5	actuator saturation	0.000	0.184	1.318	0.000	0.276	0.016
token full v5	combined extreme stress	0.000	0.162	1.318	0.000	0.227	0.072
token full v5	combined stress	0.000	0.180	1.360	0.000	0.116	0.041
token full v5	stick slip cycle	0.000	0.219	0.994	0.000	0.211	0.003

G FULL STRESS SWEEP METRICS

Table 13: Full stress sweep metrics.

Method	Split	Lvl	Succ.	Err.	Over.	Safe	Slip	Chat.	Prog.
adaptive impedance control	combined extreme stress	0.000	0.583	0.127	1.270	0.000	0.598	0.056	0.854
adaptive impedance control	combined extreme stress	0.250	0.792	0.132	1.394	0.000	0.583	0.071	0.840
adaptive impedance control	combined extreme stress	0.500	1.000	0.127	1.497	0.001	0.558	0.109	0.824
adaptive impedance control	combined extreme stress	0.750	1.000	0.134	1.585	0.003	0.533	0.138	0.801
adaptive impedance control	combined extreme stress	1.000	0.958	0.144	1.645	0.005	0.506	0.159	0.785
adaptive impedance control	combined stress	0.000	0.500	0.124	1.215	0.000	0.598	0.046	0.856
adaptive impedance control	combined stress	0.250	0.917	0.124	1.386	0.000	0.586	0.059	0.839
adaptive impedance control	combined stress	0.500	1.000	0.119	1.496	0.000	0.567	0.088	0.820
adaptive impedance control	combined stress	0.750	1.000	0.126	1.604	0.002	0.545	0.124	0.797
adaptive impedance control	combined stress	1.000	0.917	0.142	1.694	0.007	0.490	0.127	0.768
adaptive impedance control	friction slip shift	0.000	0.542	0.112	1.115	0.012	0.599	0.003	0.857
adaptive impedance control	friction slip shift	0.250	0.875	0.116	1.315	0.000	0.587	0.026	0.840
adaptive impedance control	friction slip shift	0.500	1.000	0.116	1.449	0.000	0.571	0.068	0.822
adaptive impedance control	friction slip shift	0.750	1.000	0.121	1.498	0.000	0.545	0.095	0.803
adaptive impedance control	friction slip shift	1.000	1.000	0.138	1.635	0.002	0.504	0.116	0.778
conformal safety gain	combined extreme stress	0.000	0.167	0.209	1.205	0.000	0.610	0.052	0.827
conformal safety gain	combined extreme stress	0.250	0.750	0.191	1.317	0.000	0.591	0.078	0.812
conformal safety gain	combined extreme stress	0.500	1.000	0.172	1.356	0.000	0.567	0.100	0.798
conformal safety gain	combined extreme stress	0.750	1.000	0.158	1.423	0.000	0.541	0.114	0.777
conformal safety gain	combined extreme stress	1.000	0.917	0.147	1.454	0.000	0.517	0.143	0.765
conformal safety gain	combined stress	0.000	0.083	0.210	1.130	0.000	0.612	0.031	0.828
conformal safety gain	combined stress	0.250	0.542	0.179	1.282	0.000	0.597	0.057	0.811
conformal safety gain	combined stress	0.500	0.875	0.166	1.381	0.000	0.576	0.078	0.794
conformal safety gain	combined stress	0.750	0.875	0.154	1.366	0.000	0.549	0.112	0.774
conformal safety gain	combined stress	1.000	0.583	0.145	1.430	0.001	0.480	0.142	0.746
conformal safety gain	friction slip shift	0.000	0.125	0.196	1.036	0.000	0.611	0.000	0.829
conformal safety gain	friction slip shift	0.250	0.458	0.172	1.219	0.000	0.598	0.028	0.812
conformal safety gain	friction slip shift	0.500	0.875	0.165	1.309	0.000	0.579	0.043	0.797
conformal safety gain	friction slip shift	0.750	1.000	0.144	1.288	0.000	0.552	0.080	0.779

Method	Split	Lvl	Succ.	Err.	Over.	Safe	Slip	Chat.	Prog.
conformal safety gain	friction slip shift	1.000	0.833	0.138	1.322	0.000	0.498	0.103	0.754
ensemble uncertainty gain	combined extreme stress	0.000	0.042	0.209	1.193	0.000	0.621	0.051	0.831
ensemble uncertainty gain	combined extreme stress	0.250	0.333	0.199	1.365	0.000	0.602	0.055	0.816
ensemble uncertainty gain	combined extreme stress	0.500	0.792	0.181	1.468	0.000	0.576	0.088	0.800
ensemble uncertainty gain	combined extreme stress	0.750	1.000	0.165	1.565	0.004	0.550	0.111	0.778
ensemble uncertainty gain	combined extreme stress	1.000	0.958	0.156	1.595	0.004	0.515	0.137	0.763
ensemble uncertainty gain	combined stress	0.000	0.042	0.209	1.141	0.000	0.625	0.034	0.832
ensemble uncertainty gain	combined stress	0.250	0.375	0.185	1.338	0.000	0.605	0.051	0.815
ensemble uncertainty gain	combined stress	0.500	0.750	0.170	1.488	0.000	0.588	0.072	0.797
ensemble uncertainty gain	combined stress	0.750	0.958	0.160	1.565	0.003	0.558	0.094	0.775
ensemble uncertainty gain	combined stress	1.000	0.583	0.153	1.606	0.003	0.482	0.113	0.744
ensemble uncertainty gain	friction slip shift	0.000	0.042	0.195	1.051	0.000	0.624	0.000	0.833
ensemble uncertainty gain	friction slip shift	0.250	0.250	0.179	1.275	0.000	0.610	0.021	0.816
ensemble uncertainty gain	friction slip shift	0.500	0.625	0.170	1.424	0.000	0.591	0.043	0.800
ensemble uncertainty gain	friction slip shift	0.750	0.958	0.150	1.532	0.001	0.561	0.071	0.781
ensemble uncertainty gain	friction slip shift	1.000	0.792	0.148	1.607	0.005	0.505	0.088	0.752
gain scheduled impedance	combined extreme stress	0.000	0.000	0.155	1.215	0.000	0.626	0.054	0.869
gain scheduled impedance	combined extreme stress	0.250	0.250	0.149	1.336	0.000	0.612	0.079	0.856
gain scheduled impedance	combined extreme stress	0.500	0.875	0.135	1.393	0.000	0.590	0.106	0.842
gain scheduled impedance	combined extreme stress	0.750	0.917	0.134	1.477	0.001	0.569	0.136	0.822
gain scheduled impedance	combined extreme stress	1.000	0.958	0.137	1.537	0.002	0.543	0.166	0.808
gain scheduled impedance	combined stress	0.000	0.000	0.153	1.147	0.000	0.627	0.039	0.871
gain scheduled impedance	combined stress	0.250	0.167	0.136	1.319	0.000	0.616	0.061	0.856
gain scheduled impedance	combined stress	0.500	0.500	0.127	1.426	0.000	0.599	0.095	0.839
gain scheduled impedance	combined stress	0.750	0.917	0.125	1.438	0.000	0.577	0.124	0.819
gain scheduled impedance	combined stress	1.000	1.000	0.134	1.499	0.000	0.546	0.145	0.794
gain scheduled impedance	friction slip shift	0.000	0.042	0.137	1.049	0.000	0.628	0.000	0.872
gain scheduled impedance	friction slip shift	0.250	0.167	0.132	1.249	0.000	0.616	0.028	0.857
gain scheduled impedance	friction slip shift	0.500	0.500	0.123	1.363	0.000	0.599	0.056	0.841
gain scheduled impedance	friction slip shift	0.750	0.833	0.117	1.328	0.000	0.580	0.108	0.824
gain scheduled impedance	friction slip shift	1.000	1.000	0.127	1.424	0.000	0.552	0.124	0.802
hist gradient gain regressor	combined extreme stress	0.000	0.083	0.203	1.211	0.000	0.618	0.052	0.837
hist gradient gain regressor	combined extreme stress	0.250	0.417	0.192	1.360	0.000	0.597	0.065	0.822
hist gradient gain regressor	combined extreme stress	0.500	0.917	0.181	1.454	0.000	0.572	0.094	0.806
hist gradient gain regressor	combined extreme stress	0.750	1.000	0.184	1.582	0.000	0.544	0.103	0.782
hist gradient gain regressor	combined extreme stress	1.000	0.875	0.191	1.663	0.006	0.513	0.139	0.767
hist gradient gain regressor	combined stress	0.000	0.125	0.206	1.148	0.000	0.619	0.039	0.838
hist gradient gain regressor	combined stress	0.250	0.292	0.178	1.350	0.000	0.604	0.049	0.821
hist gradient gain regressor	combined stress	0.500	0.875	0.171	1.503	0.000	0.579	0.075	0.802
hist gradient gain regressor	combined stress	0.750	0.875	0.180	1.625	0.004	0.545	0.084	0.778
hist gradient gain regressor	combined stress	1.000	0.500	0.202	1.708	0.009	0.454	0.107	0.745
hist gradient gain regressor	friction slip shift	0.000	0.083	0.192	1.040	0.000	0.620	0.000	0.840
hist gradient gain regressor	friction slip shift	0.250	0.333	0.173	1.261	0.000	0.606	0.024	0.822

Method	Split	Lvl	Succ.	Err.	Over.	Safe	Slip	Chat.	Prog.
hist gradient gain regressor	friction slip shift	0.500	0.750	0.166	1.434	0.000	0.584	0.047	0.804
hist gradient gain regressor	friction slip shift	0.750	1.000	0.176	1.552	0.001	0.551	0.061	0.783
hist gradient gain regressor	friction slip shift	1.000	0.833	0.195	1.679	0.006	0.493	0.082	0.755
impedance token policy v4	combined extreme stress	0.000	0.708	0.219	1.118	0.000	0.553	0.030	0.750
impedance token policy v4	combined extreme stress	0.250	0.042	0.182	1.214	0.000	0.480	0.054	0.729
impedance token policy v4	combined extreme stress	0.500	0.000	0.147	1.264	0.000	0.444	0.082	0.725
impedance token policy v4	combined extreme stress	0.750	0.000	0.146	1.339	0.000	0.391	0.106	0.717
impedance token policy v4	combined extreme stress	1.000	0.000	0.160	1.447	0.001	0.389	0.131	0.713
impedance token policy v4	combined stress	0.000	0.667	0.218	1.049	0.000	0.561	0.014	0.759
impedance token policy v4	combined stress	0.250	0.042	0.172	1.198	0.000	0.475	0.043	0.730
impedance token policy v4	combined stress	0.500	0.042	0.146	1.295	0.000	0.422	0.068	0.725
impedance token policy v4	combined stress	0.750	0.000	0.135	1.359	0.000	0.356	0.103	0.717
impedance token policy v4	combined stress	1.000	0.000	0.154	1.477	0.001	0.302	0.114	0.705
impedance token policy v4	friction slip shift	0.000	0.917	0.207	1.006	0.000	0.562	0.002	0.765
impedance token policy v4	friction slip shift	0.250	0.042	0.169	1.107	0.000	0.493	0.014	0.729
impedance token policy v4	friction slip shift	0.500	0.000	0.143	1.215	0.000	0.425	0.044	0.726
impedance token policy v4	friction slip shift	0.750	0.000	0.130	1.279	0.000	0.379	0.083	0.719
impedance token policy v4	friction slip shift	1.000	0.000	0.150	1.405	0.000	0.326	0.099	0.712
impedance token policy v5	combined extreme stress	0.000	0.000	0.269	1.101	0.000	0.410	0.025	0.716
impedance token policy v5	combined extreme stress	0.250	0.000	0.215	1.178	0.000	0.354	0.039	0.702
impedance token policy v5	combined extreme stress	0.500	0.000	0.177	1.209	0.000	0.322	0.056	0.687
impedance token policy v5	combined extreme stress	0.750	0.000	0.156	1.285	0.000	0.239	0.092	0.667
impedance token policy v5	combined extreme stress	1.000	0.000	0.147	1.384	0.000	0.243	0.123	0.663
impedance token policy v5	combined stress	0.000	0.000	0.271	1.018	0.000	0.393	0.009	0.719
impedance token policy v5	combined stress	0.250	0.000	0.207	1.125	0.000	0.339	0.021	0.697
impedance token policy v5	combined stress	0.500	0.000	0.176	1.200	0.000	0.228	0.054	0.675
impedance token policy v5	combined stress	0.750	0.000	0.153	1.276	0.000	0.151	0.081	0.659
impedance token policy v5	combined stress	1.000	0.000	0.138	1.338	0.000	0.095	0.125	0.637
impedance token policy v5	friction slip shift	0.000	0.000	0.271	0.900	0.000	0.374	0.000	0.709
impedance token policy v5	friction slip shift	0.250	0.000	0.207	1.036	0.000	0.339	0.004	0.694
impedance token policy v5	friction slip shift	0.500	0.000	0.177	1.132	0.000	0.210	0.021	0.673
impedance token policy v5	friction slip shift	0.750	0.000	0.145	1.181	0.000	0.170	0.061	0.663
impedance token policy v5	friction slip shift	1.000	0.000	0.132	1.320	0.000	0.094	0.101	0.642
learned gain regressor	combined extreme stress	0.000	0.000	0.195	1.238	0.000	0.634	0.054	0.851
learned gain regressor	combined extreme stress	0.250	0.125	0.187	1.380	0.000	0.615	0.064	0.836
learned gain regressor	combined extreme stress	0.500	0.750	0.176	1.496	0.000	0.591	0.083	0.822
learned gain regressor	combined extreme stress	0.750	0.958	0.158	1.572	0.004	0.568	0.110	0.802
learned gain regressor	combined extreme stress	1.000	0.958	0.152	1.648	0.012	0.544	0.138	0.788
learned gain regressor	combined stress	0.000	0.000	0.193	1.178	0.000	0.639	0.037	0.853
learned gain regressor	combined stress	0.250	0.167	0.176	1.364	0.000	0.618	0.049	0.835
learned gain regressor	combined stress	0.500	0.542	0.169	1.562	0.001	0.601	0.074	0.817
learned gain regressor	combined stress	0.750	0.917	0.152	1.557	0.004	0.571	0.096	0.796
learned gain regressor	combined stress	1.000	0.833	0.148	1.683	0.008	0.497	0.127	0.769
learned gain regressor	friction slip shift	0.000	0.000	0.186	1.098	0.000	0.639	0.000	0.854
learned gain regressor	friction slip shift	0.250	0.083	0.170	1.305	0.000	0.623	0.015	0.836
learned gain regressor	friction slip shift	0.500	0.542	0.164	1.468	0.000	0.601	0.035	0.819
learned gain regressor	friction slip shift	0.750	1.000	0.144	1.543	0.000	0.573	0.084	0.802
learned gain regressor	friction slip shift	1.000	1.000	0.145	1.595	0.007	0.521	0.098	0.777

Method	Split	Lvl	Succ.	Err.	Over.	Safe	Slip	Chat.	Prog.
oracle impedance	combined extreme stress	0.000	0.042	0.124	1.271	0.000	0.623	0.049	0.855
oracle impedance	combined extreme stress	0.250	0.500	0.120	1.404	0.000	0.598	0.073	0.834
oracle impedance	combined extreme stress	0.500	0.958	0.115	1.405	0.000	0.571	0.106	0.812
oracle impedance	combined extreme stress	0.750	0.917	0.121	1.471	0.000	0.508	0.138	0.784
oracle impedance	combined extreme stress	1.000	0.875	0.129	1.486	0.000	0.440	0.166	0.766
oracle impedance	combined stress	0.000	0.042	0.120	1.229	0.000	0.624	0.038	0.857
oracle impedance	combined stress	0.250	0.417	0.114	1.323	0.000	0.607	0.062	0.833
oracle impedance	combined stress	0.500	0.833	0.110	1.395	0.000	0.584	0.095	0.806
oracle impedance	combined stress	0.750	0.875	0.112	1.370	0.000	0.519	0.123	0.779
oracle impedance	combined stress	1.000	0.667	0.131	1.436	0.001	0.357	0.141	0.749
oracle impedance	friction slip shift	0.000	0.000	0.110	1.133	0.015	0.627	0.003	0.857
oracle impedance	friction slip shift	0.250	0.083	0.109	1.271	0.001	0.612	0.025	0.834
oracle impedance	friction slip shift	0.500	0.750	0.105	1.346	0.000	0.586	0.060	0.809
oracle impedance	friction slip shift	0.750	1.000	0.109	1.287	0.000	0.527	0.100	0.784
oracle impedance	friction slip shift	1.000	0.833	0.127	1.350	0.000	0.395	0.110	0.756
random forest gain regressor	combined extreme stress	0.000	0.708	0.207	1.214	0.000	0.588	0.055	0.806
random forest gain regressor	combined extreme stress	0.250	0.875	0.198	1.361	0.000	0.568	0.059	0.793
random forest gain regressor	combined extreme stress	0.500	1.000	0.182	1.490	0.000	0.539	0.096	0.778
random forest gain regressor	combined extreme stress	0.750	1.000	0.181	1.587	0.002	0.519	0.102	0.762
random forest gain regressor	combined extreme stress	1.000	0.708	0.184	1.635	0.007	0.491	0.128	0.748
random forest gain regressor	combined stress	0.000	0.583	0.208	1.149	0.000	0.588	0.035	0.808
random forest gain regressor	combined stress	0.250	1.000	0.183	1.360	0.000	0.568	0.045	0.792
random forest gain regressor	combined stress	0.500	0.917	0.170	1.506	0.000	0.558	0.076	0.778
random forest gain regressor	combined stress	0.750	0.958	0.175	1.626	0.004	0.530	0.093	0.763
random forest gain regressor	combined stress	1.000	0.500	0.190	1.677	0.008	0.485	0.101	0.739
random forest gain regressor	friction slip shift	0.000	0.583	0.194	1.052	0.000	0.587	0.000	0.807
random forest gain regressor	friction slip shift	0.250	0.875	0.177	1.272	0.000	0.573	0.022	0.794
random forest gain regressor	friction slip shift	0.500	0.958	0.167	1.438	0.000	0.558	0.050	0.781
random forest gain regressor	friction slip shift	0.750	0.958	0.168	1.535	0.000	0.539	0.075	0.769
random forest gain regressor	friction slip shift	1.000	0.708	0.185	1.657	0.005	0.501	0.094	0.746
risk averse impedance	combined extreme stress	0.000	0.000	0.331	1.058	0.000	0.201	0.018	0.606
risk averse impedance	combined extreme stress	0.250	0.000	0.267	1.161	0.000	0.095	0.036	0.593
risk averse impedance	combined extreme stress	0.500	0.000	0.219	1.177	0.000	0.014	0.050	0.577
risk averse impedance	combined extreme stress	0.750	0.000	0.195	1.270	0.000	0.002	0.072	0.556
risk averse impedance	combined extreme stress	1.000	0.000	0.180	1.322	0.000	0.003	0.091	0.544
risk averse impedance	combined stress	0.000	0.000	0.336	0.982	0.000	0.209	0.005	0.609
risk averse impedance	combined stress	0.250	0.000	0.256	1.118	0.000	0.077	0.028	0.592
risk averse impedance	combined stress	0.500	0.000	0.219	1.203	0.000	0.000	0.043	0.572
risk averse impedance	combined stress	0.750	0.000	0.188	1.239	0.000	0.000	0.071	0.551
risk averse impedance	combined stress	1.000	0.000	0.173	1.353	0.001	0.000	0.095	0.524
risk averse impedance	friction slip shift	0.000	0.000	0.331	0.883	0.000	0.211	0.000	0.610
risk averse impedance	friction slip shift	0.250	0.000	0.254	1.030	0.000	0.084	0.008	0.593
risk averse impedance	friction slip shift	0.500	0.000	0.217	1.123	0.000	0.000	0.017	0.575
risk averse impedance	friction slip shift	0.750	0.000	0.181	1.154	0.000	0.000	0.043	0.557
risk averse impedance	friction slip shift	1.000	0.000	0.167	1.264	0.000	0.000	0.070	0.534
robust mpc impedance	combined extreme stress	0.000	0.000	0.302	1.117	0.000	0.428	0.037	0.682
robust mpc impedance	combined extreme stress	0.250	0.000	0.249	1.212	0.000	0.403	0.054	0.673
robust mpc impedance	combined extreme stress	0.500	0.000	0.211	1.287	0.000	0.372	0.073	0.662
robust mpc impedance	combined extreme stress	0.750	0.000	0.192	1.360	0.000	0.320	0.084	0.647
robust mpc impedance	combined extreme stress	1.000	0.000	0.181	1.406	0.000	0.279	0.113	0.636
robust mpc impedance	combined stress	0.000	0.000	0.305	1.036	0.000	0.435	0.017	0.685
robust mpc impedance	combined stress	0.250	0.000	0.237	1.192	0.000	0.400	0.043	0.675
robust mpc impedance	combined stress	0.500	0.000	0.204	1.276	0.000	0.368	0.067	0.663
robust mpc impedance	combined stress	0.750	0.000	0.183	1.322	0.000	0.297	0.095	0.649
robust mpc impedance	combined stress	1.000	0.000	0.176	1.426	0.001	0.180	0.102	0.631
robust mpc impedance	friction slip shift	0.000	0.000	0.298	0.952	0.000	0.435	0.000	0.686
robust mpc impedance	friction slip shift	0.250	0.000	0.232	1.120	0.000	0.411	0.013	0.676

Method	Split	Lvl	Succ.	Err.	Over.	Safe	Slip	Chat.	Prog.
robust mpc impedance	friction slip shift	0.500	0.000	0.203	1.220	0.000	0.366	0.021	0.665
robust mpc impedance	friction slip shift	0.750	0.000	0.174	1.236	0.000	0.328	0.061	0.652
robust mpc impedance	friction slip shift	1.000	0.000	0.169	1.340	0.000	0.218	0.074	0.637

References

Fares J. Abu-Dakka and Matteo Saveriano. Variable impedance control and learning–a review. *Frontiers in Robotics and AI*, 7:590681, 2020. doi: 10.3389/frobt.2020.590681. URL `https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2020.590681/full`.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324. URL `https://link.springer.com/article/10.1023/A%3A1010933404324`.

Jonas Buchli, Freek Stulp, Evangelos Theodorou, and Stefan Schaal. Learning variable impedance control. *The International Journal of Robotics Research*, 30(7):820–833, 2011. doi: 10.1177/0278364911402527. URL `https://journals.sagepub.com/doi/10.1177/0278364911402527`.

Neville Hogan. Impedance control: An approach to manipulation: Part i–theory. *Journal of Dynamic Systems, Measurement, and Control*, 107(1):1–7, 1985a. URL `https://asmedigitalcollection.asme.org/dynamicsystems/article/107/1/1/400604/Impedance-Control-An-Approach-to-Manipulation-Part`.

Neville Hogan. Impedance control: An approach to manipulation: Part iii–applications. *Journal of Dynamic Systems, Measurement, and Control*, 107(1):17–24, 1985b. URL `https://asmedigitalcollection.asme.org/dynamicsystems/article/107/1/17/400610/Impedance-Control-An-Approach-to-Manipulation-Part`.

Klas Kronander and Aude Billard. Stability considerations for variable impedance control. *IEEE Transactions on Robotics*, 32(5):1298–1305, 2016. doi: 10.1109/TRO.2016.2593492. URL `https://dl.acm.org/doi/10.1109/TRO.2016.2593492`.

D. Q. Mayne, J. B. Rawlings, C. V. Rao, and P. O. M. Scokaert. Constrained model predictive control: Stability and optimality. *Automatica*, 36(6):789–814, 2000. doi: 10.1016/S0005-1098(99)00214-9.

Fabian Pedregosa, Gael Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Edouard Duchesnay. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011. URL `https://www.jmlr.org/papers/v12/pedregosa11a.html`.

Marc H. Raibert and John J. Craig. Hybrid position/force control of manipulators. *Journal of Dynamic Systems, Measurement, and Control*, 103(2):126–133, 1981. URL `https://asmedigitalcollection.asme.org/dynamicsystems/article/103/2/126/400298/Hybrid-Position-Force-Control-of-Manipulators`.

Glenn Shafer and Vladimir Vovk. A tutorial on conformal prediction. *Journal of Machine Learning Research*, 9:371–421, 2008. URL `https://jmlr.org/papers/v9/shafer08a.html`.

Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pp. 5026–5033, 2012. doi: 10.1109/IROS.2012.6386109. URL `https://ieeexplore.ieee.org/document/6386109`.